

Complex Moduli and Special Geometry of Calabi–Yau Manifolds

Geometric Aspects of the B-model

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Abstract

These notes study the geometry of the complex structure moduli space of Calabi–Yau manifolds from the perspective of Hodge theory.

After reviewing the necessary background on Kähler geometry and Hodge theory, we introduce Calabi–Yau manifolds and their deformation spaces. We then develop the variation of Hodge structures associated with families of Calabi–Yau manifolds, including the period map and the Gauss–Manin connection.

Finally, we discuss special geometry on the complex moduli space, with particular emphasis on the case of Calabi–Yau threefolds.

The main exposition follows Chapters 2.3–2.5 of the paper *The B-Model Approach to Topological String Theory on Calabi–Yau n -Folds* [1]. Mathematical background and complementary perspectives are drawn from Voisin’s *Hodge Theory and Complex Algebraic Geometry I* [2] and Griffiths–Harris’ *Principles of Algebraic Geometry* [3]. Additional background on mirror symmetry from the algebraic-geometric viewpoint may be found in Cox and Katz’s *Mirror Symmetry and Algebraic Geometry* [4].

Some parts of the exposition make use of terminology and results from algebraic geometry and derived categories. However, familiarity with these topics is not strictly necessary for most of the notes, and readers may safely skip certain more technical sections without affecting the overall understanding of the main ideas; these sections are included mainly for completeness. Topics such as period domains, their homogeneous structure, and intermediate Jacobians are not discussed here; for these subjects we refer the reader to Voisin [2].

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1 Compact Kähler Manifolds and the Hodge Theorem

1.1 Metric Aspects

1.1.1 Basic Definitions

Definition 1.1. Let M be a complex manifold of complex dimension n . A *hermitian metric* on M is a smooth tensor

$$G \in \Gamma(T^*M \otimes \overline{T^*M})$$

such that in local holomorphic coordinates $(z^1, \dots, z^n)$ it can be written as

$$ds^2 = \sum_{i,j=1}^n G_{i\bar{j}}(z) dz^i \otimes d\bar{z}^{\bar{j}},$$

with the following properties:

1. (Hermitian symmetry)

$$G_{j\bar{i}} = \overline{G_{i\bar{j}}}.$$

2. (Positive definiteness)

$$v^i G_{i\bar{j}} \bar{v}^{\bar{j}} > 0, \quad \forall 0 \neq v \in \mathbb{C}^n.$$

Remark 1.2. Positive definiteness implies in particular

$$\det(G_{i\bar{j}}) > 0.$$

Thus $G_{i\bar{j}}$ defines a positive-definite Hermitian matrix at every point.

Proposition 1.3. *Let (M, J) be an almost complex manifold and let G' be any real Riemannian metric on M . Then the tensor*

$$G_{mn} = \frac{1}{2}(G'_{mn} + J_m^a J_n^b G'_{ab})$$

defines a hermitian metric.

Proof. Hermiticity means

$$G(X, Y) = G(JX, JY).$$

In local coordinates this is equivalent to

$$G_{mn} = J_m^a J_n^b G_{ab}.$$

The above symmetrization of G' with respect to J satisfies this by construction. □

Proposition 1.4. *Let (M, G, J) be a hermitian manifold. Define*

$$J_{nm} = J_n^a G_{am}.$$

Then

$$J_{nm} = -J_{mn}.$$

Hence the tensor

$$\omega = J_{nm} dw^n \wedge dw^m$$

is a real 2-form on M .

Proof. Using $J_p^m J_m^a = -\delta_p^a$ and the hermiticity condition, one obtains

$$J_{nm} = -J_{mn}.$$

□

Proposition 1.5. *In holomorphic coordinates the 2-form ω takes the form*

$$\omega = i \sum_{i,j=1}^n G_{i\bar{j}} dz^i \wedge d\bar{z}^{\bar{j}}.$$

In particular, ω is of type $(1, 1)$.

Definition 1.6. The 2-form ω defined above is a real $(1, 1)$ -form and is called the *fundamental form* associated to the hermitian metric.

Proposition 1.7. *Let*

$$G := \det(G_{i\bar{j}}) > 0.$$

Then wedging ω n times yields a positive volume form

$$\text{vol} = \frac{\omega^n}{n!} = i^n \det(G_{i\bar{j}}) dz^1 \wedge d\bar{z}^{\bar{1}} \wedge \cdots \wedge dz^n \wedge d\bar{z}^{\bar{n}} = 2^n \det(G_{ij})^{1/2} dw^1 \wedge \cdots \wedge dw^{2n}.$$

In particular, M is orientable.

Remark 1.8. In the expression, the determinant appears once as $\det(G_{i\bar{j}})$ and once as $\det(G_{mn})^{1/2}$.

This difference comes from the fact that $G_{i\bar{j}}$ is the complex $n \times n$ Hermitian matrix, while G_{mn} is the associated real $2n \times 2n$ Riemannian metric in real coordinates $w^1, \dots, w^{2n}$.

1.1.2 Operations on Differential Forms

Let $A^{p,q}$ denote the space of smooth (p, q) -forms. Locally a form $\phi \in A^{p,q}$ can be written as

$$\phi = \frac{1}{p!q!} \phi_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q}(z) dz^{i_1} \wedge \cdots \wedge dz^{i_p} \wedge d\bar{z}^{\bar{j}_1} \wedge \cdots \wedge d\bar{z}^{\bar{j}_q}.$$

Definition 1.9. The hermitian metric induces a local inner product on (p, q) -forms given by

$$(\phi, \psi)(z) = \frac{1}{p!q!} \phi_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q} \bar{\psi}^{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q},$$

where

$$\bar{\psi}^{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q} = G^{\bar{k}_1 i_1} \dots G^{\bar{k}_p i_p} G^{\bar{l}_1 j_1} \dots G^{\bar{l}_q j_q} \overline{\psi_{k_1 \dots k_p \bar{l}_1 \dots \bar{l}_q}}.$$

Definition 1.10. The local inner product induces a global bilinear inner product

$$A^{p,q} \times A^{p,q} \longrightarrow \mathbb{C}$$

defined by

$$(\phi, \psi) = \int_M (\phi, \psi)(z) \text{vol}.$$

Proposition 1.11. *If M is compact, the inner product satisfies*

$$(\phi, \psi) = \overline{(\psi, \phi)}, \quad (\phi, \phi) > 0 \quad \text{unless } \phi = 0.$$

Hence $A^{p,q}$ becomes a pre-Hilbert space.

Definition 1.12. For $\psi \in A^{p,q}$ we write

$$\bar{\psi} = \frac{1}{p!q!} \bar{\psi}_{j_1 \dots j_q \bar{i}_1 \dots \bar{i}_p} dz^{j_1} \wedge \cdots \wedge dz^{j_q} \wedge d\bar{z}^{\bar{i}_1} \wedge \cdots \wedge d\bar{z}^{\bar{i}_p}.$$

Moreover,

$$\overline{\psi_{i_1 \dots i_p \bar{j}_1 \dots \bar{j}_q}} = (-1)^{pq} \bar{\psi}_{j_1 \dots j_q \bar{i}_1 \dots \bar{i}_p}.$$

Definition 1.13. The Hodge operator

$$* : A^{p,q} \longrightarrow A^{n-q,n-p}$$

is defined by the relation

$$(\phi, \psi) \text{ vol} = \phi \wedge *\bar{\psi}.$$

Proposition 1.14. The Hodge star on a (p, q) -form ψ is given explicitly by

$$*\psi = \frac{i^n (-1)^{\frac{n(n-1)}{2} + np}}{p! q! (n-p)! (n-q)!} \det(G) \epsilon^{k_1 \dots k_p}_{\bar{j}_1 \dots \bar{j}_{n-p}} \epsilon^{\bar{l}_1 \dots \bar{l}_q}_{i_1 \dots i_{n-q}} \psi_{k_1 \dots k_p \bar{l}_1 \dots \bar{l}_q} dz^{i_1} \wedge \dots \wedge dz^{i_{n-q}} \wedge d\bar{z}^{\bar{j}_1} \wedge \dots \wedge d\bar{z}^{\bar{j}_{n-p}}.$$

Proposition 1.15. For a (p, q) -form ψ one has

$$*\bar{\psi} = \overline{*\psi},$$

and

$$**\psi = (-1)^{pq} \psi.$$

Proposition 1.16. Let $p + q = n$. Then there exists a basis of $A^{p,q}$ consisting of eigenvectors of the Hodge star operator such that

$$*\psi = \pm i^{p-q} \psi.$$

Remark 1.17. To understand the previous proposition concretely, it is useful to consider a local unitary frame.

Let (X, g) be a Kähler manifold and choose local holomorphic coordinates in which the Hermitian metric satisfies

$$G_{i\bar{j}} = \delta_{ij}.$$

Equivalently, we choose a local unitary coframe

$$\{\theta^1, \dots, \theta^n\}$$

such that

$$g = \sum_i \theta^i \otimes \bar{\theta}^i.$$

In this frame the Hodge star operator can be computed explicitly. For instance, in complex dimension $n = 1$ we have

$$\theta^1 = dz, \quad \bar{\theta}^1 = d\bar{z},$$

and the real volume form is

$$\text{vol} = dx \wedge dy.$$

Using

$$*dx = dy, \quad *dy = -dx,$$

one obtains

$$*dz = -i dz, \quad *d\bar{z} = i d\bar{z}.$$

Thus the spaces $A^{1,0}$ and $A^{0,1}$ are eigenspaces of the Hodge star operator with eigenvalues $-i$ and i respectively.

More generally, when $p + q = n$, the operator

$$* : A^{p,q} \rightarrow A^{p,q}$$

is diagonalizable and its eigenvalues are $\pm i^{p-q}$. Choosing a basis consisting of eigenvectors of $*$ therefore yields

$$*\psi = \pm i^{p-q}\psi.$$

This explains the meaning of the “suitable basis” appearing in the proposition.

Definition 1.18. Using the global inner product $(\cdot, \cdot)$, define the adjoint operators

$$\partial^* : A^{p,q} \rightarrow A^{p-1,q}, \quad \bar{\partial}^* : A^{p,q} \rightarrow A^{p,q-1}$$

by the relations

$$(\partial^*\psi, \phi) := (\psi, \partial\phi), \quad (\bar{\partial}^*\psi, \phi) := (\psi, \bar{\partial}\phi).$$

Proposition 1.19. *On a compact manifold one has*

$$\bar{\partial}^* = - * \partial *, \quad \partial^* = - * \bar{\partial} *.$$

Definition 1.20. Define the $\bar{\partial}$ -Laplacian and the de Rham Laplacian by

$$\Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}, \quad \Delta_d = d^\dagger d + dd^\dagger.$$

Obviously, if the metric is Euclidean, i.e. $G_{i\bar{j}} = \delta_{ij}$, then

$$\Delta_{\bar{\partial}} = \frac{1}{2}\Delta_d.$$

1.1.3 The Kähler Condition

Definition 1.21. A hermitian metric $G_{i\bar{j}}$ is called *Kähler* if its fundamental form is closed:

$$d\omega = 0.$$

The form ω is then called the *Kähler form*.

Proposition 1.22. *The condition $d\omega = 0$ is equivalent to*

$$\partial_i G_{j\bar{k}} = \partial_j G_{i\bar{k}}, \quad \partial_{\bar{i}} G_{j\bar{k}} = \partial_{\bar{k}} G_{j\bar{i}}.$$

Proposition 1.23. *Locally, a hermitian metric is Kähler if and only if there exists a real-valued function K such that*

$$G_{i\bar{j}} = \partial_i \partial_{\bar{j}} K.$$

Equivalently,

$$\omega = i \partial \bar{\partial} K = -\frac{1}{2} d(\partial - \bar{\partial}) K.$$

Remark 1.24. Despite its local expression, the Kähler form ω cannot be exact. Indeed, if $\omega = dA$, then by Stokes' theorem one would have

$$\int_M \omega^n = 0,$$

which contradicts the fact that ω^n is a positive volume form. Actually, the equality

$$\omega = i \partial \bar{\partial} K$$

does not imply that the function K is globally defined. Rather, $(\partial - \bar{\partial})K$ need not exist globally.

Definition 1.25. The Kähler potential is defined only up to a Kähler transformation

$$K(z, \bar{z}) \longmapsto K(z, \bar{z}) - f(z) - \bar{f}(\bar{z}),$$

where f is holomorphic.

Remark 1.26. Under a Kähler transformation, e^K transforms multiplicatively. Hence in general e^K is not a global function, but rather a section of a holomorphic line bundle over M .

Proposition 1.27. *Two Kähler forms ω and ω' represent the same class in $H^2(M, \mathbb{R})$ if and only if there exists a smooth real function ϕ on M such that*

$$\omega' = \omega + \partial \bar{\partial} \phi.$$

1.1.4 Connection and Curvature in Kähler Geometry

Proposition 1.28. *In Kähler geometry the only non-vanishing Christoffel symbols have pure indices. They are given by*

$$\Gamma_{jk}^i = G^{i\bar{l}} \partial_j G_{k\bar{l}}, \quad \Gamma_{j\bar{k}}^{\bar{i}} = G^{\bar{l}i} \partial_{\bar{j}} G_{l\bar{k}} = \overline{\Gamma_{jk}^i}.$$

They define a torsion-free metric connection:

$$T_{jk}^i = \Gamma_{jk}^i - \Gamma_{kj}^i = 0, \quad \nabla_i G_{j\bar{k}} = 0.$$

The covariant derivative acts as

$$\nabla_i V_k = \partial_i V_k - \Gamma_{ik}^p V_p,$$

and analogously

$$\nabla_{\bar{i}} V_{\bar{k}} = \partial_{\bar{i}} V_{\bar{k}} - \Gamma_{\bar{i}\bar{k}}^{\bar{p}} V_{\bar{p}}.$$

Proposition 1.29. *The non-vanishing components of the Riemann tensor are*

$$R_{i\bar{j}l}{}^k = \partial_{\bar{j}} \Gamma_{il}^k, \quad R_{i\bar{j}l}{}^{\bar{k}} = -\partial_i \Gamma_{jl}^{\bar{k}},$$

and

$$R_{i\bar{j}k\bar{l}} = -\partial_i \partial_{\bar{j}} G_{k\bar{l}} + G^{\bar{m}n} (\partial_i G_{k\bar{m}}) (\partial_{\bar{j}} G_{n\bar{l}}).$$

Remark 1.30. On a Kähler manifold, the four-index Riemann tensor R_{abcd} can be nonzero only when each pair of indices (a, b) and (c, d) contains one holomorphic and one anti-holomorphic index. Equivalently, the only independent nonvanishing components may be written as

$$R_{i\bar{j}k\bar{\ell}}.$$

This follows from the Kähler condition, which implies

$$[\nabla_i, \nabla_j] = 0, \quad [\nabla_{\bar{i}}, \nabla_{\bar{j}}] = 0,$$

so the curvature two-form is of type $(1, 1)$. Hence any component with two holomorphic or two anti-holomorphic indices in either pair must vanish. All other nonzero arrangements are obtained from $R_{i\bar{j}k\bar{\ell}}$ by the usual symmetries of the Riemann tensor.

Remark 1.31 (Additional symmetry in the Kähler case). In addition to the usual symmetries of the Riemann tensor, the Kähler condition implies the further symmetry

$$R_{i\bar{j}k\bar{\ell}} = R_{k\bar{j}i\bar{\ell}}, \quad R_{i\bar{j}k\bar{\ell}} = R_{i\bar{\ell}k\bar{j}}.$$

Thus the curvature tensor is symmetric in the holomorphic indices and separately symmetric in the anti-holomorphic indices.

Equivalently,

$$R_{i\bar{j}k\bar{\ell}} = R_{k\bar{\ell}i\bar{j}}.$$

The remaining components are obtained by complex conjugation:

$$R_{i\bar{j}l}{}^{\bar{k}} = \overline{R_{i\bar{j}l}{}^{\bar{k}}}.$$

Our sign convention is

$$[\nabla_i, \nabla_j]V_k = R_{ijk}{}^p V_p.$$

Proposition 1.32. *The Ricci tensor is obtained by contracting the Riemann tensor:*

$$R_{i\bar{j}} = R_{i\bar{a}j}{}^{\bar{a}} = -\partial_i \partial_{\bar{j}} \log \det(G).$$

Remark 1.33. The Ricci tensor is Hermitian:

$$R_{i\bar{j}} = R_{j\bar{i}}, \quad R_{i\bar{j}} = \overline{R_{j\bar{i}}},$$

which follows from the fact that G is Hermitian and $\det(G)$ is real.

Definition 1.34. The Ricci form is defined by

$$\mathcal{R} = \frac{i}{2\pi} R_{i\bar{j}} dz^i \wedge d\bar{z}^{\bar{j}} = \frac{i}{2\pi} \partial \bar{\partial} \log \det(G) = \frac{i}{4\pi} d(\partial - \bar{\partial}) \log \det(G).$$

Proposition 1.35. *The Ricci form represents the first Chern class of the holomorphic tangent bundle:*

$$c_1(TM) = [\mathcal{R}] \in H^2(M, \mathbb{R}).$$

Remark 1.36. The Chern classes of a holomorphic vector bundle are independent of the choice of Hermitian metric or connection. Hence although $\mathcal{R}$ is defined using the metric, its cohomology class $c_1(TM)$ is intrinsic.

Example 1.37. $\mathbb{CP}^n$ is a Kähler manifold. The Fubini–Study metric can be obtained from an explicit Kähler potential.

On the affine patches U_i , $i = 0, \dots, n$, the Kähler potential is

$$K^{(i)}(z^{(i)}, \bar{z}^{(i)}) = \log(1 + |z^{(i)}|^2),$$

where

$$|z^{(i)}|^2 = \sum_{j \neq i} |z_j^{(i)}|^2.$$

On overlaps $U_i \cap U_j$ one has

$$K^{(i)}(z^{(i)}, \bar{z}^{(i)}) = K^{(j)}(z^{(j)}, \bar{z}^{(j)}) - \log \frac{z_i}{z_j} - \log \frac{\bar{z}_i}{\bar{z}_j}.$$

The last two terms are holomorphic and anti-holomorphic respectively, hence they do not affect the metric

$$g_{i\bar{j}} = \partial_i \partial_{\bar{j}} K,$$

which is globally well defined. Dropping the patch index, the Kähler form is

$$\omega = i \partial \bar{\partial} \log(1 + |z|^2) = i \left(\frac{dz^i \wedge d\bar{z}^{\bar{i}}}{1 + |z|^2} - \frac{\bar{z}^i z^j dz^i \wedge d\bar{z}^{\bar{j}}}{(1 + |z|^2)^2} \right).$$

This defines a positive-definite metric. Since

$$\det(g_{i\bar{j}}) = \frac{1}{(1 + |z|^2)^{n+1}},$$

one computes

$$R_{i\bar{j}} = -\partial_i \partial_{\bar{j}} \log \det(g) = (n + 1) g_{i\bar{j}}.$$

Definition 1.38. The Ricci tensor above is proportional to the Kähler metric, the metric is called *Kähler–Einstein*.

Remark 1.39. Since the Christoffel symbols have only pure indices, parallel transport preserves the complex structure. Hence the holonomy group of a Kähler manifold reduces to

$$U(n) \subset SO(2n),$$

rather than the full group $SO(2n)$ of a generic orientable Riemannian manifold. This inclusion is strict. While elements of $SO(2n)$ preserve the Riemannian metric, elements of $U(n)$ must in addition preserve the complex structure J . Equivalently,

$$U(n) = \{A \in SO(2n) \mid AJ = JA\}.$$

Thus a general rotation in $SO(2n)$ may mix holomorphic and anti-holomorphic directions, whereas unitary transformations are precisely those orthogonal transformations compatible with the complex structure.

1.2 Hodge Theorem and Hodge Decompositions

1.2.1 The Hodge Theorem

Theorem 1.40 (Symmetries of Hodge cohomology). *Let M be a compact Kähler manifold of complex dimension n . Then the cohomology groups satisfy the symmetries*

$$H^{p,q}(M) = \overline{H^{q,p}(M)},$$

and

$$H^{p,q}(M) \cong H^{n-q,n-p}(M).$$

The first is induced by complex conjugation

$$\overline{A^{p,q}} = A^{q,p},$$

while the second follows from the Hodge star operator

$$* : A^{p,q} \rightarrow A^{n-q,n-p},$$

which commutes with the Laplacian.

Theorem 1.41 (Hodge decomposition for (p,q) -forms). *Let M be a compact Kähler manifold. Then every form $\phi \in A^{p,q}$ admits a unique orthogonal decomposition*

$$\phi = h + \bar{\partial}\xi + \bar{\partial}^*\eta,$$

where

$$h \in \mathcal{H}^{p,q}, \quad \xi \in A^{p,q-1}, \quad \eta \in A^{p,q+1}.$$

Equivalently,

$$A^{p,q} = \mathcal{H}^{p,q} \oplus \bar{\partial}A^{p,q-1} \oplus \bar{\partial}^*A^{p,q+1}.$$

Remark 1.42. This decomposition is the analogue of the de Rham decomposition

$$A^p = \mathcal{H}^p \oplus dA^{p-1} \oplus d^*A^{p+1}.$$

Proposition 1.43. *If ϕ is $\bar{\partial}$ -closed, i.e.*

$$\bar{\partial}\phi = 0,$$

then the $\bar{\partial}^\eta$ component in the above decomposition vanishes. Consequently every $\bar{\partial}$ -closed form admits a unique decomposition into a harmonic part and a $\bar{\partial}$ -exact part.*

Corollary 1.44. *There is a canonical identification*

$$H_{\bar{\partial}}^{p,q}(M) \cong \mathcal{H}^{p,q}(M).$$

Theorem 1.45 (Hodge decomposition on Kähler manifolds). *Let M be a compact Kähler manifold. Then the Laplacians satisfy*

$$\Delta_d = 2\Delta_{\bar{\partial}},$$

and hence preserve the (p,q) -type. In particular,

$$H_d^{p,q}(M) = H_{\bar{\partial}}^{p,q}(M), \quad \mathcal{H}_d^{p,q}(M) = \mathcal{H}_{\bar{\partial}}^{p,q}(M).$$

Taking harmonic representatives yields the Hodge decomposition of the de Rham cohomology

$$H^r(M, \mathbb{C}) = \bigoplus_{p+q=r} H^{p,q}(M).$$

1.2.2 The $\partial\bar{\partial}$ -Lemma

Lemma 1.46 ($\partial\bar{\partial}$ -Lemma). *Let X be a Kähler manifold, and let ω be a form which is both ∂ - and $\bar{\partial}$ -closed. If ω is d -, ∂ -, or $\bar{\partial}$ -exact, then there exists a form χ such that*

$$\omega = \partial\bar{\partial}\chi.$$

Proof. We prove the case where ω is $\bar{\partial}$ -exact. Write

$$\omega = \bar{\partial}\beta,$$

and decompose β by Hodge theory as

$$\beta = \beta_{\text{harm}} + \partial\bar{\partial}^*G_{\partial}\beta + \bar{\partial}^*\partial G_{\partial}\beta.$$

where G_{∂} is the Green operator for Δ_{∂} , which is

$$\bigoplus_{p,q} \left(\Delta_{\partial}^{-1}|_{\mathcal{H}_{p,q}^{\perp}} \oplus 0|_{\mathcal{H}_{p,q}} \right).$$

By the way, it's easy to see

$$G_{\partial} = G_{\bar{\partial}}$$

We use the identity

$$\bar{\partial}\partial^* = -\partial^*\bar{\partial}.$$

Using γ to denote $G_{\partial}\beta$,

$$\omega = \bar{\partial}(\partial^*\partial + \partial\bar{\partial}^*)\gamma = -\partial^*(\bar{\partial}\partial\gamma) + \bar{\partial}\partial\bar{\partial}^*\gamma.$$

Both ω and

$$\bar{\partial}\partial\bar{\partial}^*\gamma = -\partial\bar{\partial}\partial^*\gamma$$

are ∂ -closed. Therefore,

$$\partial^*(\bar{\partial}\partial\gamma)$$

is also ∂ -closed. Since it lies in the image of the adjoint ∂^* , it must vanish. Consequently,

$$\omega = \bar{\partial}\partial(\partial^*\gamma).$$

Setting $\chi = -\partial^*\gamma$ gives

$$\omega = \partial\bar{\partial}\chi.$$

□

Lemma 1.47 ($\partial\bar{\partial}$ -Lemma for (p, q) -form). *Let X be a Kähler manifold, and let η be a (p, q) -form which is $\bar{\partial}$ -closed. If η is d -, ∂ -, or $\bar{\partial}$ -exact, then there exists a form χ such that*

$$\eta = \partial\bar{\partial}\chi.$$

Proof. Let $\eta \in \Omega^{p,q}(M)$ satisfy $\bar{\partial}\eta = 0$ and assume $\eta = d\gamma$ for some form γ . By Hodge decomposition for $\bar{\partial}$, we can write uniquely

$$\eta = \eta_{\text{harm}} + \bar{\partial}\bar{\partial}^*G_{\bar{\partial}}\eta + \bar{\partial}^*\bar{\partial}G_{\bar{\partial}}\eta,$$

and η_{harm} is the $\bar{\partial}$ -harmonic part. Since $\bar{\partial}\eta = 0$,

$$(\bar{\partial}^*\bar{\partial}G_{\bar{\partial}}\eta, \bar{\partial}^*\bar{\partial}G_{\bar{\partial}}\eta) = (\bar{\partial}G_{\bar{\partial}}\eta, \bar{\partial}\bar{\partial}^*\bar{\partial}G_{\bar{\partial}}\eta) = 0.$$

The term $\bar{\partial}^*\bar{\partial}G_{\bar{\partial}}\eta$ vanishes and we obtain

$$\eta = \eta_{\text{harm}} + \bar{\partial}\bar{\partial}^*G_{\bar{\partial}}\eta.$$

Thus η is cohomologous to η_{harm} in Dolbeault cohomology.

We claim that $\eta_{\text{harm}} = 0$. Indeed, take any $\bar{\partial}$ -harmonic form $\alpha \in \mathcal{H}_{\bar{\partial}}^{p,q}(M)$. Using d -exactness (similarly, ∂ or $\bar{\partial}$) of η and $d\alpha = 0$ (harmonic implies closed),

$$(\eta, \alpha) = (d\gamma, \alpha) = (\gamma, d^*\alpha) = 0,$$

since $d^*\alpha = 0$ for harmonic α . Hence η is L^2 -orthogonal to all harmonic forms, which forces $\eta_{\text{harm}} = 0$. Therefore

$$\eta = \bar{\partial}\bar{\partial}^*G_{\bar{\partial}}\eta.$$

Note that $\bar{\partial}^*G_{\bar{\partial}}\eta$ has pure type $(p, q-1)$. Hence

$$\partial(\bar{\partial}^*G_{\bar{\partial}}\eta) = -\bar{\partial}^*\partial G_{\bar{\partial}}\eta = -\bar{\partial}^*G_{\bar{\partial}}(\partial\eta) = 0.$$

Since the harmonic space for ∂ coincides with the harmonic space for $\bar{\partial}$, it is orthogonal to the image of $\bar{\partial}^*$. By the Hodge decomposition for ∂ , we deduce

$$\bar{\partial}^*G_{\bar{\partial}}\eta = \partial\bar{\partial}^*G_{\partial}(\bar{\partial}^*G_{\bar{\partial}}\eta).$$

We obtain

$$\eta = \bar{\partial}\partial(\bar{\partial}^*G_{\partial}\bar{\partial}^*G_{\bar{\partial}}\eta),$$

which proves the lemma. □

1.2.3 Čech–Dolbeault isomorphism

Theorem 1.48 (Čech–Dolbeault isomorphism). *Let M be a complex manifold and F a holomorphic vector bundle. Then there is a natural isomorphism*

$$H^q(M, \Omega^p(F)) \cong H_{\bar{\partial}}^{p,q}(M, F).$$

Example 1.49. Taking $F = \mathcal{O}_M$, one obtains

$$H^q(M, \Omega^p) \cong H^{p,q}(M).$$

In particular,

$$H^{p,0}(M) = H^0(M, \Omega^p).$$

Remark 1.50. Since there are no $(p, -1)$ -forms, one has

$$H^{p,0}(M) = \ker(\bar{\partial} : A^{p,0} \rightarrow A^{p,1}).$$

A $(p, 0)$ -form is $\bar{\partial}$ -closed precisely when its coefficients are holomorphic, hence

$$H^{p,0}(M) = H^0(M, \Omega^p),$$

i.e. Dolbeault $(p, 0)$ -cohomology coincides with the space of global holomorphic p -forms.

Proposition 1.51. *Let M be a compact Kähler manifold. Every holomorphic $(p, 0)$ -form is harmonic with respect to any Kähler metric.*

Proof. A holomorphic form Ω_p satisfies

$$\bar{\partial}\Omega_p = 0.$$

Since $\bar{\partial}^*$ maps to $A^{p,-1} = 0$, one has

$$\bar{\partial}^*\Omega_p = 0.$$

Hence

$$\Delta_{\bar{\partial}}\Omega_p = 0,$$

so Ω_p is harmonic. □

Proposition 1.52. *On a Kähler manifold the Kähler form ω is harmonic.*

Proof. Using the identity

$$(\bar{\partial}^*\psi)_{i_1 \dots i_p \bar{j}_2 \dots \bar{j}_q} = (-1)^{p+1} \nabla^{\bar{j}_1} \psi_{i_1 \dots i_p \bar{j}_1 \bar{j}_2 \dots \bar{j}_q},$$

one verifies that

$$\bar{\partial}\omega = 0, \quad \bar{\partial}^*\omega = 0.$$

Hence ω is harmonic. □

Remark 1.53 (Role of the Kähler condition). On a general Hermitian manifold there are two natural metric connections:

- the Levi–Civita connection ∇^{LC} , which is torsion-free and metric compatible,

$$\nabla^{LC}G = 0,$$

but does not preserve the (p, q) -type decomposition;

- the Chern connection ∇^{Ch} , which preserves the complex structure and the type decomposition,

$$\nabla^{Ch}G = 0, \quad \nabla^{Ch}J = 0,$$

but generally has nonvanishing torsion.

The local formula for $\bar{\partial}^*$ is naturally expressed using the Chern connection, since it respects the (p, q) -grading. In the non-Kähler case this formula contains additional torsion terms, so $\bar{\partial}^*$ is not simply given by a covariant divergence.

The Kähler condition $d\omega = 0$ is equivalent to the vanishing of the Chern torsion, and hence

$$\nabla^{LC} = \nabla^{Ch}.$$

Only in this situation does the identity

$$(\bar{\partial}^* \psi)_{\dots} = (-1)^{p+1} \nabla^{\bar{b}} \psi_{\dots \bar{b} \dots}$$

take its simple divergence form, which in particular implies $\bar{\partial}^* \omega = 0$.

Corollary 1.54. *Let M be a compact Kähler manifold of complex dimension n . Then*

$$h^{1,1}(M) \geq 1.$$

Moreover,

$$\omega^m \neq 0 \quad \text{in} \quad H^{m,m}(M), \quad m = 1, \dots, n.$$

1.2.4 Lefschetz Decomposition

Let M be a compact Kähler manifold with

$$n = \dim_{\mathbb{C}}(M).$$

Denote by ω the Kähler form. The exterior product with ω defines the lowering operator

$$S^- : H^r(M) \longrightarrow H^{r+2}(M), \quad S^-(\alpha) = \omega \wedge \alpha.$$

Its adjoint (with respect to the Hodge inner product) is the raising operator

$$S^+ : H^r(M) \longrightarrow H^{r-2}(M),$$

and we define the grading operator H by assigning to a class of degree r the eigenvalue

$$H \alpha = \frac{n-r}{2} \alpha.$$

Then the operators $H, S^{\pm}$ satisfy the commutation relations of the Lie algebra $\mathfrak{sl}(2, \mathbb{C})$:

$$[S^+, S^-] = 2H, \quad [H, S^{\pm}] = \pm S^{\pm}.$$

Consequently the cohomology decomposes into irreducible $\mathfrak{sl}(2, \mathbb{C})$ -representations.

Remark 1.55. The adjoint Lefschetz operator S^+ admits an explicit expression. If

$$S^-(\alpha) = L(\alpha) = \omega \wedge \alpha$$

denotes the Lefschetz operator, then

$$S^+ = \Lambda,$$

where Λ is the contraction with the Kähler form. In local unitary coordinates,

$$\Lambda = i \sum_{a=1}^n \iota_{\partial/\partial z^a} \iota_{\partial/\partial \bar{z}^a}.$$

Equivalently,

$$\Lambda = *^{-1} L *,$$

so that Λ is the adjoint of L with respect to the Hodge inner product.

Theorem 1.56 (Hard Lefschetz). *For each $k \geq 0$, the map*

$$(S^-)^k : H^{n-k}(M) \longrightarrow H^{n+k}(M)$$

is an isomorphism.

Definition 1.57. Define the primitive cohomology by

$$P^{n-k} := \text{Ker} \left((S^-)^{k+1} : H^{n-k}(M) \longrightarrow H^{n+k+2}(M) \right).$$

Theorem 1.58. *An equivalent definition of primitive cohomology is given by*

$$P^{n-k} = (\text{Ker} S^+) \cap H^{n-k}(M)$$

Proof. Let M be a compact Kähler manifold of complex dimension n , and let

$$L := S^- = \omega \wedge (\cdot), \quad \Lambda := S^+ = L^\dagger$$

be the Lefschetz operator and its adjoint with respect to the Hodge inner product. Fix $k \geq 0$ and set $r = n - k$.

Step 1: orthogonal splitting and the characterization of the orthogonal complement.

Consider the subspace

$$U_r := L(H^{r-2}(M)) \subset H^r(M).$$

Since $H^r(M)$ is finite-dimensional, we have the orthogonal decomposition

$$H^r(M) = U_r \oplus U_r^\perp.$$

We claim that

$$U_r^\perp = \text{Ker}(\Lambda|_{H^r(M)}).$$

Indeed, for $\alpha \in H^r(M)$ we have

$$\alpha \in U_r^\perp \iff (\alpha, L\beta) = 0 \quad \forall \beta \in H^{r-2}(M) \iff (\Lambda\alpha, \beta) = 0 \quad \forall \beta \in H^{r-2}(M),$$

where in the last step we used the adjointness $\Lambda = L^\dagger$. Nondegeneracy of the inner product on the finite-dimensional space $H^{r-2}(M)$ implies that the right-hand side is equivalent to $\Lambda\alpha = 0$. This proves the claim. In particular, if $\Lambda\alpha \neq 0$ then

$$(\alpha, L\Lambda\alpha) = (\Lambda\alpha, \Lambda\alpha) > 0,$$

so $\alpha \notin U_r^\perp$, i.e. α has a nonzero component coming from $U_r = L(H^{r-2})$ (“it comes from a longer Lefschetz string”).

Step 2: recursive extraction of Lefschetz strings. Start with degree 0. Choose an orthonormal basis of $H^0(M)$; each basis vector v generates the Lefschetz string

$$v, Lv, L^2v, \dots, L^n v,$$

which has length $n + 1$ by Hard Lefschetz (equivalently, $L^n : H^0 \rightarrow H^{2n}$ is an isomorphism and hence $L^j v \neq 0$ for $0 \leq j \leq n$). Let $W^{(0)}$ be the sum of the subspaces spanned by these strings. By construction $W^{(0)}$ is L -stable, and since $\Lambda = L^\dagger$ it is also Λ -stable; hence its orthogonal complement $(W^{(0)})^\perp$ is L - and Λ -stable as well. Repeating the same procedure inside $(W^{(0)})^\perp$ starting from degree 2 produces strings of length $n - 1$, then inside the next orthogonal complement starting from degree 4 produces strings of length $n - 3$, and so on. Proceeding inductively yields an orthogonal direct sum decomposition of the cohomology into Lefschetz strings, i.e. into finite-dimensional $\mathfrak{sl}_2$ -subrepresentations generated by vectors killed by Λ .

Step 3: identification of primitive classes in degree $n - k$. In the above decomposition, every element $\alpha \in H^r(M) = H^{n-k}(M)$ can be written uniquely as a sum of components lying in individual Lefschetz strings. Within any fixed string, the component in degree r is either the top element of that string (equivalently annihilated by Λ) or it is obtained by wedging with ω from a lower degree (equivalently it has a nontrivial projection to $L(H^{r-2})$). By Step 1, the condition

$$\Lambda \alpha = 0$$

is equivalent to $\alpha \in U_r^\perp$, i.e. to α being orthogonal to the image $L(H^{r-2})$ and hence to being the top element in its Lefschetz string.

Now, for classes of degree $r = n - k$, the maximal number of further wedge operations by ω before leaving the string is exactly k . Therefore, for such α being a top element is equivalent to the nilpotence condition

$$L^{k+1} \alpha = 0.$$

Consequently,

$$\text{Ker}(L^{k+1} : H^{n-k}(M) \rightarrow H^{n+k+2}(M)) = (\text{Ker} \Lambda) \cap H^{n-k}(M),$$

which is precisely the claimed equivalence of the two definitions of primitive cohomology. $\square$

As an immediate consequence of the preceding proof, we obtain the following corollary, where the primitive parts of the cohomology play the role of highest weight vectors.

Corollary 1.59 (Lefschetz decomposition). *The cohomology admits the decomposition*

$$H^r(M) = \bigoplus_k (S^-)^k P^{r-2k}(M).$$

Remark 1.60. The appearance of Lie algebra representations on cohomology is a recurrent phenomenon in geometry. Besides the Kähler $\mathfrak{sl}_2$ -action discussed above, if M is hyperkähler, it carries three Kähler forms $\omega_I, \omega_J, \omega_K$ coming from the quaternionic structure. Each determines a Lefschetz $\mathfrak{sl}_2$ -action, and these combine to a larger Lie algebra action on cohomology. A theorem of Verbitsky shows that $H^\bullet(M)$ carries a natural $\mathfrak{so}(5)$ -action extending the individual Lefschetz actions.

Hyperkähler geometry plays an important role in supersymmetric field theory. In nonlinear sigma models, extended supersymmetry severely restricts the geometry of the target space: while $\mathcal{N} = 2$ supersymmetry requires the target to be Kähler, $\mathcal{N} = 4$ supersymmetry forces it to be hyperkähler.

Example 1.61. The cohomology of $\mathbb{P}^n$ forms an irreducible $\mathfrak{sl}_2$ -representation of spin $n/2$.

The cohomology of the two-torus T^2 decomposes as

$$H^\bullet(T^2) \cong 2(0) \oplus \left(\frac{1}{2}\right),$$

where the two trivial representations (0) are generated by dz and $d\bar{z}$, while

$$\{1, dz \wedge d\bar{z}\}$$

span the representation $\left(\frac{1}{2}\right)$.

More generally, the cohomology of the torus T^{2n} admits the $\mathfrak{sl}_2(\mathbb{C})$ decomposition

$$(2(0) + \left(\frac{1}{2}\right))^{\otimes n} = \bigoplus_{r=1}^n \left(\binom{2n}{n-r} - \binom{2n}{n-r-2} \right) \left(\frac{r}{2}\right).$$

1.3 The Middle (Co)homology and the Riemann Bilinear Relations

1.3.1 Hirzebruch Signature Theorem

Let M be a compact oriented manifold of even real dimension $2n$. On the middle cohomology one has the natural intersection pairing

$$Q(\alpha, \beta) = \int_M \alpha \wedge \beta, \quad \alpha, \beta \in H^n(M, \mathbb{R}).$$

This defines a symmetric bilinear form whose most important invariant is its *signature*.

Definition 1.62. The *signature* of M is

$$\sigma(M) = h_+^n(M) - h_-^n(M),$$

where h_+^n and h_-^n denote the numbers of positive and negative eigenvalues of the intersection form on $H^n(M, \mathbb{R})$.

Theorem 1.63 (Hirzebruch Signature Theorem). *Let M^{2n} be a compact oriented manifold. Then*

$$\sigma(M) = \int_M L_{n/2}(TM),$$

where L_m is the Hirzebruch L -class.

Definition 1.64. The Hirzebruch L -class is the multiplicative characteristic class associated to the power series

$$Q(x) = \frac{x}{\tanh x} = 1 + \sum_{k=0}^{\infty} (-1)^k \frac{2^{2k}}{(2k)!} B_k x^{2k},$$

where B_k are the Bernoulli numbers

$$B_1 = \frac{1}{6}, \quad B_2 = \frac{1}{30}, \quad B_3 = \frac{1}{42}, \quad B_4 = \frac{1}{30}, \quad \dots$$

Expanding in Pontryagin classes (p_k) or equivalently in Chern classes (c_k) gives

$$L_{m/2} = 0 \quad \text{for } m \text{ odd,}$$

and

$$\begin{aligned} L_1 &= \frac{1}{3}p_1 = \frac{1}{3}(c_1^2 - 2c_2), \\ L_2 &= \frac{1}{45}(7p_2 - p_1^2) = \frac{1}{45}(4c_1^2c_2 - c_1^4 - 14c_1c_3 + 3c_2^2 + 14c_4). \end{aligned}$$

Remark 1.65 (Calabi–Yau specialization). For Calabi–Yau manifolds one has $c_1 = 0$. Hence

$$L_1 = -\frac{2}{3}c_2,$$

and

$$L_2 = \frac{1}{45}(3c_2^2 + 14c_4).$$

After integration over M , these become numerical invariants entering the signature formula.

Remark 1.66. If the complex dimension is odd, the intersection form admits a symplectic basis and therefore has equal numbers of positive and negative eigenvalues, implying

$$\sigma(M) = 0.$$

For even complex dimension further constraints arise. In particular, for Calabi–Yau fourfolds one may use the Hirzebruch–Riemann–Roch theorem, which gives the arithmetic genus

$$\chi_0 = h_{00} + h_{40} = 2 = \frac{1}{720} \int_M 3(c_2^2 - c_4).$$

These relations impose strong restrictions on the possible intersection forms.

1.3.2 Intersection Form on the Middle Cohomology

Let M be a compact Kähler manifold of complex dimension n . On the middle cohomology it is customary to specialize the bilinear form to the intersection pairing

$$\langle \alpha, \beta \rangle : H^n(M) \otimes H^n(M) \longrightarrow \mathbb{C}, \quad \langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta.$$

Remark 1.67. The form is odd (respectively even) when n is odd (respectively even), since

$$\alpha \wedge \beta = (-1)^n \beta \wedge \alpha.$$

Proposition 1.68. *Let $\omega^{p,q} \in H^{p,q}(M)$ and $\omega^{r,s} \in H^{r,s}(M)$ with $p + q = r + s = n$. Then*

$$\langle \omega^{p,q}, \omega^{r,s} \rangle = 0, \quad \text{unless } p = s \text{ and } q = r.$$

Proposition 1.69. *If $\alpha \in H^{p,q}(M)$ is primitive with $p + q = n$, then*

$$*\alpha = (-1)^{\frac{n(n+1)}{2}} i^{p-q} \alpha.$$

Remark 1.70. In fact, there is a stronger statement describing the action of the Hodge star operator on primitive forms in arbitrary degree.

Let $\omega \in \Omega_X^{p,q}$ be a primitive form with $k = p + q$. Then

$$*\omega = (-1)^{\frac{k(k+1)}{2}} i^{p-q} \frac{L^{n-k}}{(n-k)!} \omega,$$

where L denotes the Lefschetz operator $L(\alpha) = \omega_X \wedge \alpha$.

In the special case $k = n$, we have $L^{n-k} = \text{id}$, and the formula reduces to

$$*\omega = (-1)^{\frac{n(n+1)}{2}} i^{p-q} \omega,$$

which agrees with the statement of the previous proposition.

Corollary 1.71. *If $\alpha \in H^{p,q}(M)$ is primitive with $p + q = n$, then*

$$R(\alpha) = \int \alpha \wedge *\bar{\alpha} = (-1)^{\frac{n(n+1)}{2}} i^{-p+q} \langle \alpha, \bar{\alpha} \rangle > 0.$$

Remark 1.72. The quantity $R(\alpha)$ defines a positive real structure on the primitive middle cohomology. Although the intersection pairing itself is not positive definite, after restriction to primitive classes and multiplication by the phase factor one obtains a positive definite Hermitian form.

Remark 1.73 (Weil–Petersson metric). Applying this construction to the unique holomorphic $(n, 0)$ -form Ω on a Calabi–Yau manifold produces the Kähler Weil–Petersson metric on the complex moduli space via

$$i^{n^2} \int_M \Omega \wedge \bar{\Omega}.$$

We are interested in how the Hodge decomposition of the middle cohomology varies when a family of Calabi–Yau n -folds moves over its complex moduli space $\mathcal{M}$.

By the Tian–Todorov theorem, complex structure deformations of Calabi–Yau manifolds are globally unobstructed. Consequently, the moduli space $\mathcal{M}$ exists as a complex manifold, and in fact carries a natural Kähler structure.

Moreover, the complex dimension of $\mathcal{M}$ is given by the Hodge number

$$h = h_{n-1,1}(M) = \dim_{\mathbb{C}}(\mathcal{M}).$$

We therefore parametrize $\mathcal{M}$ locally by complex coordinates

$$z^i, \quad i = 1, \dots, h.$$

Remark 1.74 (Horizontal cohomology). Although the vector space $H^n(M, \mathbb{C})$ is topologically invariant under deformations of the complex structure, its Hodge decomposition

$$H^n(M) = \bigoplus_{p+q=n} H^{p,q}(M)$$

varies with the complex structure.

If a polarisation (i.e. a fixed Kähler class) is chosen, the Hard Lefschetz decomposition shows that only the primitive part of the middle cohomology varies; the components obtained by wedging with the Kähler form remain fixed. This varying subspace is called the (horizontal) cohomology.

In the Calabi–Yau threefold case one has $H^1(M) = 0$, hence the entire middle cohomology is primitive and therefore horizontal:

$$H_{\text{hor}}^3(M) = H^3(M).$$

From now on we denote the (horizontal) middle cohomology simply by

$$H^n(M) = \bigoplus_{k=0}^n H^{n-k,k}(M).$$

Strictly speaking one should write $H_{\text{hor}}^n(M)$, but we suppress both the qualifier “horizontal” and the index on M_n in the following discussion.

2 Calabi-Yau Manifolds and Their Deformation Spaces

2.1 Properties of Calabi-Yau Manifolds

2.1.1 Properties and the Definition

We now describe Calabi–Yau spaces by listing several important, partly equivalent properties that turn a Kähler manifold into a Calabi–Yau manifold M . The precise relations among these properties will be discussed below.

- (a) The canonical class K_M is trivial.
- (b) The first Chern class of the tangent bundle vanishes,

$$c_1(TM) = 0.$$

- (c) In each Kähler class on M there exists a unique Kähler metric g whose Ricci tensor vanishes,

$$R_{i\bar{j}}(g) = 0.$$

- (d) There exists — up to multiplication by a constant — a unique nowhere-vanishing holomorphic $(n, 0)$ -form, denoted by Ω .
- (e) The holonomy group $\text{Hol}(g)$ of M is a subgroup of $SU(n)$.
- (f) M admits a pair of globally defined covariantly constant (parallel) spinors ξ and $\bar{\xi}$, of opposite chirality if n is odd and of the same chirality if n is even.

Remark 2.1. On a Calabi–Yau manifold one therefore has two distinguished forms: the Kähler $(1, 1)$ -form ω and the holomorphic $(n, 0)$ -form Ω . The properties listed above play different roles in different physical contexts and are only partially equivalent; consequently, different definitions of Calabi–Yau manifolds appear in the physics literature. For reference, we now give a definition of a compact Calabi–Yau manifold.

Definition 2.2. For $n \geq 2$, a *Calabi–Yau manifold* is a quadruple

$$(M, J, g, \Omega),$$

where

- (M, J) is a compact complex manifold,
- g is a Kähler metric with Kähler form ω ,
- Ω is a nowhere-vanishing holomorphic $(n, 0)$ -form, unique up to multiplication by a phase.

These structures are related by the condition that the volume form defined by Ω agrees with the Kähler volume form, namely

$$\frac{\omega^n}{n!} = (-1)^{\frac{n(n-1)}{2}} \left(\frac{i}{2}\right)^n \Omega \wedge \overline{\Omega}.$$

Remark 2.3. It is important to clarify the role of the Ricci–flat condition in the discussion of holonomy. For a given compact complex manifold (M, J) with $c_1(M) = 0$, an arbitrary Kähler metric on M does *not* necessarily satisfy $R_{i\bar{j}} = 0$, and hence its holonomy need only be contained in $U(n)$.

Yau’s theorem asserts, however, that within any fixed Kähler class there exists a unique Ricci–flat Kähler metric g_{RF} . The statement that a Calabi–Yau manifold has holonomy contained in $SU(n)$ refers to the holonomy group of this distinguished Ricci–flat metric, not to that of an arbitrary Kähler metric on M .

We now discuss the relations between the statements (a)–(e). In order to connect (a)–(d) with (e), we assume that M is simply connected and not of product form.

The implications among the conditions (a)–(c) are summarized as follows:

- $(a) \Leftrightarrow (b)$ follows from

$$-2\pi c_1(K_M) := -2\pi c_1\left(\bigwedge^n T^{*1,0}M\right) = -2\pi c_1(T^*M) = 2\pi c_1(TM).$$

- $(c) \Rightarrow (b)$ follows from the independence of Chern classes on the choice of the Kähler metric. Indeed, the existence of a Ricci–flat metric implies

$$c_1(TM) = 0,$$

and hence this holds for every Kähler metric.

- $(b) \Rightarrow (c)$ is a corollary of Yau’s theorem, proving the conjecture of Calabi (1956).

Theorem 2.4 (Calabi–Yau theorem, geometric form). *Given:*

(C.a) a Kähler metric g , Kähler form ω , Ricci form $\mathcal{R}$ on M , and a real closed $(1, 1)$ -form $\mathcal{R}'$ representing the same Chern class

$$[\mathcal{R}] = [\mathcal{R}'] = 2\pi c_1(TM),$$

there exists

(C.b) a unique Kähler metric g' with associated Kähler form ω' such that

$$[\omega'] = [\omega] \in H^2(M, \mathbb{R}), \quad \text{Ric}(g') = \mathcal{R}'.$$

Remark 2.5. In particular, if $c_1(TM) = 0$, one may choose $\mathcal{R}' = 0$. The theorem then produces a unique Ricci-flat Kähler metric in the Kähler class $[\omega]$.

One can reformulate (C.a) and (C.b) in terms of scalar functions.

Since $\mathcal{R} - \mathcal{R}'$ is a real closed $(1, 1)$ -form which is $\bar{\partial}$ -exact, the $\partial\bar{\partial}$ -lemma implies the existence of a real function f on M such that

$$\mathcal{R} - \mathcal{R}' = i\partial\bar{\partial}f$$

up to an additive constant κ .

Recalling that the Ricci form is determined by

$$\mathcal{R} = \frac{i}{2\pi} R_{i\bar{j}} dz^i \wedge d\bar{z}^j = \frac{i}{2\pi} \partial\bar{\partial} \log \det(G),$$

and we have

$$\text{vol} = \frac{\omega^n}{n!} = i^n \det(G) dz^1 \wedge d\bar{z}^1 \wedge \cdots \wedge dz^n \wedge d\bar{z}^n,$$

one finds that f appears through

$$(\omega')^n = e^f \omega^n.$$

The constant κ can be fixed by normalizing the total volume:

$$\int_M e^f \omega^n = \int_M \omega^n.$$

Thus, instead of prescribing the Ricci form, one may equivalently prescribe the volume form.

Similarly, the condition $[\omega'] = [\omega]$ can be expressed by searching for a real function ϕ such that

$$\omega' = \omega + i\partial\bar{\partial}\phi.$$

The function ϕ becomes unique after imposing

$$\int_M \phi \text{vol}_g = 0.$$

Theorem 2.6 (Calabi–Yau theorem, scalar formulation). *For every Kähler metric g with Kähler form ω and every smooth real function f satisfying*

$$\int_M e^f \omega^n = \int_M \omega^n,$$

there exists a unique smooth real function ϕ such that

1. $\omega + i\partial\bar{\partial}\phi$ is a positive $(1, 1)$ -form,

$$2. \int_M \phi \operatorname{vol}_g = 0,$$

3.

$$(\omega + i\partial\bar{\partial}\phi)^n = e^f \omega^n.$$

Yau proved that the non-linear PDE for ϕ admits a unique solution satisfying the two conditions above. This is an existence result; up to now no explicit solutions for ϕ , and hence no explicit Ricci-flat metric on a general compact Calabi–Yau manifold, are known.

The implications among the conditions (a)–(e) are summarized as follows:

- (c) $\Rightarrow$ (e): From the special form of the curvature tensor on Kähler manifolds, one finds that the holonomy group of a Kähler manifold is generically $U(n)$. Since the Ricci tensor is obtained as the trace of the curvature, it generates precisely the $U(1)$ factor in the decomposition

$$U(n) \simeq SU(n) \times U(1).$$

If the metric is Ricci-flat, this $U(1)$ component is absent and the holonomy reduces to $SU(n)$.

- (e) $\Rightarrow$ (d): An $(n, 0)$ -form can locally be written as

$$\Omega_{i_1, \dots, i_n} = f(x) \epsilon_{i_1, \dots, i_n}.$$

Hence Ω transforms in the totally antisymmetric representation of the holonomy group $SU(n)$, i.e. it is a singlet under holonomy. Consequently,

$$\nabla \Omega = 0.$$

Since the Christoffel symbols have no mixed indices, one obtains

$$\bar{\partial}_i \Omega = \nabla_{\bar{i}} \Omega = 0,$$

so Ω is holomorphic.

On the other hand, the local expression

$$\Omega_{i_1, \dots, i_n} = f(x) \epsilon_{i_1, \dots, i_n}$$

is written with respect to a local unitary frame obtained by parallel transport. Such frames are related on overlaps by elements of the holonomy group.

Since the holonomy satisfies

$$\operatorname{Hol}(M) \subset SU(n),$$

the transition matrices between parallel frames lie in $SU(n)$ and therefore have determinant equal to 1. Under a change of frame $e_i \mapsto A_i^j e_j$ with $A \in SU(n)$, the totally antisymmetric tensor transforms as

$$\epsilon_{i_1 \dots i_n} \mapsto \det(A) \epsilon_{i_1 \dots i_n} = \epsilon_{i_1 \dots i_n}.$$

Hence the coefficient function $f(x)$ does not change on overlaps of coordinate patches. Consequently the local functions glue together to define a globally well-defined holomorphic function

$$f : M \rightarrow \mathbb{C}.$$

The coefficient $f(x)$ therefore defines a global holomorphic function on the compact manifold M , and hence must be constant.

Locally,

$$\omega = \frac{i}{2} \sum_{i=1}^n dx^i \wedge d\bar{x}^i, \quad g = \sum_{i=1}^n |dx^i|^2,$$

which are also covariantly constant. The normalization condition imposed at one point implies $|f| = 1$, and since all quantities are covariantly constant, this holds everywhere on M .

Moreover, Ω is harmonic:

$$\Delta_{\bar{\partial}} \Omega = 0,$$

because $\bar{\partial}\Omega = 0$ and

$$\bar{\partial}^* \Omega = - * \partial * \Omega = 0,$$

using that

$$* : A^{n,0} \rightarrow A^{n,0}, \quad \partial : A^{n,0} \rightarrow A^{n+1,0} = \{0\}.$$

- (d) $\Rightarrow$ (a): The form Ω provides a nowhere-vanishing covariantly constant section of the canonical bundle

$$K_M = \bigwedge^n T^{*1,0} M,$$

showing that the canonical bundle is trivial.

- (d) $\Rightarrow$ (b): Assume that there exists a nowhere vanishing holomorphic $(n,0)$ -form Ω . One then obtains a globally defined scalar function

$$\|\Omega\|^2 = \frac{1}{n!} \Omega_{i_1 \dots i_n} \bar{\Omega}^{i_1 \dots i_n}$$

where indices are raised using the Hermitian metric $g^{i\bar{j}}$.

Locally, Ω can be written as

$$\Omega_{i_1 \dots i_n} = f(x) \epsilon_{i_1 \dots i_n},$$

where $f(x)$ is a nowhere-vanishing holomorphic function on each patch. Raising indices gives

$$\bar{\Omega}^{i_1 \dots i_n} = \frac{\bar{f}}{g} \epsilon^{i_1 \dots i_n},$$

and therefore

$$g = \det(g_{i\bar{j}}) = \frac{|f|^2}{\|\Omega\|^2}.$$

Therefore

$$2\pi c_1(M) = [\mathcal{R}] = \frac{i}{2\pi} \partial \bar{\partial} \log \det(g_{k\bar{l}}) = -\frac{i}{2\pi} \partial \bar{\partial} \log \|\Omega\|^2.$$

which is exact since $\log \|\Omega\|^2$ is a globally defined scalar function. Hence

$$c_1(TM) = 0 \quad \text{in cohomology.}$$

A very general tool in Čech cohomology is *Serre duality*, which states that for any sheaf E on M one has

$$H^k(E)^* \cong H^{n-k}(E^* \otimes K_M).$$

Using the Čech–Dolbeault isomorphism

$$H^k(E) \cong H_{\bar{\partial}}^k(M, E), \quad H^r(M, \wedge^s T^* M) = H^{s,r}(M),$$

and the fact that on a Calabi–Yau manifold

$$K_M = \mathcal{O}_M,$$

one obtains relations among the cohomology groups

$$H^{0,r}(M) \cong H^{0,n-r}(M)$$

by taking $E = \mathcal{O}_M$, and by complex conjugation also

$$H^{r,0}(M) \cong H^{n-r,0}(M).$$

Remark 2.7. This result can also be seen more directly. Contracting a $(p, 0)$ -form

$$\omega = \omega_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$$

with the unique $(0, n)$ -form $\bar{\Omega}$ defines a $(0, n-p)$ -form

$$\hat{\omega}_{\bar{j}_{p+1} \dots \bar{j}_n} = \frac{1}{p!} \bar{\Omega}_{\bar{j}_1 \dots \bar{j}_n} \omega^{j_1 \dots j_p}.$$

One easily verifies that this map is invertible and commutes with the Laplacian Δ . Hence

$$H^{p,0}(M) \cong H^{0,n-p}(M) \cong H^{n-p,0}(M).$$

Remark 2.8. A general Kähler manifold has holonomy contained in $U(n)$. By Yau’s theorem, if the first Chern class vanishes one can *choose* a Ricci–flat Kähler metric g satisfying

$$R_{i\bar{j}}(g) = 0.$$

In this sense a Calabi–Yau manifold may be viewed as a Kähler manifold (with this canonically chosen metric) whose holonomy reduces from $U(n)$ to $SU(n)$.

Unless stated otherwise, one usually assumes that the holonomy is the *full* group $SU(n)$. This excludes geometries with reduced holonomy, such as complex tori $T_{\mathbb{C}}^n$ (which are flat and have trivial holonomy) or products of lower-dimensional Calabi–Yau manifolds, possibly with torus factors. More exotic examples, such as Enriques Calabi–Yau manifolds with holonomy $\mathbb{Z}_2 \times SU(2)$, are likewise excluded.

Remark 2.9. The assumption of full $SU(n)$ holonomy has important cohomological consequences. On a compact Kähler manifold, harmonicity of a $(p, 0)$ -form implies holomorphicity by type considerations. Generally, we have

$$(\Delta \alpha_p)_{m_1 \dots m_p} = -\nabla^k \nabla_k \alpha_{m_1 \dots m_p} - p R_{k[m_1} \alpha^k_{m_2 \dots m_p]} - \frac{1}{2} p(p-1) R_{jk[m_1 m_2} \alpha^{jk}_{m_3 \dots m_p]}.$$

Specializing it to the Kähler case

$$R_{ijk\bar{l}} = 0,$$

and using Ricci-flatness

$$R_{i\bar{j}} = 0,$$

the harmonicity condition becomes

$$\nabla^\nu \nabla_\nu \omega_{i_1 \dots i_p} = 0.$$

One may use the L^2 pairing together with integration by parts to obtain

$$\int_M |\nabla \omega|^2 \text{vol} = \int_M \langle \omega, \nabla^* \nabla \omega \rangle \text{vol} = 0.$$

This implies

$$\nabla \omega = 0.$$

In components one therefore obtains

$$\nabla_j \omega_{i_1 \dots i_p} = 0.$$

Thus every harmonic $(p, 0)$ -form is covariantly constant. And hence also

$$\bar{\partial} \omega = 0.$$

Such forms are therefore invariant under the holonomy group $SU(n)$. However, for $0 < p < n$ this is impossible, since the only $SU(n)$ -invariant tensors in exterior powers are the trivial representation ($p = 0$) and the totally antisymmetric representation ($p = n$). Consequently,

$$h^{p,0}(M) = 0, \quad 0 < p < n.$$

Thus the familiar vanishing pattern in the Hodge diamond of a Calabi–Yau manifold is not an additional defining condition but a consequence of full $SU(n)$ holonomy.

2.1.2 Calibration and Supersymmetric Cycles

Definition 2.10 (Calibration). Let (M, g) be a Riemannian manifold. A differential k -form $\varphi \in \Omega^k(M)$ is called a *calibration* if

1. $d\varphi = 0$,
2. for every point $p \in M$ and every oriented k -dimensional subspace $\xi \subset T_p M$,

$$\varphi|_\xi \leq \text{vol}_\xi,$$

where vol_ξ denotes the Riemannian volume form induced by g on ξ .

Remark 2.11 (Interpretation of the calibration inequality). Choose any oriented orthonormal basis $(e_1, \dots, e_k)$ of ξ with respect to the restricted metric $g|_\xi$. By definition of the induced volume form,

$$\text{vol}_\xi(e_1, \dots, e_k) = 1,$$

and therefore the calibration condition is equivalent to

$$\varphi(e_1, \dots, e_k) \leq 1.$$

Hence the definition of a calibration is purely a linear–algebraic condition on each tangent space: the k -form φ has *comass* bounded by one,

$$\|\varphi\|_{\text{comass}} \leq 1.$$

Definition 2.12 (Calibrated submanifold). Let φ be a calibration on (M, g) . An oriented k -dimensional submanifold $L \subset M$ is called *calibrated by φ* if

$$\varphi|_{T_p L} = \text{vol}_L \quad \text{for all } p \in L.$$

Proposition 2.13 (Minimal volume property). *If L is calibrated by φ , then L minimizes volume in its homology class, i.e.*

$$\text{Vol}(L) \leq \text{Vol}(L')$$

for every submanifold L' homologous to L .

Proof. Let φ be a calibration on (M, g) , and let $L^k \subset M$ be a k -dimensional oriented submanifold calibrated by φ , i.e.

$$\varphi|_L = \text{vol}_L.$$

Let L' be any other oriented k -dimensional submanifold homologous to L . Choose a $(k+1)$ -chain B such that

$$\partial B = L - L'$$

(with orientations). Since $d\varphi = 0$, Stokes' theorem gives

$$\int_L \varphi - \int_{L'} \varphi = \int_{\partial B} \varphi = \int_B d\varphi = 0,$$

hence

$$\int_L \varphi = \int_{L'} \varphi.$$

On the one hand, because L is calibrated,

$$\int_L \varphi = \int_L \text{vol}_L = \text{Vol}(L).$$

On the other hand, for every $p \in L'$ and every oriented k -plane $\xi = T_p L' \subset T_p M$, the calibration inequality implies

$$\varphi|_\xi \leq \text{vol}_\xi,$$

so restricting to L' we obtain pointwise

$$\varphi|_{L'} \leq \text{vol}_{L'}.$$

Integrating over L' yields

$$\int_{L'} \varphi \leq \int_{L'} \text{vol}_{L'} = \text{Vol}(L').$$

Combining the equalities and inequality above, we conclude

$$\text{Vol}(L) = \int_L \varphi = \int_{L'} \varphi \leq \text{Vol}(L').$$

Thus L minimizes volume in its homology class. □

Definition 2.14 (Special Lagrangian submanifold). Let (M, ω, Ω) be a Calabi–Yau n -fold. A real n -dimensional submanifold $L \subset M$ is called *special Lagrangian* if there exists a phase $\theta \in \mathbb{R}$ such that

$$\operatorname{Re}(e^{i\theta}\Omega)|_L = \operatorname{vol}_L, \quad \omega|_L = 0.$$

Equivalently, L is calibrated by the form $\operatorname{Re}(e^{i\theta}\Omega)$.

Definition 2.15 (Holomorphic calibration). Let (M, ω) be a Kähler manifold. For each k , the form

$$\frac{\omega^k}{k!}$$

is a calibration. Complex k -dimensional submanifolds $Z_k \subset M$ satisfy

$$\frac{\omega^k}{k!} \Big|_{Z_k} = \operatorname{vol}_{Z_k}, \quad \Omega|_{Z_k} = 0,$$

and hence are calibrated submanifolds.

Remark 2.16. In the physics literature calibrated cycles, usually denoted by Σ_{p+1} , are known as *supersymmetric cycles*. The requirement of maximal supersymmetry on the worldvolume of a cycle, defined by an embedding

$$\Phi : \Sigma_{p+1} \hookrightarrow M,$$

is satisfied if the global ten- or eleven-dimensional supersymmetry transformation on the worldvolume can be compensated by a worldvolume κ -symmetry transformation. This leads to the condition

$$P_- \eta = \frac{1}{2} \left(1 - \frac{i}{(p+1)!} h^{-1/2} \varepsilon^{\sigma_1 \dots \sigma_{p+1}} \partial_{\sigma_1} \phi^{m_1} \dots \partial_{\sigma_{p+1}} \phi^{m_{p+1}} \Gamma_{m_1 \dots m_{p+1}} \right) \eta = 0,$$

where η is a ten- or eleven-dimensional constant spinor, $P_{\pm}$ are projection operators, and h denotes the induced metric on the worldvolume Σ_{p+1} .

This condition specializes to the calibration conditions 2.7 and 2.8 above.

It is useful to note that all differential forms appearing in Berger’s list define calibrations and hence equivalently define supersymmetric cycles. If the complex dimension n of a Calabi–Yau manifold is even, one can also define mixed calibration conditions. For example, on a Calabi–Yau fourfold the calibration defined by

$$\frac{\omega^2}{2} + \operatorname{Re}(e^{i\theta}\Omega)$$

leads to the so-called *Cayley cycles*.

2.2 Complex Deformation Space

2.2.1 Local Tangent Space of the Complex Structure Moduli Space

Consider a $2n$ -dimensional real manifold M together with a covering by coordinate patches

$$\mathcal{U}_i, \quad i = 1, \dots, r,$$

each homeomorphic to a neighborhood

$$U_i \subset \mathbb{C}^n$$

with coordinates $z_\alpha^{(i)}(p)$, $\alpha = 1, \dots, n$. The manifold M is complex if the transition functions

$$f^{(jk)} : z^{(k)}(p) \mapsto z^{(j)}(p), \quad p \in U_j \cap U_k,$$

are biholomorphic.

One attempts to define a family of complex manifolds M_u by introducing transition functions depending holomorphically on parameters u ,

$$z_\alpha^{(j)} = f_\alpha^{(jk)}(z^{(k)}, u).$$

The difficulty is that some u -dependence corresponds merely to different choices of local coordinate systems on the same complex manifold.

To determine whether the functions $f^{(jk)}(z^{(k)}, u)$ really change the complex structure, Kodaira considers in each patch $\mathcal{U}_k$ an infinitesimal coordinate change generated by a holomorphic vector field

$$V^{(k)}(u) = \sum_{\alpha=1}^n \frac{\partial f_\alpha^{(k)}(z^{(k)}, u)}{\partial u} \frac{\partial}{\partial z_\alpha^{(k)}}.$$

Next one forms the composition of transition functions on triple overlaps $U_i \cap U_j \cap U_k$. By definition,

$$f_\alpha^{(ik)}(z^{(k)}, u) = f_\alpha^{(ij)}(f_1^{(jk)}(z^{(k)}, u), \dots, f_n^{(jk)}(z^{(k)}, u), u).$$

Differentiating with respect to u gives

$$\frac{\partial f_\alpha^{(ik)}(z^{(k)}, u)}{\partial u} = \frac{\partial f_\alpha^{(ij)}(z^{(j)}, u)}{\partial u} + \sum_{\beta=1}^n \frac{\partial z_\alpha^{(i)}}{\partial z_\beta^{(j)}} \frac{\partial f_\beta^{(jk)}(z^{(k)}, u)}{\partial u}.$$

Introduce vector fields

$$A^{(jk)}(u) = \sum_{\alpha=1}^n \frac{\partial f_\alpha^{(jk)}(z^{(k)}, u)}{\partial u} \frac{\partial}{\partial z_\alpha^{(j)}}, \quad z^{(k)} = f^{(kj)}(z_j, u).$$

Note that

$$A^{(kk)}(u) = 0,$$

since $f_\alpha^{(kk)} = z_\alpha^{(k)}$ is independent of u . Moreover, by the definition of $A^{(jk)}$,

$$A^{(kj)}(u) = -A^{(jk)}(u).$$

For general indices i, j, k , the compatible condition on $U_i \cap U_j \cap U_k$ becomes the Čech 1-cocycle condition

$$A^{(ij)}(u) + A^{(jk)}(u) + A^{(ki)}(u) = 0.$$

Remark 2.17. In order to understand the object $A^{(jk)}$, we consider an infinitesimal coordinate transformation

$$\delta z_\alpha^{(j)} = \frac{\partial f_\alpha^{(jk)}(z^{(k)}, u)}{\partial u} \delta u.$$

Thus, under such an infinitesimal deformation, the variation of the coordinates $z^{(j)}$ is controlled by the vector field $A^{(jk)}$. Concretely, given a point in U_j together with its coordinates, if the point also lies in U_k , then an infinitesimal change of the parameter u determines how the glued point in U_j , corresponding to the same point viewed in U_k , moves; namely, it moves in the direction determined by $A^{(jk)}$.

From this definition one immediately sees that

$$A^{(jk)} = -A^{(kj)}.$$

Moreover, consider three overlapping patches $U_i \cap U_j \cap U_k$. Fixing the point described in the U_i coordinates, the change of the corresponding glued point coming from the U_k description is governed by $A^{(ik)}$. On the other hand, the change of the point obtained via the U_j description is governed by $A^{(ij)}$, while the change relating the U_k and U_j descriptions is governed by $A^{(jk)}$. To first order these variations add linearly. Therefore, the compatibility of the three coordinate systems implies the Čech cocycle condition

$$A^{(ij)} + A^{(jk)} + A^{(ki)} = 0.$$

The exact 1-cocycles arise precisely from infinitesimal coordinate changes. Indeed, if on each patch U_j one chooses a holomorphic vector field

$$V^{(j)}(u),$$

then the induced change of transition functions gives

$$A^{(jk)}(u) = V^{(j)}(u) - V^{(k)}(u).$$

Such cocycles are therefore exact and correspond merely to changes of local coordinates rather than genuine deformations of the complex structure.

True deformations of the complex structure correspond to 1-cocycles which are not exact. Hence they define elements of the Čech cohomology group

$$H^1(M, \mathcal{O}(TM)),$$

By the Čech–Dolbeault theorem (Theorem 1.48), applied with

$$F = \mathcal{O}(TM),$$

one obtains an isomorphism

$$H^1(M, \mathcal{O}(TM)) \cong H^{0,1}(M, TM).$$

Therefore infinitesimal complex structure deformations are parametrized by elements of

$$H^{0,1}(M, TM),$$

which we denote by A .

2.2.2 Explicit Construction of The Čech–Dolbeault Isomorphism

In order to understand more clearly the relation between general complex structure deformations and their linearized (infinitesimal) counterparts, we need an explicit construction of the Čech–Dolbeault isomorphism. In the present discussion we restrict ourselves to the case

$$H^{0,1}(M, TM) \cong H^1(M, \mathcal{O}(TM)).$$

Let $\{U_i\}$ be an open cover of the complex manifold M , and write $U_{ij} = U_i \cap U_j$ and $U_{ijk} = U_i \cap U_j \cap U_k$. Denote by $\mathcal{O}(TM)$ the sheaf of holomorphic vector fields and by $\Omega^{0,q}(TM)$ the sheaf of smooth $(0, q)$ -forms valued in the holomorphic tangent bundle TM . The Dolbeault operator acts as

$$\bar{\partial} : \Omega^{0,q}(TM) \rightarrow \Omega^{0,q+1}(TM).$$

A fundamental fact is that the Dolbeault complex gives a fine resolution of $\mathcal{O}(TM)$:

$$0 \rightarrow \mathcal{O}(TM) \rightarrow \Omega^{0,0}(TM) \xrightarrow{\bar{\partial}} \Omega^{0,1}(TM) \xrightarrow{\bar{\partial}} \Omega^{0,2}(TM) \rightarrow \dots,$$

and therefore induces canonical isomorphisms

$$H^1(M, \mathcal{O}(TM)) \cong H^{0,1}(M, TM).$$

We now describe this isomorphism explicitly.

Let $\{\rho_k\}$ be a smooth partition of unity subordinate to the cover $\{U_k\}$, so that $\text{supp}(\rho_k) \subset U_k$ and

$$\sum_k \rho_k = 1.$$

Let $\{A^{(ij)}\}$ be a Čech 1-cocycle with values in $\mathcal{O}(TM)$:

$$A^{(ij)} \in \Gamma(U_{ij}, \mathcal{O}(TM)), \quad A^{(ij)} + A^{(jk)} + A^{(ki)} = 0 \quad \text{on } U_{ijk}.$$

Step 1: Construction of a smooth 0-cochain.

Define on each U_i a smooth TM -valued function

$$b_i := \sum_k \rho_k A^{(ik)} \in \Gamma(U_i, \Omega^{0,0}(TM)).$$

This expression is well-defined: for any point $x \in U_i$, only finitely many ρ_k are nonzero near x , and whenever $\rho_k(x) \neq 0$ one has $x \in U_k$, so $A^{(ik)}$ is defined in a neighborhood of x inside $U_i \cap U_k$.

Step 2: The Čech coboundary of b .

On the overlap U_{ij} ,

$$b_i - b_j = \sum_k \rho_k (A^{(ik)} - A^{(jk)}).$$

Using the cocycle condition, one obtains on U_{ijk}

$$A^{(ik)} - A^{(jk)} = A^{(ij)}.$$

Hence

$$b_i - b_j = \sum_k \rho_k A^{(ij)} = A^{(ij)},$$

since $\sum_k \rho_k = 1$. Thus $\{A^{(ij)}\}$ becomes a Čech coboundary after passing from the holomorphic sheaf $\mathcal{O}(TM)$ to the fine sheaf $\Omega^{0,0}(TM)$.

Step 3: Construction of a global $(0,1)$ -form.

Define

$$\alpha_i := \bar{\partial} b_i \in \Gamma(U_i, \Omega^{0,1}(TM)).$$

On U_{ij} ,

$$\alpha_i - \alpha_j = \bar{\partial}(b_i - b_j) = \bar{\partial} A^{(ij)}.$$

Since $A^{(ij)}$ is holomorphic, $\bar{\partial} A^{(ij)} = 0$, and therefore

$$\alpha_i = \alpha_j \quad \text{on } U_{ij}.$$

Hence the local forms glue to a global $(0,1)$ -form

$$\alpha \in \Gamma(M, \Omega^{0,1}(TM)), \quad \alpha|_{U_i} = \alpha_i.$$

Moreover,

$$\bar{\partial} \alpha = 0,$$

so α determines a Dolbeault cohomology class

$$[\alpha] \in H^{0,1}(M, TM).$$

Step 4: Independence of choices.

If the Čech cocycle is modified by a coboundary,

$$A^{(ij)} \mapsto A^{(ij)} + V^{(i)} - V^{(j)}, \quad V^{(i)} \in \Gamma(U_i, \mathcal{O}(TM)),$$

then the resulting α changes by a $\bar{\partial}$ -exact term. Using $\sum \rho = 1$ we get

$$\Delta \alpha_i = \Delta \bar{\partial} b_i = \sum_k (\bar{\partial} \rho_k)(V^{(i)} - V^{(k)}) = -\bar{\partial} \left(\sum_k \rho_k V^{(k)} \right).$$

Changing the partition of unity also alters α only by an exact term. Therefore the assignment

$$\{A^{(ij)}\} \mapsto [\alpha]$$

depends only on the cohomology class $[A] \in H^1(M, \mathcal{O}(TM))$ and yields the desired isomorphism

$$H^1(M, \mathcal{O}(TM)) \cong H^{0,1}(M, TM).$$

2.2.3 Deformation of the Dolbeault Operator and the Maurer-Cartan Equation

The first order deformation of the complex structure is described by an element

$$A(u) = A_{\bar{j}}^i(x, u) d\bar{z}^j \otimes \frac{\partial}{\partial z^i} \in H^{0,1}(M, TM).$$

Such a deformation modifies the Dolbeault operator to

$$\bar{\partial}_u = \bar{\partial} + A(u) \lrcorner \partial,$$

where $\lrcorner$ means contraction in the fiber of this vector bundle TM .

Equivalently, one may regard the vector field directly as a differential operator,

$$\bar{\partial}_u = \bar{\partial} + A(u),$$

in which case the contraction symbol can be omitted.

The condition $\bar{\partial}_u f(z) = 0$ defines holomorphic functions on M with respect to the deformed complex structure.

Remark 2.18 (From infinitesimal changes of transition functions to the variation of $\bar{\partial}$). We explain explicitly how an infinitesimal change of transition functions in Section 2.2.1 induces a first-order variation of the $\bar{\partial}$ -operator by using technique we have used in Section 2.2.2.

Step 1. Infinitesimal change of transition functions. Let $\{U_i\}$ be a holomorphic atlas with coordinates $z^{(i)}$. An infinitesimal deformation of the complex structure is encoded by holomorphic vector fields

$$A^{(ij)} \in \Gamma(U_{ij}, \mathcal{O}(TM))$$

satisfying the Čech 1-cocycle condition. Choose smooth $(1, 0)$ vector fields

$$b_i \in \Gamma(U_i, C^\infty(TM))$$

such that on overlaps

$$b_i - b_j = A^{(ij)} \quad \text{on } U_{ij}.$$

Step 2. Definition of deformed local coordinates. On each patch U_i define new coordinates (to first order in a parameter u)

$$w_{(i)}^a(u) := z_{(i)}^a - u b_i^a(z_{(i)}, \bar{z}_{(i)}) \quad (\text{mod } u^2),$$

where

$$b_i = \sum_a b_i^a \frac{\partial}{\partial z^a}.$$

On overlaps one checks (*don't forget to consider the difference between b_j^a and the a -th component of b_j in the coordinate U_i !!*)

$$w_{(i)}^a(u) = f_{(ij)}^a \left(w_{(j)}^a(u), 0 \right) \quad (\text{mod } u^2),$$

since

$$z_{(i)}^a = f_{(ij)}^a \left(z_{(j)}^a, u \right) = f_{(ij)}^a \left(z_{(j)}^a, 0 \right) + u A_{(ij)}^a,$$

which reproduces precisely the infinitesimal change of transition functions.

Step 3. Definition of the deformed $\bar{\partial}$ operator. The new complex structure is characterized by requiring the new coordinates to be holomorphic:

$$\bar{\partial}_u w_{(i)}^a(u) = 0 \quad (\text{mod } u^2).$$

Assume the deformation has the form

$$\bar{\partial}_u = \bar{\partial} + uD \quad (\text{mod } u^2),$$

where D is a first-order differential operator.

Substituting $w = z - ub_i$ gives

$$0 = (\bar{\partial} + uD)(z - ub_i) = \bar{\partial}z + uD(z) - u\bar{\partial}b_i.$$

Since the original coordinates are holomorphic,

$$\bar{\partial}z = 0,$$

hence

$$D(z) = \bar{\partial}b_i.$$

Step 4. Action on general functions. A first-order differential operator is determined by its action on coordinate functions. Therefore for any smooth function f ,

$$D(f) = \sum_a D(z^a) \frac{\partial f}{\partial z^a} = (\bar{\partial}b_i) \lrcorner \partial f,$$

Consequently,

$$\bar{\partial}_u f = \bar{\partial}f + u(\bar{\partial}b_i) \lrcorner \partial f \quad (\text{mod } u^2).$$

Step 5. Appearance of a global $(0, 1)$ -form. Define

$$\alpha_i := \bar{\partial}b_i \in \Gamma(U_i, \Omega^{0,1}(TM)).$$

On overlaps,

$$\alpha_i - \alpha_j = \bar{\partial}(b_i - b_j) = \bar{\partial}A^{(ij)} = 0,$$

because $A^{(ij)}$ is holomorphic. Hence the α_i glue to a global form

$$\alpha \in \Gamma(M, \Omega^{0,1}(TM)).$$

Conclusion. The infinitesimal variation of the complex structure is therefore

$$\delta\bar{\partial}(f) = \alpha \lrcorner \partial f,$$

which means

$$\delta\bar{\partial} = \alpha \lrcorner \partial,$$

so an infinitesimal deformation of transition functions $A(u)$ determines, and is determined by, a corresponding vector-valued $(0, 1)$ -form

$$\alpha \in \Omega^{0,1}(TM).$$

Henceforth, we use the same symbol A for an element of $H^1(M, \mathcal{O}(TM)) \cong H^{0,1}(M, TM)$.

The integrability condition for the new complex structure requires

$$\bar{\partial}_u^2 = 0,$$

which leads to the Maurer–Cartan equation:

$$\bar{\partial}A(u) + \frac{1}{2}[A(u), A(u)] = 0.$$

The verification is a straightforward computation and will be omitted.

Here $[\cdot, \cdot]$ denotes the Lie bracket extended to $(0, *)$ -form valued vector fields as follows. For

$$\phi(z) = \phi^i(z) \partial_{z_i} \in \mathcal{L}^{0,p}(T), \quad \omega(z) \in \mathcal{L}^{0,q}(T),$$

where

$$\phi^i = \frac{1}{p!} \phi_{\bar{i}_1 \dots \bar{i}_p}^i d\bar{z}^{\bar{i}_1} \wedge \dots \wedge d\bar{z}^{\bar{i}_p},$$

the bracket is defined by

$$[\phi, \omega] = (\phi^i \wedge \partial_i \omega^j - (-1)^{pq} \omega^i \wedge \partial_i \phi^j) \partial_j.$$

In particular, the bracket of two $(0, 1)$ -form valued vector fields produces a $(0, 2)$ -form valued vector field.

The Maurer–Cartan equation above is equivalent to the vanishing of the Nijenhuis tensor, hence characterizes the integrability of the deformed complex structure.

2.2.4 Unobstructedness of the Complex Moduli Space

Now let M be a compact Calabi–Yau manifold of complex dimension n , i.e. a compact Kähler manifold admitting a nowhere vanishing holomorphic $(n, 0)$ -form

$$\Omega \in H^{n,0}(M).$$

We explain why deformations of the complex structure are unobstructed, following the Tian–Todorov argument.

Writing a formal deformation

$$A(u) = \sum_{k \geq 1} A_k u^k,$$

the first-order condition gives

$$\bar{\partial}A_1 = 0,$$

so infinitesimal deformations are parametrized by

$$H^{0,1}(M, TM).$$

Higher orders satisfy the recursive equations

$$\bar{\partial}A_k = -\frac{1}{2} \sum_{i=1}^{k-1} [A_i, A_{k-i}], \quad k > 1.$$

The right-hand side represents the obstruction class in $H^{0,2}(M, TM)$. In order to solve these equations, we should first prove that the obstruction classes are the zero classes.

Definition 2.19. Contraction with Ω defines an isomorphism

$$\Omega^{0,p}(M, TM) \longrightarrow \Omega^{n-1,p}(M), \quad A \mapsto \widehat{A} := \iota_A \Omega.$$

Explicitly, contraction with the holomorphic $(n, 0)$ -form Ω associates to

$$A = A_{j_1, \dots, \bar{j}_p}^i d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_p} \frac{\partial}{\partial z^i} \in H^{0,p}(M, TM)$$

a form $\widehat{A} \in H^{n-1,p}(M)$ defined by

$$\widehat{A} = \frac{1}{(n-1)!} A_{j_1, \dots, \bar{j}_p}^i \Omega_{ji_2 \dots i_n} dz^{i_2} \wedge \dots \wedge dz^{i_n} \wedge d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_p}.$$

The inverse map is given by

$$(\widehat{A})^\vee = \frac{1}{(n-1)!|\Omega|^2} \bar{\Omega}^{i, i_2, \dots, i_n} \widehat{A}_{i_2, \dots, i_n, \bar{j}_1, \dots, \bar{j}_p} d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_p} \frac{\partial}{\partial z^i}.$$

Since Ω is holomorphic, this operation commutes with $\bar{\partial}$, hence induces an isomorphism on cohomology

$$H^{0,p}(M, TM) \cong H^{n-1,p}(M).$$

Under this identification the Maurer–Cartan equation becomes

$$\bar{\partial} \widehat{A} = -\frac{1}{2} \widehat{[A, A]} =: -\frac{1}{2} [\widehat{A}, \widehat{A}].$$

Expanding in powers of u gives

$$\bar{\partial} \widehat{A}_k = \frac{1}{2} \sum_{i=1}^{k-1} [\widehat{A}_i, \widehat{A}_{k-i}], \quad k > 1.$$

Lemma 2.20 (Tian–Todorov). *Let $A, B \in \Omega^{0,*}(M, TM)$, then the induced bracket satisfies*

$$[\widehat{A}, \widehat{B}] := \widehat{[A, B]} = \partial(\widehat{A \wedge B}) - (D \cdot A) \wedge \widehat{B} + \widehat{A} \wedge (D \cdot B),$$

where $A \wedge B \in \Omega^{0,*}(M, TM \otimes TM)$, $\widehat{A \wedge B} \in \Omega^{n-2,*}(M)$, and the operator D is the contraction

$$D \cdot A = (\partial_i A_{j_1 \dots \bar{j}_p}^i) d\bar{z}^{j_1} \wedge \dots \wedge d\bar{z}^{j_p}.$$

Remark 2.21. The identity expresses the Kodaira–Spencer bracket, after contraction with the holomorphic volume form, as a ∂ -exact term up to corrections involving the divergence operator D . In particular, after choosing representatives satisfying the gauge condition $D \cdot A = D \cdot B = 0$, the bracket becomes ∂ -exact, which is the key input in the Tian–Todorov unobstructedness argument.

Theorem 2.22 (Tian–Todorov). *Let M be a compact Calabi–Yau manifold of complex dimension n . Then deformations of the complex structure of M are unobstructed: every first order deformation class*

$$[A_1] \in H^{0,1}(M, TM)$$

extends to an actual (analytic) family of complex structures. Equivalently, the complex moduli space is smooth near $[M]$ and

$$T_{[M]} \mathcal{M}_{\text{cpx}} \cong H^{0,1}(M, TM) \cong H^{n-1,1}(M).$$

Proof. We describe the standard power series construction as before. A deformation of complex structure may be encoded by a formal series

$$A(u) = \sum_{k \geq 1} A_k u^k, \quad A_k \in \Omega^{0,1}(M, TM),$$

required to satisfy

$$\bar{\partial} \hat{A}_1 = 0, \quad \bar{\partial} \hat{A}_k = \frac{1}{2} \sum_{i=1}^{k-1} [\hat{A}_i, \hat{A}_{k-i}], \quad k > 1.$$

Choice of gauge and the first step. We may represent the first order class by a harmonic form: choose $\hat{A}_1 \in \mathcal{H}^{n-1,1}(M)$. On a compact Kähler manifold, harmonicity implies

$$\bar{\partial} \hat{A}_1 = 0 \quad \text{and} \quad \partial \hat{A}_1 = 0.$$

In particular by $\partial \hat{A}_1 = 0$ the divergence condition $D \cdot A_1 = 0$ holds, and the Tian–Todorov lemma yields

$$[\hat{A}_1, \hat{A}_1] = \partial(\widehat{A_1 \wedge A_1}),$$

hence $[\hat{A}_1, \hat{A}_1]$ is ∂ -exact. On the other hand, since $\bar{\partial} \hat{A}_1 = 0$, one checks from the definition of the bracket that

$$\bar{\partial}[\hat{A}_1, \hat{A}_1] = 0.$$

By the $\partial\bar{\partial}$ -lemma there exists a form ψ_1 such that

$$[\hat{A}_1, \hat{A}_1] = \bar{\partial} \partial \psi_1.$$

Setting

$$\hat{A}_2 := \frac{1}{2} \partial \psi_1$$

gives

$$\bar{\partial} \hat{A}_2 + \frac{1}{2} [\hat{A}_1, \hat{A}_1] = 0,$$

so the recursion is solved up to order u^2 .

Inductive step. Assume for some $N \geq 2$ that we have constructed $\hat{A}_1, \dots, \hat{A}_N$ satisfying

$$\partial \hat{A}_i = 0, \quad \bar{\partial} \hat{A}_i = \frac{1}{2} \sum_{j=1}^{i-1} [\hat{A}_j, \hat{A}_{i-j}], \quad i = 1, \dots, N,$$

(with the convention that the sum is 0 for $i = 1$). Consider the obstruction term at order $N + 1$:

$$\eta := \sum_{j=1}^N [\hat{A}_j, \hat{A}_{N+1-j}].$$

(i) η is ∂ -exact. Using the Tian–Todorov lemma and the gauge condition $D \cdot A_i = 0$ (which is ensured by $\partial \hat{A}_i = 0$ in this setup), we obtain

$$\eta = \partial \sum_{j=1}^N (A_j \wedge A_{N+1-j})^\wedge.$$

(ii) η is $\bar{\partial}$ -closed. Using that $\bar{\partial}$ is a derivation and the induction hypothesis for $\bar{\partial}\hat{A}_i$, together with the Jacobi identity for the bracket, one checks

$$\bar{\partial}\eta = 0.$$

Since η is both $\bar{\partial}$ -closed and ∂ -exact, the $\partial\bar{\partial}$ -lemma gives a form ψ_N such that

$$\eta = \bar{\partial}\partial\psi_N.$$

Define

$$\hat{A}_{N+1} := \frac{1}{2}\partial\psi_N.$$

Then

$$\bar{\partial}\hat{A}_{N+1} = \frac{1}{2}\eta = \frac{1}{2}\sum_{j=1}^N[\hat{A}_j, \hat{A}_{N+1-j}],$$

which is precisely the desired recursion at order $N+1$. Moreover $\partial\hat{A}_{N+1} = 0$ holds by construction, so the induction continues.

Thus we construct a formal solution $A(u)$ to the Maurer–Cartan equation with prescribed first order term A_1 . Standard analytic arguments (not reproduced here) show convergence for small u , hence every class in $H^{0,1}(M, TM)$ integrates to an actual family of complex structures. This proves unobstructedness and the identification of the tangent space. $\square$

Remark 2.23. The Tian–Todorov theorem implies that there always exists a family of Calabi–Yau manifolds with varying complex structures. The dimension of the complex moduli space is therefore given by

$$\dim_{\mathbb{C}} \mathcal{M}_{\text{cs}} = h^{0,1}(M, TM) = h^{n-1,1}(M).$$

This result is also important from the viewpoint of worldsheet physics. In particular, it ensures that some deformation is exactly marginal, and hence generates a family of $\mathcal{N} = 2$ superconformal field theories.

2.3 The Variation of Hodge Structures

2.3.1 Hodge Filtration and Hodge Structure

The primitive horizontal subspace in the cohomology of a Calabi–Yau n -fold M carries a polarized Hodge structure (Section 1.3.2). In order to study its variation over the complex moduli space $\mathcal{M}_{\text{cs}}$, one fixes an integral structure

$$H^n(M, \mathbb{Z}),$$

relative to which the variation of the Hodge components $H^{p,q}(M_z)$ can be measured inside the fixed real vector space

$$H^n(M, \mathbb{R}),$$

as the complex structure varies with parameter z .

However, the spaces $H^{p,q}(M_z)$ themselves do not assemble into holomorphic vector bundles over $\mathcal{M}_{\text{cs}}$. Instead, the variation of Hodge structure is captured by the Hodge filtration associated to the family

$$\pi : \mathcal{M}_n \longrightarrow \mathcal{M}_{\text{cs}}, \quad \pi^{-1}(t) = M_t.$$

Definition 2.24 (Hodge filtration). Let

$$H^n(M, \mathbb{C}) = \bigoplus_{k=0}^n H^{n-k,k}(M), \quad \overline{H^{p,q}(M)} = H^{q,p}(M),$$

be the Hodge decomposition with respect to the real structure $H^n(M, \mathbb{R})$. The associated Hodge filtration is the decreasing filtration

$$F^\bullet = \{F^p\}_{p=0}^n,$$

defined by

$$F^p = \bigoplus_{l \geq p} H^{l, n-l}.$$

The Hodge filtration defines a decreasing sequence of subspaces

$$H^n = F^0 \supset F^1 \supset \dots \supset F^n \supset F^{n+1} = 0.$$

Proposition 2.25. *The Hodge decomposition can be recovered from the filtration $F^\bullet$ together with complex conjugation by the relations*

$$F^p \oplus \overline{F^{n-p+1}} = H^n(M, \mathbb{C}), \quad H^{p,q}(M) = F^p(M) \cap \overline{F^q(M)}.$$

Equivalently, one has the identification

$$H^{p,q}(M) \cong F^p / F^{p+1}.$$

Definition 2.26. The Hodge filtration together with the lattice structure

$$H^n(M, \mathbb{Z}) \subset H^n(M, \mathbb{C})$$

determines the (polarized) Hodge structure on the middle cohomology.

Remark 2.27. Unlike the spaces $H^{p,q}(M)$, the subspaces $F^p(M)$ vary holomorphically with the complex structure. Since we work locally on the moduli space, one may locally trivialize the bundle whose fibers are $H^n(M, \mathbb{C})$. In this local picture, the holomorphic variation of the filtration can be interpreted as the statement that the assignment

$$U \longrightarrow \mathrm{Gr}(p, h^n),$$

which sends a point to the corresponding subspace F^p , defines a holomorphic map into the Grassmannian. A more global interpretation of this fact will be discussed in the next section.

Consequently, the spaces F^p assemble into locally free sheaves over the complex moduli space $\mathcal{M}_{\mathrm{cs}}$, forming a decreasing filtration

$$\mathcal{F}^p \subset \mathcal{F}^{p-1}.$$

2.3.2 Construction of the Hodge Filtration via Hypercohomology

Let X be a compact complex manifold of complex dimension n . Denote by Ω_X^p the sheaf of holomorphic p -forms on X .

Definition 2.28 (Holomorphic de Rham complex). The *holomorphic de Rham complex* of X is the complex of sheaves

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \Omega_X^1 \longrightarrow \Omega_X^2 \longrightarrow \cdots \longrightarrow \Omega_X^n.$$

Definition 2.29 (Hypercohomology). Let $\mathcal{F}^\bullet$ be a bounded complex of sheaves on a topological space X

$$\cdots \longrightarrow \mathcal{F}^p \longrightarrow \mathcal{F}^{p+1} \longrightarrow \cdots.$$

Choose an injective resolution for each $\mathcal{F}^p$

$$\mathcal{F}^p \longrightarrow \mathcal{I}^{p,0} \longrightarrow \mathcal{I}^{p,1} \longrightarrow \cdots.$$

This produces a double complex of sheaves $\mathcal{I}^{p,q}$. Taking global sections gives a double complex of vector spaces

$$\Gamma(X, \mathcal{I}^{p,q}).$$

Let $\text{Tot}^\bullet$ denote the associated total complex

$$\text{Tot}^k = \bigoplus_{p+q=k} \Gamma(X, \mathcal{I}^{p,q}).$$

The k -th hypercohomology of $\mathcal{F}^\bullet$ is defined by

$$\mathbb{H}^k(X, \mathcal{F}^\bullet) = H^k(\text{Tot}^\bullet).$$

Remark 2.30. If $\mathcal{F}^\bullet$ is concentrated in degree 0, namely

$$\mathcal{F}^\bullet = \mathcal{F}[0],$$

then the hypercohomology reduces to the usual sheaf cohomology

$$\mathbb{H}^k(X, \mathcal{F}^\bullet) = H^k(X, \mathcal{F}).$$

Definition 2.31 (Quasi-isomorphism). Let $\mathcal{A}^\bullet$ and $\mathcal{B}^\bullet$ be complexes of sheaves. A morphism of complexes

$$\mathcal{A}^\bullet \longrightarrow \mathcal{B}^\bullet$$

is called a quasi-isomorphism if it induces isomorphisms on all cohomology sheaves

$$\mathcal{H}^k(\mathcal{A}^\bullet) \cong \mathcal{H}^k(\mathcal{B}^\bullet).$$

Proposition 2.32 (Holomorphic de Rham complex resolves $\mathbb{C}_X$). *Let X be a complex manifold. Then the natural morphism of complexes*

$$\mathbb{C}_X \longrightarrow \Omega_X^\bullet$$

is a quasi-isomorphism.

Proof. View $\mathbb{C}_X$ as a complex concentrated in degree 0

$$\cdots \rightarrow 0 \rightarrow \mathbb{C}_X \rightarrow 0 \rightarrow \cdots.$$

There is a natural morphism of complexes

$$\mathbb{C}_X \longrightarrow \Omega_X^\bullet$$

given by viewing a locally constant function as a holomorphic function.

We compute the cohomology sheaves of $\Omega_X^\bullet$.

First consider degree 0. The kernel of

$$d : \mathcal{O}_X \rightarrow \Omega_X^1$$

consists of holomorphic functions satisfying $df = 0$. Such functions are locally constant, hence

$$\ker d = \mathbb{C}_X.$$

Since the image from the previous term is zero, we obtain

$$\mathcal{H}^0(\Omega_X^\bullet) = \mathbb{C}_X.$$

For $p > 0$, the holomorphic Poincaré lemma states that locally every closed holomorphic p -form is exact. Hence

$$\ker(d : \Omega_X^p \rightarrow \Omega_X^{p+1}) = \text{im}(d : \Omega_X^{p-1} \rightarrow \Omega_X^p),$$

so

$$\mathcal{H}^p(\Omega_X^\bullet) = 0.$$

Thus the cohomology sheaves of $\Omega_X^\bullet$ are

$$\mathcal{H}^0(\Omega_X^\bullet) = \mathbb{C}_X, \quad \mathcal{H}^p(\Omega_X^\bullet) = 0 \ (p > 0).$$

These coincide with the cohomology sheaves of $\mathbb{C}_X$, so the morphism

$$\mathbb{C}_X \rightarrow \Omega_X^\bullet$$

is a quasi-isomorphism. □

Proposition 2.33. *Let $\mathcal{A}^\bullet$ and $\mathcal{B}^\bullet$ be bounded complexes of sheaves on a topological space X . If*

$$\mathcal{A}^\bullet \longrightarrow \mathcal{B}^\bullet$$

is a quasi-isomorphism, then it induces isomorphisms on hypercohomology

$$\mathbb{H}^k(X, \mathcal{A}^\bullet) \cong \mathbb{H}^k(X, \mathcal{B}^\bullet)$$

for all k .

Proof. Choose injective resolutions

$$\mathcal{A}^\bullet \rightarrow \mathcal{I}^{p,\bullet}, \quad \mathcal{B}^\bullet \rightarrow \mathcal{J}^{p,\bullet}.$$

These resolutions form double complexes of sheaves. Taking global sections produces double complexes

$$\Gamma(X, \mathcal{I}^{p,q}), \quad \Gamma(X, \mathcal{J}^{p,q}).$$

The morphism $\mathcal{A}^\bullet \rightarrow \mathcal{B}^\bullet$ induces a morphism between these double complexes and hence a morphism between their associated total complexes.

Since the original morphism is a quasi-isomorphism, it induces isomorphisms on the cohomology sheaves of the complexes. Standard homological algebra then implies that the induced morphism between the total complexes is a quasi-isomorphism.

Therefore the cohomology groups of the total complexes coincide, which means

$$\mathbb{H}^k(X, \mathcal{A}^\bullet) \cong \mathbb{H}^k(X, \mathcal{B}^\bullet).$$

□

Remark 2.34. This result shows that hypercohomology depends only on the quasi-isomorphism class of the complex of sheaves. In particular, if a complex $\mathcal{F}^\bullet$ is quasi-isomorphic to a single sheaf $\mathcal{F}[0]$, then

$$\mathbb{H}^k(X, \mathcal{F}^\bullet) \cong H^k(X, \mathcal{F}).$$

Remark 2.35 (Derived category interpretation). The statement of Proposition 2.33 has a more intrinsic formulation in the language of derived categories.

Recall that for a complex of sheaves $A^\bullet$ on X , the hypercohomology is defined as

$$\mathbb{H}^k(X, A^\bullet) = H^k(R\Gamma(X, A^\bullet)),$$

where $R\Gamma(X, -)$ denotes the derived functor of the global section functor

$$\Gamma(X, -) : \text{Sh}(X) \rightarrow \text{Ab}.$$

Equivalently, if $A^\bullet \rightarrow I^\bullet$ is an injective resolution, then

$$R\Gamma(X, A^\bullet) \simeq \Gamma(X, I^\bullet).$$

Thus hypercohomology can be computed as

$$\mathbb{H}^k(X, A^\bullet) = H^k(\Gamma(X, I^\bullet)).$$

Now suppose

$$A^\bullet \rightarrow B^\bullet$$

is a quasi-isomorphism. By definition of the derived category $D^+(X)$, this means that

$$A^\bullet \cong B^\bullet$$

as objects of $D^+(X)$.

Since $R\Gamma(X, -)$ is a functor

$$R\Gamma(X, -) : D^+(X) \rightarrow D^+(\text{Ab}),$$

it preserves isomorphisms. Therefore we obtain an isomorphism

$$R\Gamma(X, A^\bullet) \cong R\Gamma(X, B^\bullet)$$

in $D^+(\text{Ab})$.

Taking cohomology groups gives

$$\mathbb{H}^k(X, A^\bullet) \cong \mathbb{H}^k(X, B^\bullet)$$

for all k .

In this formulation the proposition becomes a formal consequence of the fact that hypercohomology is defined via the derived functor $R\Gamma$. The use of injective resolutions in the proof is merely a concrete model for computing this derived functor.

Corollary 2.36. *Let X be a complex manifold. Then there is a natural isomorphism*

$$\mathbb{H}^k(X, \Omega_X^\bullet) \cong H^k(X, \mathbb{C}).$$

If moreover X is compact Kähler, then the Hodge-de Rham spectral sequence degenerates at E_1 , and hence

$$H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^q(X, \Omega_X^p).$$

Proof. There is a natural morphism of complexes of sheaves

$$\mathbb{C}_X \longrightarrow \Omega_X^\bullet,$$

where a locally constant function is viewed as a holomorphic function. By the holomorphic Poincaré lemma, this is a quasi-isomorphism. Therefore

$$\mathbb{H}^k(X, \Omega_X^\bullet) \cong \mathbb{H}^k(X, \mathbb{C}_X).$$

Since $\mathbb{C}_X$ is concentrated in degree 0, its hypercohomology is just the usual sheaf cohomology:

$$\mathbb{H}^k(X, \mathbb{C}_X) = H^k(X, \mathbb{C}).$$

This proves the first assertion.

If X is compact Kähler, consider the spectral sequence associated to the stupid filtration on $\Omega_X^\bullet$. This spectral sequence arises from the double complex used to compute hypercohomology and converges to the cohomology of the associated total complex.

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow \mathbb{H}^{p+q}(X, \Omega_X^\bullet).$$

By the $\partial\bar{\partial}$ -lemma, this spectral sequence degenerates at E_1 . Combining this with the first part yields

$$H^k(X, \mathbb{C}) \cong \bigoplus_{p+q=k} H^q(X, \Omega_X^p).$$

□

Remark 2.37. Thus the complex cohomology of X may be computed as the hypercohomology of the holomorphic de Rham complex. This description is particularly useful when studying the Hodge filtration.

Definition 2.38 (Truncated de Rham complex). For each integer p define the truncated complex

$$\Omega_X^{\geq p} : 0 \longrightarrow \Omega_X^p \longrightarrow \Omega_X^{p+1} \longrightarrow \cdots \longrightarrow \Omega_X^n.$$

Definition 2.39 (Hodge filtration). The Hodge filtration on the cohomology $H^k(X, \mathbb{C})$ is defined by

$$F^p H^k(X, \mathbb{C}) = \text{Im} \left(\mathbb{H}^k(X, \Omega_X^{\geq p}) \rightarrow \mathbb{H}^k(X, \Omega_X^\bullet) \right).$$

Proposition 2.40. *If X is compact Kähler, then the filtration defined above coincides with the usual Hodge filtration*

$$F^p H^k(X, \mathbb{C}) = \bigoplus_{r \geq p} H^{r, k-r}(X).$$

Remark 2.41. From this point of view the Hodge filtration is obtained by filtering the holomorphic de Rham complex by the degree of differential forms. The associated filtration on cohomology is induced via hypercohomology.

This description is particularly useful in families of complex manifolds, where the complexes $\Omega_{X/B}^\bullet$ form complexes of sheaves on the base and the corresponding hypercohomology groups assemble into the Hodge bundles.

Remark 2.42. In the language of derived categories the above construction may be rephrased by saying that the filtered complex

$$\Omega_X^\bullet$$

defines an object in the filtered derived category and the Hodge filtration arises from the truncation functor. We will not pursue this point of view here since the hypercohomological definition already suffices for our purposes.

2.3.3 Families of Complex Manifolds

We formalize varying complex structures using families of complex manifolds.

Definition 2.43. Let U be a complex manifold with coordinates

$$u = (u^1, \dots, u^N).$$

A *family of complex manifolds* consists of a complex manifold $\mathcal{M}$ together with a holomorphic map

$$\pi : \mathcal{M} \longrightarrow U$$

such that π is a proper holomorphic submersion. The fibre over $u \in U$ is

$$M_u := \pi^{-1}(u),$$

which is a complex manifold.

If U is connected and $0 \in U$ is a reference point, we say that $\mathcal{M}$ is a *family of deformations* of the fibre M_0 . Each fibre M_u is then called a deformation of M_0 .

The following fundamental result explains the topology of smooth families.

Theorem 2.44 (Ehresmann). *Let*

$$\pi : \mathcal{M} \longrightarrow U$$

be a proper submersion between differentiable manifolds, and assume U is contractible with base point 0 . Then there exists a diffeomorphism

$$T : \mathcal{M} \xrightarrow{\simeq} M_0 \times U$$

over U , i.e.

$$\text{pr}_2 \circ T = \pi.$$

Remark 2.45. Giving such a trivialisation is equivalent to giving its first component

$$T_0 : \mathcal{M} \rightarrow M_0,$$

which induces diffeomorphisms

$$M_u \simeq M_0$$

for all $u \in U$. Up to composing T with $(T_0|_{M_0})^{-1}$, we may assume

$$T_0|_{M_0} = \text{Id},$$

so that T_0 is a retraction of $\mathcal{M}$ onto M_0 .

Proposition 2.46. *Let*

$$\pi : \mathcal{M} \rightarrow U$$

be a family of complex manifolds and let $0 \in U$. Then, after possibly replacing U by a neighbourhood of 0, there exists a C^∞ trivialisation

$$T = (T_0, \pi) : \mathcal{M} \longrightarrow M_0 \times U$$

such that the fibres of

$$T_0 : \mathcal{M} \rightarrow M_0$$

are complex submanifolds of $\mathcal{M}$.

Remark 2.47. The fibres of T_0 are submanifolds of $\mathcal{M}$ which are diffeomorphic to U . The fact that they are complex does *not* imply that T_0 is holomorphic, since these submanifolds do not vary holomorphically with the point $x \in M_0$.

However, this statement means that the induced family of complex structures on M_0 , parametrised by U , varies holomorphically. More precisely, for each $u \in U$ the identification

$$M_u \xrightarrow{\sim} M_0$$

induced by $T_0|_{M_u}$ equips M_0 with a complex structure depending holomorphically on u .

Remark 2.48 (Calabi–Yau deformation theory). For Calabi–Yau manifolds, deformation theory predicts that the dimension of the deformation space is

$$\dim_{\mathbb{C}} U = h^{0,1}(M, TM) = h^{n-1,1}(M).$$

It is often useful to consider a compactification of U to the complex moduli space $\mathcal{M}_{\text{cs}}$, which may contain singular fibres. Away from these singular loci (which occur in complex codimension at least one), the fibres M_u are smooth and mutually diffeomorphic.

Remark 2.49 (Local product structure). Away from singular fibres, the projection π is locally trivial. Hence for a neighbourhood $V \subset \mathcal{M}_{\text{cs}}$ there exists a local product structure

$$\pi^{-1}(V) \simeq M \times V.$$

In particular, derivatives with respect to the moduli coordinates u^i commute with derivatives along the fibre directions.

Remark 2.50 (Why variation of Hodge structure appears). By Ehresmann's theorem, a smooth proper family

$$\pi : \mathcal{M} \rightarrow U$$

is locally trivial as a C^∞ fibre bundle: for sufficiently small $V \subset U$,

$$\pi^{-1}(V) \cong_{C^\infty} M_0 \times V.$$

Hence all fibres are canonically identified as smooth manifolds, and the cohomology groups $H^n(M_u, \mathbb{Z})$ form a locally constant system.

However, this trivialisation is in general *not* biholomorphic. If there existed a holomorphic product structure

$$\pi^{-1}(V) \cong_{\text{hol}} M_0 \times V,$$

then all fibres would be biholomorphic to M_0 , and the complex structure would be locally constant. Consequently the Hodge decomposition would also be locally constant, yielding a trivial variation of Hodge structure.

Therefore, the essential geometric situation is that the underlying C^∞ manifold is locally constant while the complex structure varies holomorphically with $u \in U$. The resulting change of the Hodge filtration inside the fixed cohomology space is precisely what is encoded by the variation of Hodge structure.

Definition 2.51 (Locally constant cycles). Let $C_u \subset M_u$ be a real k -dimensional submanifold. We say that $\{C_u\}$ is *locally constant* if it represents the same integral homology class

$$[C_u] \in H_k(M_u, \mathbb{Z})$$

for all u in a neighbourhood. Since homology is integral, this class cannot jump under smooth deformations as long as we avoid singular fibres.

Remark 2.52 (Local identification and monodromy). The definition implicitly uses an identification between the homology groups of nearby fibres. Away from singular fibres, over a sufficiently small neighbourhood $V \subset U$ the family is smoothly trivial,

$$\pi^{-1}(V) \cong M \times V,$$

and all fibres M_u are canonically diffeomorphic. This induces natural isomorphisms

$$H_k(M_u, \mathbb{Z}) \cong H_k(M_{u_0}, \mathbb{Z}),$$

under which a locally constant family of cycles corresponds to a flat section of the homology local system

$$\mathcal{H}_k := R^k \pi_* \mathbb{Z}.$$

Globally such identifications need not be single-valued. Parallel transport of cycles along loops in U may produce a nontrivial automorphism of $H_k(M, \mathbb{Z})$, known as the *monodromy action*. The definition above therefore relies only on local trivializations, where the base is taken sufficiently small (in particular, simply connected) so that monodromy does not appear.

2.3.4 The Kodaira–Spencer Map

Let

$$\phi : \mathcal{X} \rightarrow B$$

be a holomorphic family of complex manifolds, and let

$$X = \phi^{-1}(0)$$

be the fibre over $0 \in B$.

Proposition 2.53. *The differential*

$$\phi_* : T_{\mathcal{X}} \longrightarrow \phi^* T_B$$

is a morphism of holomorphic vector bundles. Its restriction to X has kernel equal to the holomorphic tangent bundle T_X . Consequently there is a short exact sequence of holomorphic vector bundles on X

$$0 \longrightarrow T_X \longrightarrow T_{\mathcal{X}}|_X \longrightarrow \phi^* T_B|_X \longrightarrow 0.$$

Remark 2.54. Since X is the fibre over 0, the pullback bundle $\phi^* T_B|_X$ is a trivial holomorphic vector bundle with fibre $T_{B,0}$. Thus

$$\phi^* T_B|_X \cong T_{B,0} \otimes \mathcal{O}_X.$$

Hence the above short exact sequence can be viewed as an extension of the bundle T_X by the trivial bundle $T_{B,0} \otimes \mathcal{O}_X$.

Definition 2.55. Let X be a complex manifold and let E, F be $\mathcal{O}_X$ -modules. The groups

$$\mathrm{Ext}^i(E, F)$$

are defined as the right derived functors of the functor

$$\mathrm{Hom}(E, -).$$

Equivalently, if

$$0 \longrightarrow F \longrightarrow I^0 \longrightarrow I^1 \longrightarrow \dots$$

is an injective resolution of F , then

$$\mathrm{Ext}^i(E, F) = H^i(\mathrm{Hom}(E, I^\bullet)).$$

Definition 2.56. Let X be a complex manifold and let E, F be $\mathcal{O}_X$ -modules. The sheaves

$$\mathcal{E}xt^i(E, F)$$

are defined as the right derived functors of the sheaf functor

$$\mathcal{H}om(E, -).$$

Equivalently, if

$$0 \longrightarrow F \longrightarrow I^0 \longrightarrow I^1 \longrightarrow \dots$$

is an injective resolution of F , then

$$\mathcal{E}xt^i(E, F) = \mathcal{H}^i(\mathcal{H}om(E, I^\bullet)).$$

Remark 2.57. There is a spectral sequence

$$H^p(X, \mathcal{E}xt^q(E, F)) \Rightarrow \text{Ext}^{p+q}(E, F).$$

In particular, if E is locally free, then

$$\mathcal{E}xt^q(E, F) = 0 \quad (q > 0),$$

and therefore

$$\text{Ext}^1(E, F) \cong H^1(X, \mathcal{H}om(E, F)).$$

We have the following proposition.

Proposition 2.58. *Let E, G be $\mathcal{O}_X$ -modules. Then $\text{Ext}^1(G, E)$ is in natural bijection with the set of isomorphism classes of extensions*

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0.$$

Despite the above proposition in the general setting, we are more interested in the following special case, whose proof provides useful insight into the problem we are considering.

Proposition 2.59. *Let X be a complex manifold, let E be an $\mathcal{O}_X$ -module, and let G be a locally free $\mathcal{O}_X$ -module. Then the isomorphism classes of extensions*

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0$$

are classified by

$$H^1(X, \mathcal{H}om(G, E)).$$

Equivalently, since G is locally free,

$$H^1(X, \mathcal{H}om(G, E)) \cong \text{Ext}^1(G, E).$$

Proof. We first construct a map from extensions to $H^1(X, \mathcal{H}om(G, E))$.

Let

$$0 \longrightarrow E \xrightarrow{\iota} F \xrightarrow{p} G \longrightarrow 0$$

be an extension. Since G is locally free, it is locally projective. Hence there exists an open cover $\{U_i\}$ of X such that on each U_i the above sequence splits. Choose a local splitting

$$s_i : G|_{U_i} \longrightarrow F|_{U_i}$$

satisfying

$$p \circ s_i = \text{id}_{G|_{U_i}}.$$

On an overlap $U_{ij} = U_i \cap U_j$, the difference

$$s_i - s_j : G|_{U_{ij}} \longrightarrow F|_{U_{ij}}$$

takes values in $\text{Ker}(p) = E$, since

$$p \circ (s_i - s_j) = \text{id}_G - \text{id}_G = 0.$$

Thus $s_i - s_j$ may be regarded as a section

$$a_{ij} \in \mathcal{H}om(G, E)(U_{ij}).$$

We claim that $\{a_{ij}\}$ is a Čech 1-cocycle. Indeed, on a triple overlap U_{ijk} ,

$$a_{ij} + a_{jk} + a_{ki} = (s_i - s_j) + (s_j - s_k) + (s_k - s_i) = 0.$$

Hence

$$\{a_{ij}\} \in Z^1(\{U_i\}, \mathcal{H}om(G, E)).$$

Next we show that the resulting cohomology class is independent of the chosen local splittings. Suppose $\{s'_i\}$ is another family of local splittings. Then

$$b_i := s'_i - s_i$$

is a section of $\mathcal{H}om(G, E)(U_i)$, and

$$a'_{ij} = (s'_i - s'_j) = (s_i - s_j) + (b_i - b_j) = a_{ij} + b_i - b_j.$$

Thus $\{a'_{ij}\}$ differs from $\{a_{ij}\}$ by a Čech coboundary, so both define the same class in

$$H^1(X, \mathcal{H}om(G, E)).$$

Therefore every extension determines a well-defined class

$$[F] \in H^1(X, \mathcal{H}om(G, E)).$$

We now construct the inverse map. Let

$$\{a_{ij}\} \in Z^1(\{U_i\}, \mathcal{H}om(G, E))$$

be a Čech 1-cocycle. On each U_i , consider the sheaf

$$E|_{U_i} \oplus G|_{U_i}.$$

On U_{ij} , define an isomorphism

$$\psi_{ij} : E|_{U_{ij}} \oplus G|_{U_{ij}} \longrightarrow E|_{U_{ij}} \oplus G|_{U_{ij}}$$

by

$$\psi_{ij}(e, g) = (e + a_{ij}(g), g).$$

The cocycle condition on $\{a_{ij}\}$ implies

$$\psi_{ij} \circ \psi_{jk} = \psi_{ik}$$

on triple overlaps, so these local sheaves glue to a sheaf F on X . Since the transition maps preserve the submodule E and induce the identity on the quotient G , the glued sheaf F fits into a short exact sequence

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0.$$

It remains to check that changing $\{a_{ij}\}$ by a coboundary yields an isomorphic extension. Suppose

$$a'_{ij} = a_{ij} + b_i - b_j$$

for some $b_i \in \mathcal{H}om(G, E)(U_i)$. Then the local automorphisms

$$\theta_i : E|_{U_i} \oplus G|_{U_i} \longrightarrow E|_{U_i} \oplus G|_{U_i}, \quad \theta_i(e, g) = (e + b_i(g), g)$$

identify the gluing defined by $\{a_{ij}\}$ with that defined by $\{a'_{ij}\}$. Hence cohomologous cocycles determine isomorphic extensions.

Finally, if the class of $\{a_{ij}\}$ vanishes, then after replacing a_{ij} by a cohomologous cocycle we may assume $a_{ij} = 0$. In that case the extension is globally split:

$$F \cong E \oplus G.$$

Conversely, if the extension is split, one may choose the local splittings s_i to be the restriction of a global splitting, and then $a_{ij} = 0$. Thus the extension is split if and only if its class vanishes.

We have therefore established a bijection between the set of isomorphism classes of extensions

$$0 \longrightarrow E \longrightarrow F \longrightarrow G \longrightarrow 0$$

and

$$H^1(X, \mathcal{H}om(G, E)).$$

Since G is locally free, one has

$$\mathcal{E}xt^q(G, E) = 0 \quad \text{for } q > 0,$$

and hence

$$\text{Ext}^1(G, E) \cong H^1(X, \mathcal{H}om(G, E)).$$

This proves the result. □

Remark 2.60. If $G = V \otimes \mathcal{O}_X$ is a trivial bundle with fibre a finite-dimensional vector space V , then

$$\text{Ext}^1(G, E) \cong V^\vee \otimes H^1(X, E) \cong \text{Hom}(V, H^1(X, E)).$$

Proposition 2.61. *The short exact sequence*

$$0 \longrightarrow T_X \longrightarrow T_{\mathcal{X}}|_X \longrightarrow \phi^*T_B|_X \longrightarrow 0$$

determines an extension class in

$$\text{Ext}^1(\phi^*T_B|_X, T_X).$$

Via the natural identification

$$\phi^*T_B|_X \cong T_{B,0} \otimes \mathcal{O}_X$$

this class corresponds to a linear map

$$\rho : T_{B,0} = H^0(X, \phi^*T_B|_X) \longrightarrow H^1(X, T_X).$$

Proof. Taking cohomology of the short exact sequence of sheaves yields a long exact sequence

$$H^0(X, \phi^*T_B|_X) \longrightarrow H^1(X, T_X).$$

The map is the connecting morphism associated with the short exact sequence. Since

$$H^0(X, \phi^*T_B|_X) = T_{B,0},$$

this connecting morphism defines a linear map

$$\rho : T_{B,0} \rightarrow H^1(X, T_X).$$

Under the identification

$$\mathrm{Ext}^1(T_{B,0} \otimes \mathcal{O}_X, T_X) \cong \mathrm{Hom}(T_{B,0}, H^1(X, T_X)),$$

this map represents precisely the extension class of the above short exact sequence. $\square$

Definition 2.62. The map

$$\rho : T_{B,0} \longrightarrow H^1(X, T_X)$$

is called the *Kodaira–Spencer map* of the family

$$\phi : \mathcal{X} \rightarrow B$$

at the point $0 \in B$.

Definition 2.63. Let $0 \in B$ and let $\mathcal{M}_0$ be the maximal ideal of $\mathcal{O}_{B,0}$. The *first order neighbourhood* of 0 in B is the subscheme

$$B_\varepsilon = (\{0\}, \mathcal{O}_{B,0}/\mathcal{M}_0^2).$$

Thus B_ε is a single point equipped with the ring

$$\mathcal{O}_{B,0}/\mathcal{M}_0^2 \cong \mathbb{C}[\varepsilon]/(\varepsilon^2)$$

when $\dim B = 1$.

Definition 2.64. Let $\phi : \mathcal{X} \rightarrow B$ be a deformation of $X = \phi^{-1}(0)$. The *first order deformation* of X induced by $\mathcal{X}$ is

$$X_\varepsilon = \phi^{-1}(B_\varepsilon).$$

Remark 2.65. As a topological space we have $X_\varepsilon = X$, but the structure sheaf contains additional functions $\phi^* z_i \bmod \phi^* \mathcal{M}_0^2$, which encode the directions in $\mathcal{X}$ normal to X .

Proposition 2.66. Assume $\dim B = 1$. There exists an open covering $\{U_i\}$ of X and isomorphisms

$$\theta_i : \mathcal{O}_{X_\varepsilon}|_{U_i} \cong \mathcal{O}_{U_i}[\varepsilon]/(\varepsilon^2).$$

Proof. By the local holomorphic inversion theorem the family $\mathcal{X} \rightarrow B$ is locally a product along X . Hence for each $U_i \subset X$ there exists $V_i \subset \mathcal{X}$ with

$$V_i \cap X = U_i$$

and a holomorphic isomorphism compatible with ϕ

$$V_i \cong U_i \times B_i.$$

Restricting the structure sheaf to X_ε amounts to quotienting by the ideal generated by ε^2 , where ε is a coordinate on B . This gives the stated isomorphism. $\square$

Proposition 2.67. *On overlaps $U_{ij} = U_i \cap U_j$ the change of trivialisation*

$$\theta_{ij} := \theta_i \circ \theta_j^{-1}$$

is an automorphism

$$\theta_{ij} : \mathcal{O}_{U_{ij}}[\varepsilon]/(\varepsilon^2) \longrightarrow \mathcal{O}_{U_{ij}}[\varepsilon]/(\varepsilon^2)$$

compatible with the exact sequence

$$0 \rightarrow \mathcal{O}_{U_{ij}}\varepsilon \rightarrow \mathcal{O}_{U_{ij}}[\varepsilon]/(\varepsilon^2) \rightarrow \mathcal{O}_{U_{ij}} \rightarrow 0.$$

Proposition 2.68. *Every such automorphism is determined by a derivation*

$$\chi_{ij} : \mathcal{O}_{U_{ij}} \rightarrow \mathcal{O}_{U_{ij}}$$

via

$$\theta_{ij}(\phi) = \phi + \varepsilon \chi_{ij}(\phi).$$

Hence χ_{ij} is a holomorphic vector field on U_{ij} , i.e. a section of T_X .

Proposition 2.69. *The derivations χ_{ij} satisfy*

$$\chi_{ij} + \chi_{jk} + \chi_{ki} = 0$$

on triple overlaps U_{ijk} .

Proposition 2.70. *The collection $\{\chi_{ij}\}$ defines a Čech cocycle*

$$\{\chi_{ij}\} \in Z^1(\{U_i\}, T_X).$$

Its cohomology class

$$[\{\chi_{ij}\}] \in H^1(X, T_X)$$

is independent of the choice of trivialisation.

Remark 2.71. Changing the trivialisation θ_i modifies χ_{ij} by

$$\chi_{ij} \mapsto \chi_{ij} + (\chi_i - \chi_j),$$

which corresponds to adding a Čech coboundary. Hence the resulting class in $H^1(X, T_X)$ is well defined. This class is the Kodaira–Spencer class of the deformation.

Proposition 2.72. *First order deformations of X parametrised by*

$$B_\varepsilon = \text{Spec } \mathbb{C}[\varepsilon]/(\varepsilon^2)$$

are classified by the cohomology group

$$H^1(X, T_X).$$

Remark 2.73. In general, not every first order deformation arises as the first order neighbourhood of X inside a deformation $\mathcal{X} \rightarrow B$. Such deformations are called *obstructed*.

If a universal family of deformations of X exists over a germ $(B, 0)$, then the first order deformations are parametrised by the tangent space

$$T_{B,0}.$$

The existence of obstructions is equivalent to the fact that B is singular at 0.

Proposition 2.74. *Let*

$$X_\varepsilon = \phi^{-1}(B_\varepsilon)$$

be the first order deformation associated to a family $\phi : \mathcal{X} \rightarrow B$.

The Kodaira–Spencer map

$$\rho : T_{B,0} \rightarrow H^1(X, T_X)$$

is determined by the scheme X_ε .

Definition 2.75. Define the sheaf

$$T_{X_\varepsilon|X}$$

to be the sheaf of derivations of $\mathcal{O}_{X_\varepsilon}$ with values in $\mathcal{O}_X$.

Remark 2.76. The tangent space of B_ε at 0 is generated by

$$\frac{\partial}{\partial \varepsilon}.$$

Moreover, locally the sheaf $T_{X_\varepsilon|X}$ is freely generated over $\mathcal{O}_X$ by

$$T_X \quad \text{and} \quad \frac{\partial}{\partial \varepsilon}.$$

Proposition 2.77. *There is an exact sequence of sheaves*

$$0 \rightarrow T_X \rightarrow T_{X_\varepsilon|X} \xrightarrow{\phi_*} T_{B_\varepsilon,0} \otimes \mathcal{O}_X \rightarrow 0.$$

Remark 2.78. The map ϕ_* is obtained by evaluating a derivation on the function ε . When X_ε is the first order neighbourhood of X in $\mathcal{X}$, this exact sequence coincides with the fundamental exact sequence of deformation theory.

Proposition 2.79. *Let θ_i be the local trivialisations*

$$\theta_i : \mathcal{O}_{X_\varepsilon}|_{U_i} \cong \mathcal{O}_{U_i}[\varepsilon]/(\varepsilon^2).$$

They induce local splittings of the above exact sequence given by

$$\chi_i = \theta_i^* \left(\frac{\partial}{\partial \varepsilon} \right) = \frac{\partial}{\partial \varepsilon} \circ \theta_i$$

which are sections of $T_{X_\varepsilon}|_{U_i}$.

Proposition 2.80. *On overlaps U_{ij} one has*

$$\chi_{ij} = \theta_j^* \left(\theta_{ij}^* \left(\frac{\partial}{\partial \varepsilon} \right) \right) - \frac{\partial}{\partial \varepsilon} = \theta_i^* \left(\frac{\partial}{\partial \varepsilon} \right) - \theta_j^* \left(\frac{\partial}{\partial \varepsilon} \right) = \chi_i - \chi_j.$$

Thus $\{\chi_{ij}\}$ defines a Čech cocycle with values in T_X .

Proposition 2.81. *The Kodaira–Spencer map applied to the tangent vector*

$$\frac{\partial}{\partial \varepsilon}$$

is represented by the Čech cocycle

$$\{\chi_{ij}\} \in Z^1(\{U_i\}, T_X).$$

2.3.5 Naïve ODE for Period Functions

Definition 2.82 (Period function). Let ω be a k -form on $\mathcal{M}$ and denote by

$$\omega_u := \omega|_{M_u}$$

its restriction to the fibre. If each ω_u is closed, we define the associated *period function*

$$\Pi(u) = \int_{C_u} \omega_u.$$

Remark 2.83. Cohomology classes vary naturally over $\mathbb{C}$. Because of the local product structure, derivatives with respect to moduli commute with exterior differentiation along the fibre. Hence if ω_u is closed, then so are all derivatives

$$\partial_{u^i}^p \omega_u, \quad p = 0, \dots, b_k(M_u).$$

Proposition 2.84 (Exact linear relation). *Since the dimension of $H^k(M_u, \mathbb{R})$ equals the Betti number $b_k(M_u)$, the closed forms $\partial_{u^i}^p \omega_u$ cannot all be independent up to an exact term. There exist functions $a_p(u)$ such that a linear combination becomes exact:*

$$\sum_{p=0}^{b_k} a_p(u) \partial_{u^i}^p \omega_u = d\eta_u.$$

Corollary 2.85. *Let $\{M_u\}_{u \in U}$ be a smooth family of compact manifolds and let ω_u be a family of closed k -forms on the fibres. Fix a locally constant cycle $C_u \subset M_u$ and define the period function*

$$\pi(u) = \int_{C_u} \omega_u.$$

Then $\pi(u)$ satisfies a linear differential equation of finite order. More precisely, there exist functions $a_i(u)$ such that

$$\sum_{i=0}^{b_k(M_u)} a_i(u) \partial_{u^\alpha}^i \pi(u) = 0,$$

where $b_k(M_u) = \dim H^k(M_u, \mathbb{R})$ is the k -th Betti number and u^α denotes a local coordinate on the moduli space.

Proof. We can just integrate over the locally constant cycle C_u and apply Stokes' theorem. $\square$

Remark 2.86. In the Calabi–Yau case the shortcomings of the naïve derivation become particularly evident. The periods of interest are those of the holomorphic volume form

$$\Omega_u \in H^{n,0}(M_u),$$

and one has

$$h^{n,0}(M_u) = 1.$$

In a local holomorphic trivialization, ordinary derivatives with respect to the moduli parameters preserve the Hodge type, hence

$$\partial_{u^\alpha} \Omega_u \in A^{n,0}(M_u).$$

Since $H^{n,0}(M_u)$ is one-dimensional, the cohomology class of $\partial_{u^\alpha}\Omega_u$ is necessarily proportional to that of Ω_u , namely

$$\partial_{u^\alpha}\Omega_u = f_\alpha(u)\Omega_u + d(\cdots).$$

Consequently, the induced differential equation satisfied by the period function is at most first order. The resulting differential equation therefore has a one-dimensional space of solutions.

However, periods are defined by integrating Ω_u over different cycles,

$$\Pi_C(u) = \int_C \Omega_u, \quad C \in H_n(M_u, \mathbb{Z}),$$

and in general these period functions are not proportional to each other by constant factors. Hence they cannot simultaneously satisfy the same one-dimensional differential equation. It follows that the naïve construction cannot produce the genuine Picard–Fuchs equations, whose solution space must accommodate all periods at once.

Remark 2.87. The shortcoming of this argument comes from its lack of geometric invariance. The derivative ∂_u depends on the chosen local trivialization of the family, and different choices lead to different differential operators and hence different equations. Therefore the resulting differential equation is not canonically associated to the family of complex manifolds.

The geometrically natural object is instead the Gauss–Manin connection on the cohomology bundle

$$\mathcal{H}^k := R^k\pi_*\mathbb{C},$$

whose flat transport identifies cohomology groups independently of any choice of trivialization. The genuine Picard–Fuchs equations arise from linear relations among iterated covariant derivatives with respect to this connection, rather than ordinary derivatives.

2.3.6 The Hodge Bundle and the Gauss–Manin Connection

Definition 2.88 (Higher direct image sheaf). Let

$$\pi : X \longrightarrow S$$

be a continuous map of topological spaces (or a morphism of complex manifolds), and let $\mathbb{Z}_X$ denote the constant sheaf on X . The n -th higher direct image of $\mathbb{Z}_X$ is defined by

$$R^n\pi_*\mathbb{Z} := \mathcal{H}^n(R\pi_*\mathbb{Z}_X),$$

where

$$R\pi_* : D^+(X) \longrightarrow D^+(S)$$

is the derived pushforward functor and $\mathcal{H}^n$ denotes the n -th cohomology sheaf.

Proposition 2.89 (Computation formula for the higher direct image). *Let $\pi : X \rightarrow S$ be a continuous map of topological spaces and let $\mathbb{Z}_X$ denote the constant sheaf on X .*

For every open subset $U \subset S$ there is a natural identification

$$(R^n\pi_*\mathbb{Z}_X)(U) \cong H^n(\pi^{-1}(U), \mathbb{Z}).$$

In particular, for every point $s \in S$ one has

$$(R^n \pi_* \mathbb{Z}_X)_s \cong H^n(X_s, \mathbb{Z}),$$

where

$$X_s = \pi^{-1}(s)$$

is the fiber over s .

Proof. By definition, the higher direct image $R^n \pi_* \mathbb{Z}_X$ is the n -th right derived functor of the direct image functor π_* . If

$$0 \rightarrow \mathbb{Z}_X \rightarrow \mathcal{I}^\bullet$$

is an injective resolution of $\mathbb{Z}_X$, then

$$R^n \pi_* \mathbb{Z}_X = \mathcal{H}^n(\pi_* \mathcal{I}^\bullet).$$

For an open subset $U \subset S$ one has

$$(\pi_* \mathcal{I}^k)(U) = \mathcal{I}^k(\pi^{-1}(U)).$$

Hence the complex computing the sections of $R\pi_* \mathbb{Z}_X$ over U is precisely

$$\mathcal{I}^\bullet(\pi^{-1}(U)).$$

Since $\mathcal{I}^\bullet$ is an injective resolution of $\mathbb{Z}_X$, its cohomology computes the sheaf cohomology of $\mathbb{Z}_X$ on $\pi^{-1}(U)$, and therefore

$$(R^n \pi_* \mathbb{Z}_X)(U) \cong H^n(\pi^{-1}(U), \mathbb{Z}).$$

Taking stalks and using the identification

$$(R^n \pi_* \mathbb{Z}_X)_s = \varinjlim_{s \in U} (R^n \pi_* \mathbb{Z}_X)(U),$$

one obtains the fiber description

$$(R^n \pi_* \mathbb{Z}_X)_s \cong H^n(X_s, \mathbb{Z}).$$

□

Remark 2.90 (Geometric meaning). The higher direct image sheaf organizes the cohomology groups of the fibers into a single sheaf over the base space. When π is a proper smooth family, the topology of the fibers is locally constant, and therefore $R^n \pi_* \mathbb{Z}$ forms a locally constant sheaf (a local system). Geometrically, this means that the cohomology groups $H^n(X_s, \mathbb{Z})$ can be canonically identified along small paths in the base.

Definition 2.91 (Complexified local system and Hodge bundle). Tensoring the local system with $\mathbb{C}$ gives

$$R^n \pi_* \mathbb{C} = R^n \pi_* \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{C}.$$

The associated holomorphic vector bundle (the *Hodge bundle*) is defined by

$$\mathcal{H} := R^n \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_S.$$

Its fiber over $s \in S$ is naturally identified with

$$\mathcal{H}_s = H^n(X_s, \mathbb{C}).$$

Remark 2.92 (Geometric interpretation). The local system $R^n\pi_*\mathbb{Z}$ records the topological cohomology of the fibers, which varies locally constantly. The Hodge bundle $\mathcal{H}$ is obtained by extending scalars to holomorphic functions on the base, allowing holomorphic variation with respect to the moduli parameters. Thus the Hodge bundle encodes how cohomology classes vary with the complex structure.

Definition 2.93 (Gauss–Manin connection). Let

$$\pi : X \longrightarrow S$$

be a smooth proper family of complex manifolds, and let

$$\mathbb{H}_{\mathbb{C}} := R^n\pi_*\mathbb{C}$$

denote the associated local system. The *Hodge bundle* is defined by

$$\mathcal{H} := \mathbb{H}_{\mathbb{C}} \otimes_{\mathbb{C}} \mathcal{O}_S.$$

The *Gauss–Manin connection* is the unique $\mathbb{C}$ -linear map

$$\nabla^{GM} : \mathcal{H} \longrightarrow \mathcal{H} \otimes \Omega_S^1$$

satisfying the Leibniz rule

$$\nabla^{GM}(f\sigma) = df \otimes \sigma + f \nabla^{GM}\sigma,$$

and such that every locally constant section $s \in \mathbb{H}_{\mathbb{C}}$ is flat (here the $s \otimes 1$ is only well-defined locally because of the necessary procedure of sheafification when we tensor two sheaves together):

$$\nabla^{GM}(1 \otimes s) = 0.$$

Remark 2.94 (Local description). On a contractible open set $U \subset S$, the local system $\mathbb{H}_{\mathbb{C}}$ admits a flat trivialization

$$\mathcal{H}|_U \cong H^n(X_{s_0}, \mathbb{C}) \otimes_{\mathbb{C}} \mathcal{O}_U.$$

With respect to this identification, a section can be written as

$$\sigma = \sum_i f_i \otimes \alpha_i, \quad \alpha_i \in H^n(X_{s_0}, \mathbb{C}),$$

and the Gauss–Manin connection is given by

$$\nabla^{GM}\sigma = \sum_i df_i \otimes \alpha_i.$$

Thus the connection differentiates only the holomorphic coefficients while keeping the cohomology classes fixed under flat transport.

Remark 2.95 (Geometric meaning). The Gauss–Manin connection measures how cohomology classes vary with respect to parameters while the underlying topology of the fibers remains unchanged. Although flat sections are constant in the topological sense, the Hodge decomposition

$$H^n(X_s, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X_s)$$

depends on the complex structure. Consequently a flat section need not preserve its (p, q) -type as s varies, which leads to the Griffiths transversality property

$$\nabla^{GM}\mathcal{F}^p \subset \mathcal{F}^{p-1} \otimes \Omega_S^1.$$

We now describe the Gauss–Manin connection in a form convenient for later use. Our notation here differs from that used previously.

Recall that the Gauss–Manin connection acts on the cohomology bundle

$$\mathcal{H}^n = R^n \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{M}_{cs}}.$$

Definition 2.96. The sheaf $R^n \pi_* \mathbb{C}$ is a local system. Hence locally on the moduli space one can choose a basis of flat sections

$$\{e_I\},$$

on which we define

$$\nabla^{GM} e_I = 0.$$

Proposition 2.97. *In such a flat frame every section s of $\mathcal{H}^n$ can be written as*

$$s = s^I(z) e_I.$$

The Gauss–Manin connection acts by

$$\nabla_i^{GM} s = (\partial_i s^I) e_I.$$

Equivalently, in this flat trivialization the Gauss–Manin connection coincides with the ordinary differential acting on the coefficient functions.

Proof. Since the e_I are flat sections, one has

$$\nabla_i^{GM} e_I = 0.$$

Using the Leibniz rule for the connection,

$$\nabla_i^{GM} (s^I e_I) = (\partial_i s^I) e_I + s^I \nabla_i^{GM} e_I,$$

and the second term vanishes. Hence

$$\nabla_i^{GM} s = (\partial_i s^I) e_I.$$

□

Proposition 2.98. *Let $\{\tilde{e}_A(z)\}$ be another frame of $\mathcal{H}^n$ related to the flat frame by*

$$\tilde{e}_A = M_A^I(z) e_I.$$

Then the Gauss–Manin connection takes the form

$$\nabla_i^{GM} = \partial_i + A_i(z)$$

with

$$A_i = (\partial_i M) M^{-1}.$$

Proof. Using the relation $\tilde{e} = Me$, one obtains

$$\nabla^{GM}\tilde{e} = \nabla^{GM}(Me) = (dM)e.$$

Rewriting this in the $\tilde{e}$ frame gives

$$\nabla^{GM}\tilde{e} = (dM)M^{-1}\tilde{e}.$$

Hence the connection matrix in this frame is

$$A = dM M^{-1}.$$

□

Remark 2.99. Thus the Gauss–Manin connection is locally the ordinary differential when expressed in a flat trivialization of the local system $R^n\pi_*\mathbb{C}$. In any other frame the connection acquires a matrix term A_i , which arises from the variation of the frame with respect to the moduli.

Corollary 2.100. *Let $\{e^I\}$ be a flat frame of the local system $R^n\pi_*\mathbb{C}$, so that*

$$\nabla^{GM}e^I = 0.$$

In the corresponding trivialization one has

$$\partial_i e^I = 0.$$

For any section

$$s = s_I(z) e^I = \tilde{s}_I(z) \tilde{e}^I(z)$$

of the cohomology bundle, the Gauss–Manin connection satisfies

$$\nabla_i^{GM}s = \nabla_i^{GM}(s_I e^I) = \partial_i(s_I e^I).$$

In particular, writing the same section in another frame $\{\tilde{e}^I\}$ (which may not be flat),

$$s = \tilde{s}_I \tilde{e}^I,$$

one still has

$$\nabla_i^{GM}s = \partial_i(\tilde{s}_I \tilde{e}^I),$$

where the derivative now acts both on the coefficients $\tilde{s}_I$ and on the frame $\tilde{e}^I$.

2.3.7 Grassmannians and Flag Varieties

In this section, we briefly review some basic facts about Grassmannians and flag varieties.

Proposition 2.101. *The Grassmannian $\text{Grass}(k, W)$ naturally carries the structure of a compact complex (and in fact projective) manifold of dimension $k(w - k)$, where $w = \dim W$.*

Proof. Let $w = \dim W$ and set $G := GL(W) \simeq GL(w, \mathbb{C})$. Fix a reference k -plane $E_0 \subset W$ (for instance, the span of the first k standard basis vectors after choosing a basis of W). The natural action of G on W induces a transitive action on k -planes, hence on $\text{Grass}(k, W)$. Let

$$P := \text{Stab}_G(E_0) = \{g \in G \mid g(E_0) = E_0\}.$$

Then P is a closed complex Lie subgroup of G , and we have a G -equivariant identification

$$\text{Grass}(k, W) \cong G/P.$$

Since G is a complex Lie group and P is a closed complex Lie subgroup, the quotient G/P carries a unique structure of a complex manifold such that the projection $G \rightarrow G/P$ is a holomorphic submersion. In particular, $\text{Grass}(k, W)$ is a complex manifold.

To compute its complex dimension, note that the Lie algebra of P consists of block upper triangular matrices relative to the decomposition $W = E_0 \oplus (W/E_0)$:

$$\mathfrak{p} = \left\{ \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \right\}, \quad A \in M_{k \times k}(\mathbb{C}), \quad B \in M_{k \times (w-k)}(\mathbb{C}), \quad D \in M_{(w-k) \times (w-k)}(\mathbb{C}).$$

Hence

$$\dim_{\mathbb{C}} \mathfrak{p} = k^2 + k(w-k) + (w-k)^2, \quad \dim_{\mathbb{C}} \mathfrak{g} = w^2,$$

and therefore

$$\dim_{\mathbb{C}} \text{Grass}(k, W) = \dim_{\mathbb{C}}(G/P) = \dim_{\mathbb{C}} \mathfrak{g} - \dim_{\mathbb{C}} \mathfrak{p} = k(w-k).$$

Finally, by the Plücker embedding

$$\text{Grass}(k, W) \hookrightarrow \mathbb{CP}(\wedge^k W),$$

the Grassmannian is a projective algebraic variety; in particular it is compact. $\square$

Remark 2.102. Not every homogeneous space of a complex Lie group is projective. If G is a complex semisimple algebraic group, then G/P is projective precisely when P is a parabolic subgroup. Such homogeneous spaces are called flag varieties. The Grassmannian is a basic example of this phenomenon.

Definition 2.103. Let V be a complex vector space of dimension n , and let

$$0 < f_r < \cdots < f_1 < n$$

be a sequence of integers. The *flag variety*

$$\text{Flag}(f_1, \dots, f_r; V)$$

is the set of all filtrations (flags) of subspaces

$$V = F^0 \supset F^1 \supset \cdots \supset F^r \supset 0$$

such that

$$\dim F^i = f_i \quad \text{for all } i.$$

Proposition 2.104. *The flag variety $\text{Flag}(f_1, \dots, f_r; V)$ carries a natural structure of a compact complex manifold. Moreover, it is a projective algebraic variety.*

Proof. Each subspace $F^i \subset V$ defines a point in the Grassmannian $\text{Grass}(f_i, V)$. This gives an embedding

$$\text{Flag}(f_1, \dots, f_r; V) \hookrightarrow \prod_{i=1}^r \text{Grass}(f_i, V),$$

whose image consists of tuples satisfying the incidence relations $F^{i+1} \subset F^i$. Since Grassmannians are complex projective manifolds and the incidence conditions are algebraic, the flag variety inherits the structure of a complex projective manifold. $\square$

Remark 2.105. Equivalently, the flag variety can be realized as a homogeneous space

$$\text{Flag}(f_1, \dots, f_r; V) \simeq GL(V)/P,$$

where $P \subset GL(V)$ is a parabolic subgroup stabilizing a reference flag. In particular, the complex structure arises naturally from the structure of a complex Lie group quotient.

The following lemma will be useful later. Here G denotes the Grassmannian, rather than an algebraic group.

Lemma 2.106. *The identification*

$$T_{G,K} \cong \text{Hom}(K, W/K)$$

obtained as the composition

$$T_{G,K} \cong T_{\text{Hom}(K,V),0} \cong \text{Hom}_{\mathbb{C}}(K, W/K)$$

is canonical, i.e. independent of the choice of V .

2.3.8 The Period Map and Griffiths Transversality

In this section, we introduce the period map which provide a local geometric framework for describing the variation of the Hodge filtration on the Hodge bundle. It will serve as fundamental local tools for the later global study of the Hodge filtration sheaf.

Let $\pi : \mathcal{X} \rightarrow B$ be a proper holomorphic submersion of complex manifolds, here B is contractible.

Definition 2.107. *The period map*

$$\mathcal{P}^{p,k} : B \longrightarrow \text{Grass}(b^{p,k}, H^k(X, \mathbb{C}))$$

is the map which to $b \in B$ associates the subspace

$$F^p H^k(X_b, \mathbb{C}) \subset H^k(X_b, \mathbb{C}) \simeq H^k(X, \mathbb{C}).$$

To prove results about period map, we need more information about the relative geometry on the family of complex manifolds.

Let $\pi : \mathcal{X} \rightarrow B$ be a proper submersion, and let $\mathcal{H}^k$ be the (C^∞ or holomorphic according to the context) bundle associated to the local system $H_{\mathbb{Z}}^k$. Let ∇ be the Gauss–Manin connection. Let Ω be a complex differential form of degree k on $\mathcal{X}$ such that for all $b \in B$, the restriction

$$\Omega_b = \Omega|_{X_b}$$

is closed. Then we have a section

$$\omega : b \longmapsto [\Omega_b] \in H^k(X_b, \mathbb{C}) \simeq H^k(\mathcal{X}_U, \mathbb{C}),$$

where the last equality holds in a contractible neighbourhood U of b .

Proposition 2.108. *If $u \in T_{B,0}$ and $v \in \Gamma(T_{\mathcal{X}}|_{X_0})$ is such that $\phi_*(v) = u$, we have*

$$\nabla(\omega)|_0(u) = \text{class}(\text{int}(v)(d\Omega)|_{X_0}).$$

Remark 2.109. Let $\pi : \mathcal{X} \rightarrow B$ be a proper submersion and let $u \in T_{B,0}$. The vector field $v \in \Gamma(T_{\mathcal{X}}|_{X_0})$ satisfying $\pi_*(v) = u$ should be understood as a *lift* of the base tangent vector u to the total space.

More precisely, since $\pi|_{X_0}$ is constant, the differential

$$d\pi : T_{\mathcal{X}}|_{X_0} \longrightarrow \pi^*T_B|_{X_0} \simeq X_0 \times T_{B,0}$$

takes a vector field v along X_0 to a section of the trivial bundle $X_0 \times T_{B,0}$. The condition $\pi_*(v) = u$ means that

$$d\pi_x(v(x)) = u \quad \text{for all } x \in X_0,$$

i.e. the projection of v to the base direction is the constant section determined by u .

There is in general no canonical way to identify $T_{B,0}$ with a subspace of $T_{\mathcal{X},x}$; hence one cannot simply take “ $v(x) = u$ ”. Choosing v amounts to choosing a C^∞ splitting of the exact sequence

$$0 \longrightarrow T_{X_0} \longrightarrow T_{\mathcal{X}}|_{X_0} \longrightarrow \pi^*T_{B,0} \longrightarrow 0,$$

and different choices of lifts differ by vertical vector fields. The resulting cohomology class is nevertheless independent of this choice.

Remark 2.110. The expression

$$\text{class}(\iota_v(d\Omega)|_{X_0})$$

should *not* be interpreted as first restricting the form $d\Omega$ to X_0 and then contracting with v . Indeed, the vector field $v \in \Gamma(T_{\mathcal{X}}|_{X_0})$ is in general not tangent to X_0 , so the contraction $\iota_v((d\Omega)|_{X_0})$ would not even be well defined.

The correct meaning is the following. Let $i_0 : X_0 \hookrightarrow \mathcal{X}$ denote the inclusion. One first extends v locally to a vector field $\tilde{v}$ on $\mathcal{X}$, forms the contraction $\iota_{\tilde{v}}(d\Omega)$ on $\mathcal{X}$, and then pulls back the resulting form to the fibre:

$$\iota_v(d\Omega)|_{X_0} := i_0^*(\iota_{\tilde{v}}(d\Omega)) \in \Omega^k(X_0).$$

This definition is independent of the chosen extension of v .

Note that the symbol $|_{X_0}$ therefore denotes the pullback of differential forms to the submanifold X_0 , not merely the restriction of coefficients. The pullback automatically removes all components involving directions transverse to X_0 , so the result is a genuine differential form on the fibre and defines a class in $H^k(X_0, \mathbb{C})$.

Remark 2.111. The formula

$$\nabla(\omega)_0(u) = \text{class}(i_0^*(\iota_v(d\Omega)))$$

may be viewed as a fibrewise version of the Cartan–Lie formula

$$L_{\tilde{v}}\Omega = d(\iota_{\tilde{v}}\Omega) + \iota_{\tilde{v}}(d\Omega),$$

where $i_0 : X_0 \hookrightarrow \mathcal{X}$ denotes the inclusion and $\tilde{v}$ is any local extension of $v \in \Gamma(T_{\mathcal{X}}|_{X_0})$ to a vector field on $\mathcal{X}$.

The Gauss–Manin derivative is obtained by differentiating the cohomology classes $[\Omega_b]$ after identifying nearby fibres via the flow generated by $\tilde{v}$, which yields

$$\nabla(\omega)_0(u) = \text{class}(i_0^*(L_{\tilde{v}}\Omega)).$$

Applying the Cartan–Lie formula and using the naturality of the exterior derivative,

$$i_0^*(d(\iota_{\tilde{v}}\Omega)) = d(i_0^*(\iota_{\tilde{v}}\Omega)),$$

we see that the first term is exact on X_0 and therefore vanishes in de Rham cohomology. Consequently,

$$\nabla(\omega)_0(u) = \text{class}(i_0^*(\iota_v(d\Omega))).$$

Since contraction depends pointwise only on the value of the vector field along X_0 , the resulting form is independent of the chosen extension $\tilde{v}$. Thus the variation of the cohomology class is governed precisely by the term $\iota_v(d\Omega)$ restricted to the fibre.

Remark 2.112. We need to convince ourselves that $\text{class}(i_0^*(\iota_v(d\Omega)))$ is well defined.

Since d commutes with pullback, it suffices to show $d\iota_{\tilde{v}}(d\Omega)$ restricts to zero on X_0 . By Cartan’s formula,

$$d\iota_{\tilde{v}}(d\Omega) = L_{\tilde{v}}(d\Omega),$$

and since $\Omega|_{X_b}$ is closed we have $i_0^*(d\Omega) = 0$. Hence

$$i_0^*(d\iota_{\tilde{v}}(d\Omega)) = 0,$$

so $i_0^*(\iota_{\tilde{v}}(d\Omega))$ is closed.

Proposition 2.113 (Kodaira 1986). *Let $\Delta = (\Delta_b)_{b \in B}$ be a relative differential operator acting on a vector bundle $F \rightarrow \mathcal{X}$, such that each induced operator Δ_b on F_b is elliptic of fixed order. Then if $\dim \text{Ker} \Delta_b$ is independent of b , the subspace*

$$\text{Ker} \Delta_b \subset C^\infty(F_b)$$

varies in a C^∞ way with b .

This means that, up to shrinking B near b , there exist C^∞ sections $(\eta_b^i)_{b \in B}$ of F over $\mathcal{X}$ whose restrictions to X_b for fixed b form a basis of $\text{Ker} \Delta_b$.

Theorem 2.114. *The period map $\mathcal{P}^{p,k}$ is holomorphic for all p, k with $p \leq k$.*

Proof. The map $\mathcal{P}^{p,k}$ is at least C^∞ , by an application of Proposition 2.113. To see that it is holomorphic, it now suffices to show that its differential is $\mathbb{C}$ -linear, which is equivalent to saying that the $\mathbb{C}$ -linear extension of its differential to $T_{B,b} \otimes \mathbb{C}$ vanishes on the vectors of type $(0, 1)$.

Now, by the results of the preceding section, the differential

$$d\mathcal{P}^{p,k} : T_{B,b} \longrightarrow \text{Hom}(F^p H^k(X_b), H^k(X, \mathbb{C})/F^p H^k(X_b, \mathbb{C}))$$

is obtained by choosing for $\sigma \in F^p H^k(X_b, \mathbb{C})$ a differentiable function $\tilde{\sigma}$ on B with values in $H^k(X_b, \mathbb{C})$, satisfying $\tilde{\sigma}(b) = \sigma$ and $\tilde{\sigma}(b') \in F^p H^k(X_{b'}, \mathbb{C})$ for all $b' \in B$. We then have

$$d\mathcal{P}^{p,k}(u)(\sigma) = u(\tilde{\sigma}) \mod F^p H^k(X_b, \mathbb{C}).$$

(More intrinsically, $\tilde{\sigma}$ must be viewed as a section of the bundle $\mathcal{H}^k$, and $u(\tilde{\sigma})$ as $\nabla_u \tilde{\sigma}$.)

Now, applying Proposition 2.113 and using the fact that the cohomology of the complex $F^p A^k(X_b)$ is of constant rank, we see that there exists a differential form Ω on $\mathcal{X}$ in a neighbourhood of X_b satisfying the conditions

1. $\Omega \in F^p A^k(\mathcal{X})$,
2. $\Omega|_{X_{b'}}$ is closed and its class in $F^p H^k(X_{b'}, \mathbb{C})$ is equal to $\tilde{\sigma}(b')$ for b' near b .

More precisely, Proposition 2.113 gives the existence of such a relative form (i.e. a form in $F^p A^k_{\mathcal{X}/B}$), and it suffices to take a lifting in $F^p A^k(\mathcal{X})$.

Let us take a C^∞ decomposition

$$T_{\mathcal{X}}|_{X_b} \simeq T_{X_b} \oplus M$$

along X_b , where M is a complex subbundle isomorphic to $\phi^* T_{B,b}$. We can also see this decomposition as a decomposition of the real tangent bundle, compatible with the complex structure. Let $u \in T_{B,b}$, and let v be the section of M such that $\phi_*(v) = u$. Proposition 2.108 then gives

$$d\mathcal{P}^{p,k}(u)(\sigma) = \nabla_u(\tilde{\sigma}) \mod F^p H^k(X_b) = [(\text{int}(v)(d\Omega)|_{X_b})] \mod F^p H^k(X_b).$$

This formula obviously remains valid when u is a complexified tangent vector, since the subspace $M \subset T_{\mathcal{X}}$ is a complex subspace at every point.

Now, if u is a vector of type $(0, 1)$, then the vector field v is of type $(0, 1)$ along X_b . As $d\Omega \in F^p A^{k+1}(\mathcal{X})$, we have

$$\text{int}(v)(d\Omega) \in F^p A^k(\mathcal{X}).$$

Thus (restriction don't change (p, q) -type),

$$\text{int}(v)(d\Omega)|_{X_b} \in F^p A^k(X_b),$$

and as it is a closed form, the class of $\text{int}(v)(d\Omega)|_{X_b}$ lies in $F^p H^k(X_b, \mathbb{C})$. So for such a u , we have

$$d\mathcal{P}^{p,k}(u)(\sigma) = \nabla_u(\tilde{\sigma}) \mod F^p H^k(X_b, \mathbb{C}) = 0.$$

□

Theorem 2.115 (Griffiths transversality). *The map*

$$d\mathcal{P}^{p,k} : T_{B,b} \longrightarrow \text{Hom}(F^p H^k(X_b), H^k(X_b, \mathbb{C})/F^p H^k(X_b))$$

has values in

$$\text{Hom}(F^p H^k(X_b), F^{p-1} H^k(X_b)/F^p H^k(X_b)).$$

Proof. With the notation of the preceding proof, we have

$$d\mathcal{P}^{p,k}(u)(\sigma) = \nabla_u(\tilde{\sigma}) \mod F^p H^k(X_b),$$

and,

$$\nabla_u(\tilde{\sigma}) = \text{class}(\text{int}(v)(d\Omega)|_{X_b}).$$

Now, $\Omega \in F^p A^k(\mathcal{X})$, hence

$$d\Omega \in F^p A^{k+1}(\mathcal{X}), \quad \text{int}(v)(d\Omega) \in F^{p-1} A^k(\mathcal{X}).$$

Thus $\nabla_u(\tilde{\sigma})$ is representable by a closed form in $F^{p-1} A^k(X_b)$, and therefore

$$\nabla_u(\tilde{\sigma}) \in F^{p-1} H^k(X_b, \mathbb{C}).$$

Consequently,

$$d\mathcal{P}^{p,k}(u)(\sigma) \in F^{p-1} H^k(X_b, \mathbb{C})/F^p H^k(X_b, \mathbb{C}).$$

□

2.3.9 The Hodge Filtration Sheaf

We begin by reviewing some notions of holomorphic subbundles.

Definition 2.116 (Holomorphic subbundle: local frame characterization). Let $E \rightarrow X$ be a holomorphic vector bundle over a complex manifold X . A subset $F \subset E$ is called a *holomorphic vector subbundle* of rank k if the following conditions hold:

1. For every $x \in X$, the fiber

$$F_x := F \cap E_x$$

is a k -dimensional complex linear subspace of E_x ;

2. For every point $x \in X$, there exists an open neighbourhood $U \subset X$ and holomorphic sections

$$s_1, \dots, s_k \in \mathcal{O}(E|_U)$$

such that for all $y \in U$,

$$F_y = \text{Span}_{\mathbb{C}}\{s_1(y), \dots, s_k(y)\}.$$

In other words, a holomorphic subbundle is locally generated by holomorphic sections of E .

Definition 2.117 (Holomorphic subbundle: Grassmannian characterization). Let $E \rightarrow X$ be a holomorphic vector bundle of rank r , and suppose $F_x \subset E_x$ is a k -dimensional subspace for each $x \in X$. This assignment defines a map

$$\Phi : X \longrightarrow \mathrm{Gr}(k, r), \quad x \longmapsto F_x,$$

to the Grassmannian of k -planes in $\mathbb{C}^r$ (after choosing a local trivialization of E).

The family $\{F_x\}$ defines a holomorphic vector subbundle $F \subset E$ if and only if the map Φ is holomorphic.

Remark 2.118. We observe that the holomorphic variation of the Hodge structure over the complex structure moduli space is precisely reflected in the fact that the Hodge filtration forms holomorphic subbundles of the Hodge bundle.

Definition 2.119 (Holomorphic subbundle: sheaf-theoretic characterization). Let $E \rightarrow X$ be a holomorphic vector bundle over a complex manifold X , and let $\mathcal{O}(E)$ denote the sheaf of holomorphic sections of E .

A holomorphic vector subbundle $F \subset E$ of rank k is equivalently described by a locally free $\mathcal{O}_X$ -module of rank k subsheaf

$$\mathcal{F} \subset \mathcal{O}(E)$$

In this case, the fibers are recovered as

$$F_x = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathbb{C},$$

and F defines a holomorphic vector subbundle of E .

Remark 2.120. The sheaf-theoretic formulation is particularly convenient when studying families, since holomorphicity is encoded in the $\mathcal{O}_X$ -module structure rather than in pointwise linear algebra.

Definition 2.121 (Relative holomorphic de Rham complex). Let $\pi : \mathcal{X} \rightarrow S$ be a proper holomorphic submersion. The relative holomorphic de Rham complex is

$$\Omega_{\mathcal{X}/S}^\bullet = [\mathcal{O}_{\mathcal{X}} \rightarrow \Omega_{\mathcal{X}/S}^1 \rightarrow \cdots \rightarrow \Omega_{\mathcal{X}/S}^n].$$

For each p , its stupid truncation is the subcomplex

$$\Omega_{\mathcal{X}/S}^{\geq p} := [0 \rightarrow \cdots \rightarrow 0 \rightarrow \Omega_{\mathcal{X}/S}^p \rightarrow \Omega_{\mathcal{X}/S}^{p+1} \rightarrow \cdots].$$

Definition 2.122 (Hodge bundle and Hodge filtration). Let

$$\mathcal{H}^m := R^m \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_S.$$

The Hodge filtration on $\mathcal{H}^m$ is defined by

$$F^p \mathcal{H}^m := \mathrm{Im} \left(\mathbb{R}^m \pi_* \Omega_{\mathcal{X}/S}^{\geq p} \longrightarrow \mathbb{R}^m \pi_* \Omega_{\mathcal{X}/S}^\bullet \right) \subset \mathcal{H}^m,$$

where $\mathbb{R}^m \pi_*$ denotes the m -th hyperdirect image.

Equivalently, $F^p \mathcal{H}^m$ is the filtration on

$$\mathbb{R}^m \pi_* \Omega_{\mathcal{X}/S}^\bullet \cong \mathcal{H}^m$$

induced by the stupid filtration of the complex $\Omega_{\mathcal{X}/S}^\bullet$.

Remark 2.123. The map

$$\mathbb{R}^m \pi_* \Omega_{X/S}^{\geq p} \longrightarrow \mathbb{R}^m \pi_* \Omega_{X/S}^\bullet$$

is induced by the inclusion of complexes

$$\Omega_{X/S}^{\geq p} \hookrightarrow \Omega_{X/S}^\bullet.$$

Thus $F^p \mathcal{H}^m$ is the image of the hypercohomology of the truncated complex inside the hypercohomology of the full relative de Rham complex.

Lemma 2.124 (Derived projection formula). *Let $\pi : X \rightarrow S$ be a morphism of ringed spaces. For any sheaf $\mathcal{F}$ of $\mathbb{C}$ -vector spaces on X and any $\mathcal{O}_S$ -module $\mathcal{G}$ on S , there is a natural isomorphism in the derived category*

$$R\pi_*(\mathcal{F} \otimes_{\mathbb{C}} \pi^{-1} \mathcal{G}) \cong (R\pi_* \mathcal{F}) \otimes_{\mathbb{C}} \mathcal{G}.$$

Proposition 2.125. *Let $\pi : X \rightarrow S$ be a smooth proper holomorphic map. Then there is a natural isomorphism*

$$R^m \pi_*(\pi^{-1} \mathcal{O}_S) \cong (R^m \pi_* \mathbb{C}) \otimes_{\mathbb{C}} \mathcal{O}_S.$$

Proof. Observe that

$$\pi^{-1} \mathcal{O}_S \cong \mathbb{C}_X \otimes_{\mathbb{C}} \pi^{-1} \mathcal{O}_S.$$

Applying the derived projection formula with $\mathcal{F} = \mathbb{C}_X$ and $\mathcal{G} = \mathcal{O}_S$ gives

$$R\pi_*(\pi^{-1} \mathcal{O}_S) = R\pi_*(\mathbb{C}_X \otimes_{\mathbb{C}} \pi^{-1} \mathcal{O}_S) \cong (R\pi_* \mathbb{C}_X) \otimes_{\mathbb{C}} \mathcal{O}_S.$$

Taking the m -th cohomology sheaf yields

$$R^m \pi_*(\pi^{-1} \mathcal{O}_S) \cong (R^m \pi_* \mathbb{C}) \otimes_{\mathbb{C}} \mathcal{O}_S.$$

□

Corollary 2.126. *There is a natural identification*

$$R^m \pi_* \Omega_{X/S}^\bullet \cong R^m \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_S = \mathcal{H}^m.$$

Under this identification one has

$$F^0 \mathcal{H}^m = \mathcal{H}^m.$$

Proof. By the relative Poincaré lemma, the relative de Rham complex is quasi-isomorphic to $\pi^{-1} \mathcal{O}_S$:

$$\Omega_{X/S}^\bullet \simeq \pi^{-1} \mathcal{O}_S.$$

Hence applying the derived pushforward gives

$$R^m \pi_* \Omega_{X/S}^\bullet \cong R^m \pi_* \pi^{-1} \mathcal{O}_S.$$

By Proposition 2.125 we have

$$R^m \pi_* \pi^{-1} \mathcal{O}_S \cong (R^m \pi_* \mathbb{C}) \otimes_{\mathbb{C}} \mathcal{O}_S.$$

Thus

$$R^m \pi_* \Omega_{X/S}^\bullet \cong R^m \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_S = \mathcal{H}^m.$$

Finally, since

$$\Omega_{X/S}^{\geq 0} = \Omega_{X/S}^{\bullet},$$

the definition of the filtration immediately gives

$$F^0 \mathcal{H}^m = \mathcal{H}^m.$$

□

Proposition 2.127 (Fiberwise description of the Hodge filtration). *For each $s \in S$, the induced filtration on the fiber*

$$(\mathcal{H}^m)_s \cong H^m(X_s, \mathbb{C})$$

coincides with the classical Hodge filtration:

$$(F^p \mathcal{H}^m)_s \cong F^p H^m(X_s, \mathbb{C}).$$

If moreover X_s is compact Kähler, then

$$F^p H^m(X_s, \mathbb{C}) = \bigoplus_{r \geq p} H^{r, m-r}(X_s).$$

Proof. By definition, $F^p \mathcal{H}^m$ is induced from the stupid filtration on the relative holomorphic de Rham complex. Restricting to the fiber X_s gives the corresponding stupid filtration on

$$\Omega_{X_s}^{\bullet}.$$

Hence the induced filtration on

$$(\mathcal{H}^m)_s \cong \mathbb{H}^m(X_s, \Omega_{X_s}^{\bullet}) \cong H^m(X_s, \mathbb{C})$$

is exactly

$$\text{Im} \left(\mathbb{H}^m(X_s, \Omega_{X_s}^{\geq p}) \rightarrow \mathbb{H}^m(X_s, \Omega_{X_s}^{\bullet}) \right),$$

which is the usual Hodge filtration.

If X_s is compact Kähler, the Hodge-to-de Rham spectral sequence degenerates at E_1 , and therefore the filtration is identified with

$$F^p H^m(X_s, \mathbb{C}) = \bigoplus_{r \geq p} H^{r, m-r}(X_s).$$

□

Definition 2.128 (Relative Hodge-to-de Rham spectral sequence). The stupid filtration on $\Omega_{\mathcal{X}/S}^{\bullet}$ gives rise to a spectral sequence

$$E_1^{p,q} = R^q \pi_* \Omega_{\mathcal{X}/S}^p \Rightarrow \mathbb{R}^{p+q} \pi_* \Omega_{\mathcal{X}/S}^{\bullet}.$$

This is called the relative Hodge-to-de Rham spectral sequence.

Proposition 2.129 (Associated graded pieces). *Assume that the relative Hodge-to-de Rham spectral sequence degenerates at E_1 . Then for every p, m there is a canonical isomorphism*

$$\text{Gr}_F^p \mathcal{H}^m := F^p \mathcal{H}^m / F^{p+1} \mathcal{H}^m \cong R^{m-p} \pi_* \Omega_{\mathcal{X}/S}^p.$$

Proof. The filtration $F^\bullet \mathcal{H}^m$ is the filtration on the abutment induced by the stupid filtration on $\Omega_{\mathcal{X}/S}^\bullet$. For any spectral sequence associated to a filtered complex, one has

$$E_\infty^{p,m-p} \cong \mathrm{Gr}_F^p \mathcal{H}^m.$$

If the spectral sequence degenerates at E_1 , then

$$E_\infty^{p,m-p} = E_1^{p,m-p} = R^{m-p} \pi_* \Omega_{\mathcal{X}/S}^p.$$

Therefore

$$\mathrm{Gr}_F^p \mathcal{H}^m \cong R^{m-p} \pi_* \Omega_{\mathcal{X}/S}^p.$$

□

Corollary 2.130 (Fiberwise graded pieces). *Assume that the relative Hodge-to-de Rham spectral sequence degenerates at E_1 . Then for each $s \in S$ one has*

$$(\mathrm{Gr}_F^p \mathcal{H}^m)_s \cong H^{m-p}(X_s, \Omega_{X_s}^p).$$

If X_s is compact Kähler, then this is naturally identified with

$$H^{p,m-p}(X_s).$$

Remark 2.131. The role of the hyperdirect image $\mathbb{R}^m \pi_*$ is precisely the same as in ordinary hypercohomology: the relative de Rham complex is a complex of sheaves, and one must first take its derived direct image before passing to the m -th cohomology sheaf. The Hodge filtration is therefore a filtration on the derived direct image of the relative de Rham complex, induced by its stupid filtration.

Proposition 2.132. *The subbundles $F^p \mathcal{H}^k$ satisfy the property*

$$\nabla F^p \mathcal{H}^k \subset F^{p-1} \mathcal{H}^k \otimes \Omega_B.$$

Proof. A holomorphic section $\sigma \in F^p \mathcal{H}^k$ is, in particular, a C^∞ section of $\mathcal{H}^k$ satisfying

$$\sigma(b) \in F^p H^k(X_b, \mathbb{C}), \quad \forall b \in B.$$

Thus this proposition is an immediate consequence of the Griffiths Transversality we have proved before, since

$$dP^{p,k}(u)(\sigma(b)) = \nabla_u(\sigma) \mod F^p H^k(X_b), \quad \forall b \in B, u \in T_{B,b}.$$

□

We know a natural identification

$$T_{G,K} \cong \mathrm{Hom}(K, W/K),$$

where $G = \mathrm{Grass}(k, W)$ and $K \in G$. This identification is obtained by differentiating the holomorphic sections of the tautological subbundle in a neighbourhood of K .

Applying this construction to the period map, and noting that differentiating the sections of the trivial bundle

$$H^k(X_0, \mathbb{C}) \otimes \mathcal{O}_B$$

is equivalent to applying the Gauss–Manin connection to sections of $\mathcal{H}^k$, we obtain the following result.

Lemma 2.133. *The differential*

$$dP_b^{p,k} : T_{B,b} \longrightarrow T_{G, F^p H^k(X_b)} = \text{Hom} \left(F^p H^k(X_b), H^k(X_b, \mathbb{C}) / F^p H^k(X_b) \right)$$

is the map constructed by adjunction starting from

$$\bar{\nabla}_b^p : F^p H^k(X_b) \longrightarrow H^k(X_b, \mathbb{C}) / F^p H^k(X_b) \otimes \Omega_{B,b},$$

which is the value at b of the morphism $\bar{\nabla}^p$ defined as the composition

$$F^p \mathcal{H}^k \xrightarrow{\nabla} \mathcal{H}^k \otimes \Omega_B \longrightarrow (\mathcal{H}^k / F^p \mathcal{H}^k) \otimes \Omega_B.$$

Remark 2.134. Applying Griffith Transversality, we obtain

$$\text{Im } dP^{p,k} \subset \text{Hom} \left(F^p H^k(X_b) / F^{p+1} H^k(X_b), F^{p-1} H^k(X_b, \mathbb{C}) / F^p H^k(X_b) \right).$$

Definition 2.135. Consider the commutative diagram defining the morphism $\bar{\nabla}$:

$$\begin{array}{ccc} \nabla : F^{p+1} \mathcal{H}^k & \longrightarrow & F^p \mathcal{H}^k \otimes \Omega_B \\ \downarrow & & \downarrow \\ \nabla : F^p \mathcal{H}^k & \longrightarrow & F^{p-1} \mathcal{H}^k \otimes \Omega_B \\ \downarrow & & \downarrow \\ \bar{\nabla}^{p,q} : \mathcal{H}^{p,q} & \longrightarrow & \mathcal{H}^{p-1,q+1} \otimes \Omega_B \\ \downarrow & & \downarrow \\ 0 & & 0 \end{array}$$

Proposition 2.136. *The map $\bar{\nabla}^{p,q}$ is $\mathcal{O}_B$ -linear.*

Proof. Since the Gauss–Manin connection satisfies the Leibniz rule

$$\nabla(f\sigma) = f\nabla(\sigma) + \sigma \otimes df,$$

for $f \in \mathcal{O}_B$ and $\sigma \in F^p \mathcal{H}^k$, we have

$$\nabla(f\sigma) = f\nabla(\sigma) \pmod{F^p \mathcal{H}^k \otimes \Omega_B}.$$

For $\sigma \in F^p \mathcal{H}^k$ this implies

$$\bar{\nabla}^{p,q}(f\sigma) = f \bar{\nabla}^{p,q}(\sigma),$$

so $\bar{\nabla}^{p,q}$ is a morphism of $\mathcal{O}_B$ -modules. □

Definition 2.137. For each point $b \in B$, the value of this morphism is

$$\bar{\nabla}_b^{p,q} : \mathcal{H}_b^{p,q} \longrightarrow \mathcal{H}_b^{p-1,q+1} \otimes \Omega_{B,b}.$$

Using the identification

$$\mathcal{H}_b^{p,q} \cong H^q(X_b, \Omega_{X_b}^p),$$

this becomes

$$\bar{\nabla}_b^{p,q} : H^q(X_b, \Omega_{X_b}^p) \longrightarrow H^{q+1}(X_b, \Omega_{X_b}^{p-1}) \otimes \Omega_{B,b}.$$

This map is called the *infinitesimal variation of Hodge structure* at the point b .

Remark 2.138. It follows from the definition of $\overline{\nabla}^{p,q}$ that

$$dP^{p,k} : T_{B,b} \longrightarrow \text{Hom} \left(\frac{F^p H^k(X_b)}{F^{p+1} H^k(X_b)}, \frac{F^{p-1} H^k(X_b)}{F^p H^k(X_b)} \right)$$

is constructed by adjunction starting from the map

$$\overline{\nabla}_b^{p,q} : \frac{F^p H^k(X_b)}{F^{p+1} H^k(X_b)} \longrightarrow \frac{F^{p-1} H^k(X_b)}{F^p H^k(X_b)} \otimes \Omega_{B,b}.$$

Thus we obtain the following cohomological description:

$$dP^{p,k}(u) \in \text{Hom} \left(H^q(X_b, \Omega_{X_b}^p), H^{q+1}(X_b, \Omega_{X_b}^{p-1}) \right).$$

Theorem 2.139 (Griffiths). *The map*

$$dP^{p,k}(u) : H^q(X_b, \Omega_{X_b}^p) \longrightarrow H^{q+1}(X_b, \Omega_{X_b}^{p-1})$$

is equal to the cup product with the class

$$\rho(u) \in H^1(X_b, T_{X_b}),$$

where ρ is the Kodaira–Spencer map. More precisely, it is the composition of the cup product with $\rho(u)$ and the map induced on cohomology by the interior product

$$T_{X_b} \otimes \Omega_{X_b}^p \longrightarrow \Omega_{X_b}^{p-1}.$$

2.3.10 The Picard-Fuchs Equation

If we pick any $\Omega \in \mathcal{H}$ then the expression

$$(\nabla^{GM})^{h^n} \Omega$$

for derivatives in any directions in $\mathcal{M}_{cs}$ must be expressible in terms of $(\nabla^{GM})^k \Omega$, $k = 0, \dots, h^n$ in cohomology. This is equivalent to the *Picard–Fuchs ideal* $\mathcal{I}$, which is the kernel of the map

$$\phi(X_1, \dots, X_r) = \nabla_{X_1} \cdots \nabla_{X_r} \Omega(t)$$

from the sheaves of linear differential operators $\mathcal{D}_i$ on $\mathcal{M}_{cs}$ to the Hodge bundle $\mathcal{F}^0$, which makes the latter into a $\mathcal{D}$ -module. Here $X_i = \partial_{z_i}$, $i = 1, \dots, r = \dim(\mathcal{M}_{cs})$ is a basis of vector fields on $\mathcal{M}_{cs}$. We have the ring of differential operators

$$\mathcal{D} = \mathbb{C}\{z_1, \dots, z_r\} \left[\frac{\partial}{\partial z_1}, \dots, \frac{\partial}{\partial z_r} \right],$$

where $\mathbb{C}\{z_1, \dots, z_r\}$ is the ring of convergent power series in $z_1, \dots, z_r$.

The *Picard–Fuchs ideal* plays an important role in the theory. For example, we will see later that knowing $\mathcal{I}$ allows us to calculate the Yukawa couplings. If the above map ϕ is surjective, which happens for Calabi–Yau threefolds, then the $\mathcal{D}$ -module structure on $\mathcal{F}^0$ is completely determined by the Picard–Fuchs ideal $\mathcal{I}$.

Remark 2.140 (Surjectivity and cyclic $\mathcal{D}$ -modules). This statement follows from a basic algebraic fact.

(a) **Meaning of surjectivity of ϕ .** The map ϕ being surjective means that every local section of $\mathcal{F}^0$ can be written in the form

$$s = D \cdot \Omega$$

for some differential operator $D \in \mathcal{D}$. Equivalently,

$$\mathcal{F}^0 = \mathcal{D} \cdot \Omega,$$

so $\mathcal{F}^0$ is a *cyclic $\mathcal{D}$ -module* generated by the single element Ω .

(b) **Classification of cyclic $\mathcal{D}$ -modules.** If a $\mathcal{D}$ -module is generated by one element Ω , then it is canonically isomorphic to a quotient

$$\mathcal{F}^0 \cong \mathcal{D}/\mathcal{A} \backslash \backslash (\Omega),$$

where

$$\mathcal{A} \backslash \backslash (\Omega) = \{ D \in \mathcal{D} \mid D \cdot \Omega = 0 \}$$

is the annihilator ideal of Ω . In the present situation this annihilator is precisely the Picard–Fuchs ideal $\mathcal{I}$.

Therefore,

$$\mathcal{F}^0 \cong \mathcal{D}/\mathcal{I}.$$

This explains the statement that, once ϕ is surjective, the $\mathcal{D}$ -module structure on $\mathcal{F}^0$ is completely determined by the Picard–Fuchs ideal $\mathcal{I}$: knowing the left ideal $\mathcal{I} \subset \mathcal{D}$ is enough to reconstruct $\mathcal{F}^0$ as a $\mathcal{D}$ -module up to isomorphism.

Remark 2.141 (Picard–Fuchs ideal and the horizontal D -module). Let $\mathcal{H} := R^n \pi_* \mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_{\mathcal{M}_{cs}}$ be the flat bundle equipped with the Gauss–Manin connection ∇^{GM} , and let

$$F^0 \subset \mathcal{H}$$

denote the Hodge bundle generated by the holomorphic $(n, 0)$ -form Ω .

The sheaf $\mathcal{D}$ of holomorphic differential operators on $\mathcal{M}_{cs}$ acts on F^0 via the Gauss–Manin connection,

$$X \cdot s := \nabla_X^{GM} s, \quad X \in T\mathcal{M}_{cs}.$$

This defines a morphism of $\mathcal{D}$ -modules

$$\phi : \mathcal{D} \longrightarrow F^0, \quad P \longmapsto P \cdot \Omega.$$

The Picard–Fuchs ideal $\mathcal{I}$ is defined as

$$\mathcal{I} := \ker \phi,$$

so that $\mathcal{D}/\mathcal{I}$ is the cyclic $\mathcal{D}$ -module generated by Ω .

The *horizontal D -module* (or horizontal subbundle) is defined by

$$\mathcal{H}_{\text{hor}} := \mathcal{D} \cdot \Omega = \text{Span}_{\mathcal{O}_{\mathcal{M}_{cs}}} \{ \Omega, \nabla^{GM} \Omega, (\nabla^{GM})^2 \Omega, \dots \} \subset \mathcal{H}.$$

Equivalently, it is the smallest $\mathcal{D}$ -submodule of $\mathcal{H}$ containing Ω .

By construction one has a canonical identification

$$\mathcal{D}/\mathcal{I} \cong \mathcal{H}_{\text{hor}}.$$

In general the horizontal submodule is a proper subbundle of F^0 . When the map ϕ is surjective (as happens for Calabi–Yau threefolds), the cyclic D -module generated by Ω coincides with the full Hodge bundle,

$$\mathcal{H}_{\text{hor}} = F^0, \quad F^0 \cong \mathcal{D}/\mathcal{I}.$$

Remark 2.142 (Algebraic structures involved). The objects appearing in the identity

$$\mathcal{F}^0 \cong \mathcal{D}/\mathcal{I}$$

carry different algebraic structures which should be clearly distinguished.

1. The sheaf $\mathcal{D} = \mathcal{D}_S$ of linear differential operators on S is a *sheaf of noncommutative rings*. Locally it is generated by $\mathcal{O}_S$ together with vector fields, and the commutation relation

$$[\partial_{z_i}, f] = \partial_{z_i}(f)$$

shows that multiplication in $\mathcal{D}$ is not commutative.

2. The Picard–Fuchs ideal $\mathcal{I} \subset \mathcal{D}$ is a *left ideal sheaf*, namely

$$\mathcal{I} = \{D \in \mathcal{D} \mid D \cdot \Omega = 0\}.$$

Since $\mathcal{D}$ is noncommutative, $\mathcal{I}$ need not be a two-sided ideal; only left multiplication by elements of $\mathcal{D}$ is required to preserve $\mathcal{I}$.

3. The Hodge bundle $\mathcal{F}^0$ is not a ring but a $\mathcal{D}$ -module. In particular, it remains a sheaf of abelian groups (indeed an $\mathcal{O}_S$ -module), equipped with an action

$$\mathcal{D} \times \mathcal{F}^0 \longrightarrow \mathcal{F}^0$$

induced by the Gauss–Manin connection.

The isomorphism $\mathcal{F}^0 \cong \mathcal{D}/\mathcal{I}$ is an isomorphism of $\mathcal{D}$ -modules.

The equations

$$D \cdot y = 0, \quad D \in \mathcal{I},$$

are the *Picard–Fuchs equations*, and a local section y of $\mathcal{O}_S$ is a solution if $D \cdot y = 0$ for all $D \in \mathcal{I}$. Since

$$\frac{\partial}{\partial z_i} \int_{\gamma_t} \Omega(t) = \int_{\gamma_t} \nabla_{\partial/\partial z_i} \Omega(t)$$

for a (locally) constant homology n -cycle γ_t , it follows that the periods

$$y = \int_{\gamma_t} \Omega(t)$$

of Ω are solutions of the Picard–Fuchs equations. We will often write periods as

$$y = \langle g(t), \Omega(t) \rangle,$$

where $g(t) \in H^n(X_t, \mathbb{C})$ is the Poincaré dual of γ_t (thus $g(t)$ is a flat section of ∇). Note that a differential operator D is in the Picard–Fuchs ideal if and only if

$$D \cdot y = 0$$

for *all* periods $y = \langle g, \Omega \rangle$.

Although we have only constructed the Picard–Fuchs equations locally, by analytic continuation they make sense as algebraic differential equations over all of S if we restrict our attention to the periods of algebraic differentials.

Remark 2.143 (Why the “ $n + 1$ ” order is a *special* case). The upper bound $h^n = \dim H^n(X_t, \mathbb{C})$ comes from finite dimensionality of $H^n(X_t, \mathbb{C})$: the vectors

$$\Omega, \nabla^{GM}\Omega, (\nabla^{GM})^2\Omega, \dots$$

must become linearly dependent after at most h^n steps, hence one obtains a Picard–Fuchs relation of order $\leq h^n$.

The frequently quoted order $n + 1$ for a one-parameter Calabi–Yau n -fold is not automatic; it uses additional input on the Hodge numbers. Assume:

1. $r = \dim \mathcal{M}_{cs} = 1$ (so we differentiate along a single vector field $X = \partial_z$),
2. the Hodge decomposition on $H^n(X_t, \mathbb{C})$ satisfies

$$h^{n-k,k} = \dim H^{n-k,k}(X_t) = 1, \quad k = 0, 1, \dots, n,$$

(equivalently, the Hodge filtration has successive quotients of dimension 1),

3. and the iterated derivatives are *maximally non-degenerate* in the sense that

$$(\nabla_X^{GM})^k \Omega \notin F^{n-k+1} \quad (k = 0, 1, \dots, n),$$

so each step actually produces a new nonzero class in the graded piece $\mathrm{Gr}_F^{n-k} = F^{n-k} / F^{n-k+1}$.

Then Griffiths transversality $\nabla_X^{GM}(F^p) \subset F^{p-1}$ implies that

$$(\nabla_X^{GM})^k \Omega \in F^{n-k} \quad \text{and its class in } \mathrm{Gr}_F^{n-k} \text{ is nonzero.}$$

Because $\dim \mathrm{Gr}_F^{n-k} = h^{n-k,k} = 1$ by (2), the classes for $k = 0, 1, \dots, n$ are forced to be linearly independent, hence they span all of $H^n(X_t, \mathbb{C})$ and one gets a *minimal* Picard–Fuchs operator of order exactly $n + 1$. If either $r > 1$ or some $h^{n-k,k} > 1$ (or the non-degeneracy in (3) fails), the minimal order can be larger than $n + 1$ and is only a priori bounded by h^n .

The local flatness implies that the non-trivial essence in this setup is to large extend in the global monodromy group Γ acting on the sections over $\mathcal{M}_{cs}$, which explains the pivotal role of Γ .

In fact one can turn the problem around and start with the monodromies at critical divisors, which determine the leading behaviour of the sections, which in turn are captured for simple examples by the Riemann–Hilbert symbol, and ask the question whether there is a differential system with regular singularities only, which then fully determines the sections. Problems of these kind are called Riemann–Hilbert problems.

Remark 2.144. It is important to distinguish carefully between the information contained in the monodromy representation and that encoded by the Picard–Fuchs equation.

The monodromy representation determines the underlying local system $R^n\pi_*\mathbb{C}$, or equivalently the flat bundle equipped with the Gauss–Manin connection $(\mathcal{H}, \nabla)$, via the Riemann–Hilbert correspondence. This data is purely topological and does not involve the Hodge decomposition.

In contrast, the Picard–Fuchs operator depends not only on the flat connection but also on the choice of the holomorphic top form $\Omega \in F^n$, i.e. the highest step of the Hodge filtration. The differential equation arises from the linear dependence relations among

$$\Omega, \nabla\Omega, \nabla^2\Omega, \dots$$

inside the finite-dimensional flat bundle $\mathcal{H}$. Consequently, the Picard–Fuchs equation reflects the interaction between the Gauss–Manin connection and the Hodge filtration rather than the local system alone.

Therefore, while the Riemann–Hilbert correspondence identifies flat connections with monodromy data, the monodromy representation by itself does not uniquely determine the Picard–Fuchs operator. Additional geometric structure, namely the variation of Hodge structure and the distinguished section Ω , is required.

2.4 Periods, Torelli Theorems and the Stringy Moduli Space

2.4.1 Integral Basis in the Middle (Co)homology

Let M_n be a compact complex manifold of complex dimension n . We consider the middle (co)homology groups

$$H_n(M_n, \mathbb{Z}), \quad H^n(M_n, \mathbb{Z}),$$

equipped with the intersection pairing.

Definition 2.145 (Integral symplectic basis, n odd). If n is odd, the intersection form on $H_n(M_n, \mathbb{Z})$ is antisymmetric. One may therefore choose an integral symplectic basis

$$A^I, B_I \in H_n(M_n, \mathbb{Z}), \quad I = 0, \dots, \frac{b_n}{2} - 1,$$

together with a dual basis

$$\alpha_I, \beta^I \in H^n(M_n, \mathbb{Z}),$$

such that the non-vanishing pairings are

$$A^J \cap B_I = -B_I \cap A^J = \delta_I^J,$$

and equivalently

$$\int_{M_n} \alpha_I \wedge \beta^J = - \int_{M_n} \beta^J \wedge \alpha_I = \delta_I^J,$$

as well as

$$\int_{A^J} \alpha_I = \delta_I^J, \quad \int_{B_I} \beta^J = \delta_I^J.$$

Definition 2.146 (Period vector). Let $\Omega(z)$ be the holomorphic $(n, 0)$ -form depending on the complex structure moduli $z \in \mathcal{M}_{cs}$. The periods with respect to the integral symplectic basis are defined by

$$X^I(z) = \int_{A^I} \Omega(z), \quad F_I(z) = \int_{B_I} \Omega(z).$$

They assemble into the period vector

$$\vec{\Pi}(z) = \begin{pmatrix} F_I(z) \\ X^I(z) \end{pmatrix}.$$

Remark 2.147 (Even-dimensional case). If n is even, the intersection form on $H_n(M_n, \mathbb{Z})$ is symmetric. Its signature is determined by the Hirzebruch signature theorem.

Remark 2.148 (Period coordinates on the complex moduli space). Let X be a Calabi–Yau n -fold and $\Omega(z)$ the holomorphic $(n, 0)$ -form varying over the complex structure moduli space $\mathcal{M}_{cs}$. The period map associates to z the projective class

$$[\Omega(z)] \in \mathbb{CP}(H^n(X, \mathbb{C})).$$

By Kodaira–Spencer theory one has

$$\dim \mathcal{M}_{cs} = h^{n-1,1},$$

and, for Calabi–Yau manifolds, the local Torelli theorem implies that the period map is locally an immersion. Consequently, in a neighbourhood where a chosen period $X^0 = \int_{\gamma_0} \Omega$ does not vanish, one may introduce local holomorphic coordinates

$$t_i = \frac{X^i}{X^0}, \quad X^i = \int_{\gamma_i} \Omega, \quad i = 1, \dots, h^{n-1,1},$$

for suitably chosen cycles $\gamma_i \in H_n(X, \mathbb{Z})$.

The existence of such coordinates does not require the cycles γ_i to be mutually non-intersecting. In the case where the middle intersection pairing is symplectic (e.g. n odd), one often chooses a Lagrangian half of a symplectic basis (the A -cycles), for which the intersection numbers vanish among themselves; this provides a canonical normalization of the period coordinates compatible with the integral symplectic structure. For even n , however, the number of mutually isotropic cycles is restricted by the signature of the intersection form, and the above coordinate construction should instead be understood as arising from the local Torelli property rather than from the existence of a large non-intersecting set of cycles.

Remark 2.149 (Integral structure and Picard–Fuchs equations). Using Griffiths’ residue representation of the holomorphic $(n, 0)$ -form, certain period integrals can be computed explicitly in suitable regions of moduli space. In general, one determines the complex moduli dependence of the period vector $\vec{\Pi}(z)$ as a basis of solutions of a system of linear differential equations (the Picard–Fuchs system).

The integral symplectic basis is then recovered by matching these solutions with the known asymptotic behaviour of integral cycles near degeneration points. The monodromy action preserves the intersection pairing and therefore acts by integral symplectic transformations,

$$\vec{\Pi} \mapsto M \vec{\Pi}, \quad M \in Sp(b_n, \mathbb{Z}).$$

2.4.2 First Order Deformations of the Ricci-flat Metric

Let (M, g) be a Calabi–Yau manifold, i.e.

$$R_{\mu\nu}(g) = 0.$$

We study infinitesimal deformations of the metric

$$g_{\mu\nu} \mapsto g_{\mu\nu} + \delta g_{\mu\nu}$$

which preserve the Calabi–Yau condition to first order:

$$R_{\mu\nu}(g + \delta g) = 0.$$

Infinitesimal coordinate transformations (diffeomorphisms) generated by a vector field V^μ induce metric variations

$$\delta g_{\mu\nu} = \nabla_\mu V_\nu + \nabla_\nu V_\mu.$$

These do not correspond to genuine geometric deformations. Hence we impose orthogonality to diffeomorphism directions:

$$\int_M \sqrt{g} \delta g^{\mu\nu} (\nabla_\mu V_\nu + \nabla_\nu V_\mu) d^m x = 0.$$

This condition is equivalent to the gauge fixing condition

$$\nabla^\mu \delta g_{\mu\nu} = 0.$$

Remark 2.150. We show that

$$\int_M \sqrt{g} \delta g^{\mu\nu} (\nabla_\mu V_\nu + \nabla_\nu V_\mu) = 0 \quad \forall V$$

is equivalent to

$$\nabla^\mu \delta g_{\mu\nu} = 0.$$

Since $\delta g^{\mu\nu}$ is symmetric,

$$\int_M \sqrt{g} \delta g^{\mu\nu} (\nabla_\mu V_\nu + \nabla_\nu V_\mu) = 2 \int_M \sqrt{g} \delta g^{\mu\nu} \nabla_\mu V_\nu.$$

Using covariant integration by parts (assuming M compact or boundary terms vanish),

$$\int_M \sqrt{g} \delta g^{\mu\nu} \nabla_\mu V_\nu = - \int_M \sqrt{g} (\nabla_\mu \delta g^{\mu\nu}) V_\nu.$$

Hence the orthogonality condition becomes

$$\int_M \sqrt{g} (\nabla_\mu \delta g^{\mu\nu}) V_\nu = 0 \quad \forall V,$$

which implies

$$\nabla_\mu \delta g^{\mu\nu} = 0.$$

Expanding the equation $R_{\mu\nu}(g + \delta g) = 0$ to linear order around a Ricci-flat background gives

$$\nabla^\rho \nabla_\rho \delta g_{\mu\nu} - 2R_\mu{}^\kappa{}_\nu{}^\sigma \delta g_{\kappa\sigma} = 0.$$

Since g is Kähler, indices split into holomorphic and anti-holomorphic components. The deformation decomposes as

$$\delta g_{ij}, \quad \delta g_{i\bar{j}}.$$

- $\delta g_{i\bar{j}}$ is real;
- δg_{ij} (together with its conjugate $\delta g_{\bar{i}\bar{j}}$) is complex.

Using the equation in Remark 2.9 one obtains:

$$\delta g_{i\bar{j}} \text{ is } \Delta_d\text{-harmonic,}$$

and

$$\delta g^{i\bar{j}} := g^{i\bar{k}} \delta g_{\bar{k}\bar{j}}$$

is $\Delta_{\bar{\partial}}$ -harmonic.

Therefore the infinitesimal Ricci-flat deformations factorize into two independent types:

Proposition 2.151. *For a Calabi-Yau 3-fold, the first order deformations of its Calabi-Yau metric correspond to*

$$H^{1,1}(M) \quad \text{and} \quad H^1(M, TM) \cong H^{2,1}(M).$$

The last identification follows from Serre duality using the nowhere-vanishing holomorphic $(3, 0)$ -form. These two sectors correspond respectively to the Kähler moduli and complex structure moduli, which appear as deformations in the A-model and B-model.

Remark 2.152. (Einstein deformations vs. Calabi-Yau moduli: a precise moduli-space interpretation) Let M be a fixed smooth $2n$ -dimensional manifold. Suppose (J_0, ω_0) is a Calabi-Yau structure on M , namely J_0 is an integrable complex structure with $c_1(M, J_0) = 0$, ω_0 is a J_0 -Kähler form, and

$$g_0(\cdot, \cdot) = \omega_0(\cdot, J_0 \cdot)$$

is the unique Ricci-flat Kähler metric in the class $[\omega_0]$ given by Yau's theorem. We clarify the relation between Einstein deformation theory and the moduli space of Calabi-Yau manifolds.

There are two a priori different geometric moduli spaces:

- The *complex structure moduli space*

$$\mathcal{M}_{\text{CS}}(M) = \{ \text{integrable complex structures } J \text{ with } c_1(M, J) = 0 \} / \text{Diff}_0(M),$$

whose tangent space is the Kodaira-Spencer space

$$T_{[J_0]} \mathcal{M}_{\text{CS}}(M) \cong H^1(M, T_M^{1,0}).$$

- The *Ricci-flat metric moduli space*

$$\mathcal{M}_{\text{RF}}(M) = \{g \in \text{Met}(M) \mid \text{Ric}(g) = 0\} / \text{Diff}_0(M),$$

a purely Riemannian object which does not in general carry any complex structure.

In general these spaces are unrelated: a Ricci-flat metric need not be Kähler nor Calabi–Yau.

(2) Linearization at a Calabi–Yau background. Fix the Calabi–Yau metric g_0 . An infinitesimal Einstein deformation is a symmetric tensor $h = \delta g$ solving the linearized equation (after gauge fixing)

$$\Delta_L h = 0,$$

where Δ_L is the Lichnerowicz Laplacian defined using g_0 .

Because the *background* metric is Kähler (i.e. $\nabla^{g_0} J_0 = 0$), the operator Δ_L preserves the type decomposition induced by J_0 :

$$S^2 T^* M \otimes \mathbb{C} = (2, 0) \oplus (1, 1) \oplus (0, 2).$$

Hence any solution decomposes as

$$h = h^{1,1} + h^{2,0} + h^{0,2}.$$

The reality and symmetry conditions imply

$$h_{i\bar{j}} = \overline{h_{j\bar{i}}}, \quad h_{i\bar{j}} = \overline{h_{ij}},$$

so the $(1, 1)$ part forms a real vector space, while $(2, 0)$ and $(0, 2)$ together form complex directions.

Elliptic theory together with Kähler identities yields the identification

$$T_{[g_0]} \mathcal{M}_{\text{RF}}(M) \cong (H^{1,1}(M, J_0) \cap H^2(M, \mathbb{R})) \oplus H^1(M, T_M^{1,0}).$$

Using the holomorphic volume form of the Calabi–Yau structure,

$$H^1(M, T_M^{1,0}) \cong H^{n-1,1}(M),$$

and for $n = 3$ this becomes $H^{2,1}(M)$.

(3) Infinitesimal vs. actual deformations. The tensor h should be interpreted as a *tangent vector* to the solution space, not as a new metric. If h lies in the kernel above, then (under the unobstructedness properties of the Calabi–Yau case) there exists a genuine family of Ricci-flat metrics g_t satisfying

$$\text{Ric}(g_t) = 0, \quad g_t = g_0 + th + O(t^2).$$

Thus Einstein deformation theory computes the *first-order approximation* to the moduli space.

(4) Why this agrees with Calabi–Yau moduli. In the Calabi–Yau setting two deep results ensure integrability:

- Tian–Todorov: deformations in $H^1(T_M^{1,0})$ integrate to actual complex structures J_t (no obstruction);

- Yau's theorem: for each pair $(J_t, [\omega_t])$ there exists a unique Ricci-flat Kähler metric g_t .

Consequently, near g_0 the Calabi–Yau branch of $\mathcal{M}_{\text{RF}}(M)$ is locally modeled by

$$\mathcal{M}_{\text{CS}}(M) \times \mathcal{M}_{\text{K}}(M, J_0),$$

and its tangent space is precisely

$$H^1(M, T_M^{1,0}) \oplus (H^{1,1}(M) \cap H^2(M, \mathbb{R})).$$

Therefore the appearance of Kodaira–Spencer deformations inside Einstein deformation theory does not rely on assuming that all Ricci-flat deformations remain Calabi–Yau; rather, it reflects the fact that at a Calabi–Yau background the Calabi–Yau moduli form a distinguished integrable branch inside the Ricci-flat metric moduli space.

We discuss more about the moduli space associated to $H^{1,1}(M)$.

In a basis of $(1, 1)$ -forms $\omega^{(k)}$, we expand a Kähler form ω

$$\omega = \sum_{k=1}^{h^{1,1}} v^k \omega^{(k)}.$$

in terms of the real Kähler parameters $v^k > 0$, which are the 2-volumes of the curves C_k dual to $\omega^{(k)}$

$$v^k = \int_{C_k} \omega.$$

Therefore the range of v^k is bounded by the inequalities, which ensure positivity of the 2-volumes of curves C and for examples for 3-fold the 4-volumes of effective divisors D and the 6-volumes of M , i.e.

$$\int_C \omega > 0, \quad \int_D \omega \wedge \omega > 0, \quad \int_M \omega \wedge \omega \wedge \omega > 0.$$

These conditions describe a real cone in $\mathbb{R}_+^{h^{1,1}}$, which is called the Kähler cone. The volume parameters v^k shrink to zero area at the boundaries of the Kähler cone.

Remark 2.153. The discussion above concerns the global structure of the Kähler moduli for a fixed complex structure on M . Although locally a Kähler form can be expanded linearly as

$$\omega = \sum_{k=1}^{h^{1,1}} v^k \omega^{(k)},$$

the parameters v^k do not range freely in $H^{1,1}(M, \mathbb{R})$ but are restricted by positivity conditions

$$\int_C \omega > 0$$

for all effective curves C (and, for a threefold, equivalently by the positivity of divisor and total volumes). These inequalities define an open convex cone

$$\mathcal{K}(M) \subset H^{1,1}(M, \mathbb{R}),$$

called the Kähler cone, which is therefore a genuinely global object.

It is important that this description assumes the complex structure is fixed. Under deformation of complex structure, the Hodge decomposition varies and the subspace

$$H^{1,1}(X_t) \subset H^2(X, \mathbb{C})$$

moves as part of the variation of Hodge structure, so the Kähler cone is naturally associated to the A-model rather than to the full moduli space.

From the viewpoint of mirror symmetry, approaching the boundary of the Kähler cone—where volumes of certain cycles shrink to zero—corresponds to degeneration loci in the complex moduli space. In contrast, the large volume limit corresponds to an infinite-distance limit inside the Kähler cone, which is mirrored by a large complex structure limit characterized by maximally unipotent monodromy.

2.4.3 The Complexified Kähler Moduli Space

The string σ -model action describes string theory perturbatively in the large volume limit. The relevant bosonic part is given by

$$S = \int_{\Sigma} d\sigma^2 \left(g_{i\bar{j}} \eta^{ab} + i b_{i\bar{j}} \epsilon^{ab} \right) \partial_a x^i \partial_b x^{\bar{j}}.$$

Here σ^a with $a, b = 0, 1$ denote worldsheet coordinates. The fields $x^i(\sigma)$ and $x^{\bar{j}}(\sigma)$, with $i, \bar{j} = 1, \dots, n$, are the pullbacks of complex coordinates on M via the embedding map

$$X : \Sigma \longrightarrow M.$$

The tensors η^{ab} and ϵ^{ab} denote respectively the flat worldsheet metric and the antisymmetric tensor on Σ .

The equations of motion for the B -field imply that $b_{i\bar{j}}$ is a harmonic $(1, 1)$ -form, and hence determines a cohomology class in $H^{1,1}(M)$. Together with the Ricci-flat metric deformations, it is therefore natural to decompose the spacetime moduli as follows: for each cohomology class in $H^{2,1}(M)$ one obtains a complex field arising from complex structure deformations, while the B -field complexifies the real Kähler modulus associated to each element of $H^{1,1}(M)$.

The natural metric on the moduli space is

$$ds^2 = \frac{1}{2V} \int_M g^{i\bar{m}} g^{j\bar{n}} [\delta g_{ij} \delta g_{\bar{m}\bar{n}} + (\delta g_{i\bar{m}} \delta g_{j\bar{n}} + \delta b_{i\bar{m}} \delta b_{j\bar{n}})] \sqrt{g} d^6 x.$$

This decomposition also arises naturally from the effective action of type IIA supergravity. In this description, Kähler deformations appear as complex moduli endowed with a special Kähler structure, while complex structure deformations lie inside a complex subspace of a quaternionic manifold. The additional degrees of freedom completing the quaternionic structure are provided by odd Ramond–Ramond form fields, which are invisible in the perturbative worldsheet action discussed above.

Furthermore, $\mathcal{N} = 2$ supergravity interactions imply that, at least for smooth Calabi–Yau manifolds, the metric on the full moduli space is block diagonal. An independent argument for the

separation of the two complex deformation spaces comes from the marginal deformations of the $(\mathcal{N}, \mathcal{N}) = (2, 2)$ worldsheet theory.

It is natural to complexify the Kähler parameters by adding the contribution of the antisymmetric Neveu–Schwarz B -field. Integrating the second term in the action over a basis of two-cycles $C_k \subset M$ and normalizing appropriately, one defines the *complexified Kähler parameters*

$$t^k = \frac{1}{2\pi i} \int_{C_k} (i b - \omega) = \frac{1}{2\pi i} (b^k - v^k),$$

where

$$b^k = \int_{C_k} b, \quad v^k = \int_{C_k} \omega$$

denote respectively the B -field periods and the volumes of the cycles C_k .

It is important to note that the second term in the action counts the winding number describing how many times the string wraps the cycle C_k . Since the action is invariant under integer phase shifts

$$S \longrightarrow S + 2\pi i n, \quad n \in \mathbb{Z},$$

a shift of the B -field period by $2\pi i$ defines an exact symmetry of the large volume expansion. Consequently, the complexified Kähler coordinates satisfy the periodic identification

$$t^k \longrightarrow t^k + 1.$$

The mirror symmetry conjecture asserts that Calabi–Yau manifolds occur in mirror pairs (M, W) such that the complex moduli space

$$\mathcal{M}_{cs}(M)$$

of M is identified with the complexified Kähler moduli space

$$\mathcal{M}_{cks}(W)$$

of the mirror manifold W , and vice versa.

In particular, this requires that the relevant moduli spaces are at least one-dimensional, since both M and W are Kähler manifolds and therefore possess at least one Kähler modulus. Consequently, rigid Calabi–Yau manifolds, which admit no complex structure deformations, do not directly fit into this simple picture.

Nevertheless, even in such cases the complexified Kähler moduli space $\mathcal{M}_{cks}(M)$ can be mapped to the complex deformation space of a mirror geometry. The mirror may then be realized either as a higher-dimensional manifold or as a space with anticommuting (super)coordinates.

Mirror symmetry furthermore provides a natural complex parametrization of the complexified Kähler moduli space $\mathcal{M}_{cks}$: namely, it is given by the complex structure parameters of the mirror manifold, denoted by

$$t_k^m.$$

2.4.4 Elliptic Curve Moduli

The simplest example illustrating both the local and global Torelli theorem, as well as the stringy moduli space, is provided by the complex moduli space of an elliptic curve $\mathcal{E}$. Locally, it is parametrized by the periods of the holomorphic $(1,0)$ -form

$$\omega := \Omega_1$$

over a symplectic basis (A, B) of $H_1(\mathcal{E}, \mathbb{Z})$:

$$\tau = \frac{\int_B \omega}{\int_A \omega} \in \mathbb{H}_+.$$

Here $\mathbb{H}_+$ denotes the complex upper half-plane

$$\mathbb{H}_+ = \{z \in \mathbb{C} \mid \Im(z) > 0\} = SU(1, 1)/U(1).$$

Remark 2.154. The identification

$$\mathbb{H}_+ = SU(1, 1)/U(1)$$

shows that the upper half-plane is naturally a Hermitian symmetric space. In particular, it can be interpreted as the *period domain* parametrizing polarized Hodge structures of weight one with Hodge numbers $(h^{1,0}, h^{0,1}) = (1, 1)$.

For an elliptic curve $\mathcal{E}$, the Hodge decomposition

$$H^1(\mathcal{E}, \mathbb{C}) = H^{1,0} \oplus H^{0,1}$$

is completely determined by the complex line $H^{1,0} \subset H^1(\mathcal{E}, \mathbb{C})$ satisfying the polarization condition coming from the intersection pairing. The space of all such allowed lines is precisely the homogeneous space $SU(1, 1)/U(1)$.

Recall that an elliptic curve can be written as

$$\mathcal{E} = \mathbb{C}/\mathbb{Z}^2(1, \tau),$$

namely as the complex plane modulo the lattice generated by $(1, \tau)$. Geometrically this corresponds to a parallelogram with opposite sides identified. The condition $\Im(\tau) > 0$ expresses the positivity of the oriented volume.

Globally, two parameters $\tau, \tau' \in \mathbb{H}_+$ determine isomorphic elliptic curves precisely when they are related by the fractional linear action of $PSL(2, \mathbb{Z})$. Hence the complex moduli space is

$$\mathcal{M}_{cs}(\mathcal{E}) = PSL(2, \mathbb{Z}) \backslash \mathbb{H}_+.$$

The monodromy action on the period vector then coincides with the natural action of the orbifold fundamental group $PSL(2, \mathbb{Z})$.

Remark 2.155. For a fixed Calabi–Yau manifold X , changing the symplectic basis of $H_n(X, \mathbb{Z})$ acts on the period vector by the full integral automorphism group preserving the polarization (for odd n , a symplectic group $Sp(H_n(X, \mathbb{Z}))$). Hence all such basis changes produce different coordinate representatives of the *same* complex structure.

However, this group of basis changes should not be confused with the monodromy group arising from the global complex moduli space. The monodromy representation

$$\pi_1(\mathcal{M}_{cs}) \longrightarrow \text{Aut}(H_n(X, \mathbb{Z}), Q)$$

factors through the mapping class group of the underlying smooth manifold, and therefore its image consists only of those lattice automorphisms that can be realized by geometric deformations of X . In general this image is a proper subgroup of the full symplectic group.

The factorization arises because parallel transport along a loop in the moduli space produces a diffeomorphism of the underlying smooth manifold (up to isotopy). Hence the monodromy representation factors through the mapping class group before acting on homology.

The elliptic curve is exceptional: for the torus one has

$$\text{MCG}(T^2) \cong SL(2, \mathbb{Z}),$$

so every symplectic change of basis arises from monodromy. Consequently, all parameters obtained by varying the symplectic basis correspond exactly to the $PSL(2, \mathbb{Z})$ -orbit of a single modulus τ . This coincidence fails for higher-dimensional Calabi–Yau manifolds.

Let $\mathcal{F}$ denote the fundamental region of the action of $\Gamma_1 = PSL(2, \mathbb{Z})$ on $\mathbb{H}_+$. Consider a family of Riemann surfaces $\mathcal{E}(s)$, typically realized as the zero locus of an algebraic equation representing the anticanonical class in a Fano surface. For generic values of the complex parameter s the curve is smooth. The actual monodromy group

$$\Gamma_{\mathcal{E}(s)} \subset \Gamma_1$$

is a subgroup of finite index m , and the complex moduli space is

$$\mathcal{M}_{cs}(\mathcal{E}(s)) = \mathbb{H}_+ / \Gamma_{\mathcal{E}(s)},$$

so that

$$\frac{\text{Vol}(\mathcal{M}_{cs}(\mathcal{E}(s)))}{\text{Vol}(\mathbb{H}_+ / \Gamma_1)} = m.$$

By Yau’s theorem, a Ricci-flat Kähler metric on a Calabi–Yau manifold is uniquely determined once the complex structure and the Kähler class $[\omega]$ are fixed. For the two-torus, since $h^{1,1} = 1$, the Kähler class is completely determined by the volume, and hence the moduli space of Ricci-flat metrics is

$$\mathcal{M}_g = \mathcal{M}_{cs}(\mathcal{E}) \times \mathbb{R}_+.$$

In string theory, or more precisely in the non-linear sigma model, the volume is complexified by the Neveu–Schwarz B -field. By mirror symmetry, the moduli space of complexified Kähler structures is another copy of the fundamental region $\mathcal{F}$. Since string theory on the two-torus is exactly solvable, the dependence on the moduli can be made explicit. The resulting stringy moduli space is

$$\mathcal{M}_{\text{string}} = \Gamma_\sigma \backslash (\mathcal{F}_{cs} \times \mathcal{F}_{ck}),$$

where $\Gamma_\sigma \cong (\mathbb{Z}_2)^3$ is generated by mirror symmetry, charge conjugation, and world-sheet parity of the non-linear sigma model.

Remark 2.156. For the two-torus sigma model the physical moduli space is obtained by further quotienting by a discrete symmetry group

$$\Gamma_\sigma \cong (\mathbb{Z}_2)^3,$$

generated by three involutions acting on the pair of moduli $(\tau, \rho) \in \mathbb{H}_+ \times \mathbb{H}_+$:

(i) *Mirror symmetry (T-duality)*. This exchanges complex and Kähler moduli,

$$(\tau, \rho) \mapsto (\rho, \tau).$$

(ii) *Charge conjugation*. This acts by complex conjugation combined with orientation reversal,

$$(\tau, \rho) \mapsto (-\bar{\tau}, -\bar{\rho}).$$

(iii) *World-sheet parity*. Reversing the orientation of the string world-sheet changes the sign of the B -field while leaving the metric invariant, hence

$$(\tau, \rho) \mapsto (\tau, -\bar{\rho}).$$

Each transformation squares to the identity, and together they generate a group isomorphic to $(\mathbb{Z}_2)^3$. The stringy moduli space is therefore

$$\mathcal{M}_{\text{string}} = \Gamma_\sigma \backslash (\mathcal{F}_{cs} \times \mathcal{F}_{ck}).$$

Remark 2.157. The symmetry between complex and Kähler moduli has a fundamentally different meaning for elliptic curves and for higher-dimensional Calabi–Yau manifolds.

For the two-torus, the moduli space of the non-linear sigma model takes the form

$$\mathbb{H}_+ \times \mathbb{H}_+,$$

parametrized by the complex structure modulus τ and the complexified Kähler modulus ρ . The two factors arise symmetrically from the structure of the torus conformal field theory, and mirror symmetry is realized as a discrete duality exchanging them,

$$(\tau, \rho) \mapsto (\rho, \tau).$$

Thus, in this case mirror symmetry is an internal symmetry of a single theory, reflecting an intrinsic symmetry of the moduli space itself.

For Calabi–Yau threefolds, however, mirror symmetry is somehow different. It does not merely exchange complex and Kähler moduli within one moduli space; rather, it asserts an equivalence between string theories compactified on two distinct Calabi–Yau manifolds X and Y , such that

$$\mathcal{M}_{cs}(X) \cong \mathcal{M}_K(Y), \quad \mathcal{M}_K(X) \cong \mathcal{M}_{cs}(Y).$$

The matching of moduli spaces is therefore only the first manifestation of a deeper correspondence identifying special Kähler geometry, quantum corrections, and ultimately the full physical theories. The torus case is exceptional in that the mirror manifold coincides with the original one, making the duality appear as a simple exchange symmetry.

2.4.5 Calabi–Yau 3-Folds

The moduli space of the metrics of Calabi–Yau threefolds is not a homogeneous space as in the examples of Riemann surfaces or K3 surfaces. Moreover, there is no global Torelli theorem. Instead, it carries a structure known as *special Kähler geometry*, which will be discussed from the geometrical point of view in the next section. Much of this structure follows from the existence of the $(n, 0)$ -form and Griffiths transversality, which hold in any dimension, although they lead to very different geometrical structures in even and odd dimensions.

We conclude this section with a description of the deformation spaces of Calabi–Yau threefolds from the physics point of view. In the low-energy effective action of type IIA/B string theory these marginal deformations appear as vacuum expectation values of complex scalar fields labeling the vacuum manifold of the $N = 2$ supergravity theory in four dimensions. The general structure of this vacuum manifold for abelian gauge groups $U(1)^{\#V}$ and $U(1)^{\#H}$ is locally of the form

$$\mathcal{M}_{2\#V} \times \mathcal{Q}_{4\#H},$$

where $\mathcal{M}$ is a complex special Kähler manifold describing the scalar fields in the vector multiplets and $\mathcal{Q}$ is a quaternionic manifold describing the scalar fields in the hypermultiplets. The subscripts indicate the real dimensions of the moduli spaces. Its relation to the perturbative sector of type II string compactifications on Calabi–Yau threefolds M and W is given by

$$\mathcal{M}_{\text{tot}}^{IIA}(M) = \mathcal{M}_{2h^{1,1}(M)}^{IIA} \times \mathcal{Q}_{4(h^{2,1}(M)+1)},$$

$$\mathcal{M}_{\text{tot}}^{IIB}(W) = \mathcal{M}_{2h^{2,1}(W)}^{IIB} \times \mathcal{Q}_{4(h^{1,1}(W)+1)}.$$

A far-reaching formulation of the mirror conjecture states that type IIA and type IIB string compactifications are completely equivalent whenever M and W form a mirror pair. In particular,

$$\mathcal{M}_{\text{tot}}^{IIA}(M) = \mathcal{M}_{\text{tot}}^{IIB}(W).$$

The best studied object is $\mathcal{M}_{2h^{2,1}(W)}^{IIB}$, since it coincides with the complex moduli space of W .

Remark 2.158. The enhancement of the Calabi–Yau metric moduli space from a complex manifold to the quaternionic space $\mathcal{Q}$ of Kähler multiplets is due to the moduli of Ramond fields. The additional quaternionic dimension in $\mathcal{Q}$ arises from the universal dilaton multiplet, whose scalar components (S, C) contain in particular the type II dilaton S .

Remark 2.159. For type II string compactifications on a Calabi–Yau threefold the scalar moduli space of the resulting four-dimensional $N = 2$ supergravity theory splits locally into two factors,

$$\mathcal{M}_{\text{tot}} = \mathcal{M}_V \times \mathcal{Q}_H,$$

where $\mathcal{M}_V$ is a complex special Kähler manifold describing the vector multiplet scalars, while $\mathcal{Q}_H$ is a quaternionic manifold describing the hypermultiplet scalars.

For type IIA compactified on a Calabi–Yau threefold M , the vector multiplet moduli space is the complexified Kähler moduli space, parametrized by

$$B + iJ \in H^{1,1}(M, \mathbb{C}),$$

whereas the hypermultiplet space $\mathcal{Q}_H$ contains the complex structure moduli together with additional Ramond–Ramond scalars and the universal dilaton multiplet.

For type IIB compactifications the roles are reversed: the vector multiplet moduli space coincides with the complex moduli space, while the hypermultiplet space contains the complexified Kähler moduli, Ramond–Ramond fields, and the dilaton.

Thus mirror symmetry between Calabi–Yau threefolds M and W interchanges the vector multiplet sectors of the two theories, rather than merely exchanging complex and Kähler moduli inside a single geometric moduli space.

3 Special Geometry on the Complex Moduli Space

3.1 Universal Part of Special Geometry

3.1.1 The Weil–Petersson metric

Let M_n be a Calabi–Yau n -fold and let $\mathcal{M}_{cs}$ denote its complex moduli space. By the Tian–Todorov theorem the moduli space of Calabi–Yau manifolds is unobstructed, and its tangent space is given by

$$T\mathcal{M}_{cs} \cong H^1(M_n, TM_n).$$

Contracting with the holomorphic volume form Ω_n identifies

$$H^1(M_n, TM_n) \cong H^{n-1,1}(M_n).$$

Definition 3.1. Let Ω_n be the nowhere vanishing holomorphic $(n, 0)$ -form on M_n . The *Kähler potential* on the complex moduli space is defined locally by

$$e^{-K} = R(\Omega_n) = i^{n^2} \langle \Omega_n, \overline{\Omega}_n \rangle.$$

Definition 3.2. The *Weil–Petersson metric* on $\mathcal{M}_{cs}$ is defined by

$$G_{i\bar{j}} := \partial_i \partial_{\bar{j}} K.$$

Remark 3.3. The expression

$$e^{-K} = i^{n^2} \langle \Omega_n, \overline{\Omega}_n \rangle$$

should *not* be interpreted as defining a globally well-defined function on the complex moduli space.

Indeed, for a Calabi–Yau n -fold the line

$$F^n = H^{n,0}$$

forms a holomorphic line bundle over $\mathcal{M}_{cs}$ (the *Hodge line bundle*). A choice of holomorphic volume form Ω_n corresponds to choosing a local holomorphic frame of this line bundle and is therefore only defined up to a holomorphic rescaling

$$\Omega_n \mapsto \Omega'_n = e^{f(z)} \Omega_n, \quad f(z) \text{ holomorphic.}$$

Under this change one has

$$i^{n^2} \langle \Omega'_n, \overline{\Omega}'_n \rangle = e^{f(z) + \overline{f(z)}} i^{n^2} \langle \Omega_n, \overline{\Omega}_n \rangle.$$

Consequently

$$K' = K - f - \bar{f}.$$

This is precisely the usual ambiguity of a Kähler potential, since

$$\partial_i \partial_{\bar{j}}(f + \bar{f}) = 0.$$

Therefore the Weil–Petersson metric

$$G_{i\bar{j}} = \partial_i \partial_{\bar{j}} K$$

is independent of the choice of local representative Ω_n .

Remark 3.4. Equivalently, the polarization pairing induces a Hermitian metric h on the Hodge line bundle F^n by setting, for a local frame Ω_n ,

$$h(\Omega_n, \Omega_n) = i^{n^2} \langle \Omega_n, \overline{\Omega_n} \rangle.$$

The Weil–Petersson Kähler form can then be written intrinsically as

$$\omega_{WP} = -i \partial \bar{\partial} \log h,$$

which is the Chern curvature of the Hermitian line bundle (F^n, h) . Thus the Kähler potential K should be viewed as a *local potential* for this curvature form rather than as a globally defined function.

Proposition 3.5. *Let Ω_n be the holomorphic $(n, 0)$ -form on M_n . Then Griffiths transversality implies*

$$\partial_i \Omega_n = \alpha_i(z) \Omega_n + \chi_i, \quad \chi_i \in H^{n-1,1}(M_n).$$

Remark 3.6. The decomposition

$$\partial_i \Omega_n = \alpha_i(z) \Omega_n + \chi_i, \quad \chi_i \in H^{n-1,1}(M_n)$$

does not require the coordinates z_i on $\mathcal{M}_{cs}$ to be flat.

Indeed Griffiths transversality states that

$$\nabla_i^{\text{GM}} \Omega_n \in F^{n-1} = H^{n,0}(M_n) \oplus H^{n-1,1}(M_n).$$

Hence one can write

$$\nabla_i^{\text{GM}} \Omega_n = a_i(z) \Omega_n + \chi_i.$$

Substituting Proposition 2.98 into the previous expression yields

$$\partial_i \Omega_n = (a_i(z) - A_i(z)) \Omega_n + \chi_i.$$

Here the connection matrices A_i preserve the Hodge type of the forms on which they act.

Defining $\alpha_i(z) = a_i(z) - A_i(z)$ gives

$$\partial_i \Omega_n = \alpha_i(z) \Omega_n + \chi_i.$$

Proposition 3.7. *Let*

$$e^{-K} = i^{n^2} \langle \Omega_n, \overline{\Omega_n} \rangle.$$

Then the coefficient in the above decomposition satisfies

$$\alpha_i = -\partial_i K.$$

Proof. Taking a derivative gives

$$-K_i e^{-K} = i^{n^2} \partial_i \langle \Omega_n, \overline{\Omega_n} \rangle.$$

Since $\partial_i \overline{\Omega_n} = 0$ and

$$\partial_i \Omega_n = \alpha_i \Omega_n + \chi_i, \quad \chi_i \in H^{n-1,1},$$

we obtain

$$\partial_i \langle \Omega_n, \overline{\Omega_n} \rangle = \alpha_i \langle \Omega_n, \overline{\Omega_n} \rangle.$$

Substituting into the previous equation yields

$$\alpha_i = -\partial_i K.$$

□

Definition 3.8. Define the covariant derivative

$$D_i \Omega_n := (\partial_i + K_i) \Omega_n.$$

Proposition 3.9. *The covariant derivative satisfies*

$$D_i \Omega_n = \chi_i \in H^{n-1,1}(M_n), \quad \overline{D_{\bar{i}} \Omega_n} = \bar{\chi}_{\bar{i}} \in H^{1,n-1}(M_n).$$

By the Tian–Todorov theorem, if $\{z_i\}$ are independent local coordinates on $\mathcal{M}_{cs}$ then

$$\chi_i, \quad i = 1, \dots, h^{n-1,1}$$

form a basis of $H^{n-1,1}(M_n)$.

Remark 3.10. The relation between χ_i and $\bar{\chi}_{\bar{i}}$ follows from complex conjugation of the covariant derivative.

Since

$$D_i = \partial_i + K_i, \quad K_i = \partial_i K,$$

and K is real-valued, we have

$$\overline{\partial_i} = \partial_{\bar{i}}, \quad \overline{K_i} = K_{\bar{i}}.$$

Hence

$$\overline{D_i} = D_{\bar{i}}.$$

Applying this to Ω_n gives

$$\bar{\chi}_{\bar{i}} = \overline{D_i \Omega_n} = D_{\bar{i}} \overline{\Omega_n}.$$

Defining

$$\bar{\chi}_{\bar{i}} := D_{\bar{i}} \overline{\Omega_n}$$

one obtains

$$\overline{\chi_i} = \bar{\chi}_{\bar{i}}.$$

Proposition 3.11. *Let*

$$\chi_i = D_i \Omega_n.$$

Then the metric $G_{i\bar{j}}$ can be written as

$$G_{i\bar{j}} = e^K (-i)^{n^2} \langle \chi_i, \bar{\chi}_{\bar{j}} \rangle.$$

Equivalently, defining

$$g_{i\bar{j}} := -i)^{n^2} \langle \chi_i, \bar{\chi}_{\bar{j}} \rangle,$$

one has

$$g_{i\bar{j}} = e^{-K} G_{i\bar{j}}.$$

Proof. Using

$$e^{-K} = i)^{n^2} \langle \Omega_n, \bar{\Omega}_n \rangle$$

together with

$$D_i \Omega_n = \chi_i$$

one obtains the stated identity by direct computation. $\square$

Remark 3.12. The tensor $G_{i\bar{j}}$ is the Weil–Petersson metric on the complex moduli space. The quantity

$$g_{i\bar{j}} = -i)^{n^2} \langle \chi_i, \bar{\chi}_{\bar{j}} \rangle$$

depends on the choice of the local holomorphic representative Ω_n of the Hodge line bundle, while $G_{i\bar{j}}$ is invariant under the rescaling $\Omega_n \mapsto e^{f(z)} \Omega_n$ and therefore defines a global Kähler metric on $\mathcal{M}_{cs}$.

Thus the Weil–Petersson metric on the complex moduli space arises from the Hodge inner product and is Kähler.

Higher derivatives of $D_i \Omega_n$ lead to a collection of differential identities collectively known as *special geometry*.

3.1.2 Covariant Derivatives on Tensor-Valued Sections

It is useful to summarize the gauge transformation properties and the corresponding covariant derivatives of the objects introduced before.

Definition 3.13 (Hodge line bundle). Let $\mathcal{M}_{cs}$ be the complex moduli space of a Calabi–Yau manifold. The nowhere vanishing holomorphic top form Ω_n defines a section of a holomorphic line bundle

$$\mathcal{L} \rightarrow \mathcal{M}_{cs}.$$

Under a change of local trivialization of $\mathcal{L}$ it transforms as

$$\Omega(z) \mapsto e^{f(z)} \Omega(z),$$

where $f(z)$ is a holomorphic function.

Definition 3.14 (Kähler transformation). Let

$$e^{-K} = i^{n^2} \langle \Omega_n, \bar{\Omega}_n \rangle.$$

Under the change of trivialization

$$\Omega \mapsto e^{f(z)} \Omega$$

the Kähler potential transforms as

$$K(z, \bar{z}) \mapsto K(z, \bar{z}) - f(z) - \bar{f}(\bar{z}).$$

Consequently

$$e^{-K}$$

defines a section of the line bundle

$$\mathcal{L} \otimes \bar{\mathcal{L}}.$$

Definition 3.15 (Covariant derivative on $\mathcal{L}$). Define the operator

$$D_i = \partial_i + K_i, \quad K_i = \partial_i K.$$

Then D_i defines a covariant derivative acting on sections of the Hodge line bundle $\mathcal{L}$.

Proof. Recall that under a change of trivialization

$$\Omega \mapsto e^{f(z)} \Omega.$$

The Kähler potential transforms as

$$K \mapsto K - f(z) - \bar{f}(\bar{z}),$$

and therefore

$$K_i \mapsto K_i - \partial_i f.$$

Applying the operator D_i to the transformed section gives

$$D_i(e^f \Omega) = (\partial_i + K_i - \partial_i f)(e^f \Omega).$$

Using

$$\partial_i(e^f \Omega) = e^f (\partial_i \Omega + (\partial_i f) \Omega),$$

we obtain

$$D_i(e^f \Omega) = e^f (\partial_i \Omega + (\partial_i f) \Omega) + (K_i - \partial_i f) e^f \Omega = e^f (\partial_i \Omega + K_i \Omega).$$

Hence

$$D_i \Omega \mapsto e^f D_i \Omega,$$

so $D_i \Omega$ transforms in the same way as Ω itself. Therefore D_i defines a covariant derivative on the line bundle $\mathcal{L}$. $\square$

Proposition 3.16. *Let*

$$V_{j\bar{j}} \in T_{1,0}^* \mathcal{M}_{cs} \otimes T_{0,1}^* \mathcal{M}_{cs} \otimes \mathcal{L}^{\otimes n} \otimes \bar{\mathcal{L}}^{\otimes m}$$

be a section. Then the covariant derivatives act as

$$D_i V_{j\bar{j}} = \partial_i V_{j\bar{j}} - \Gamma_{ij}^l V_{l\bar{j}} + n K_i V_{j\bar{j}},$$

$$D_{\bar{i}} V_{j\bar{j}} = \partial_{\bar{i}} V_{j\bar{j}} - \bar{\Gamma}_{i\bar{j}}^{\bar{l}} V_{j\bar{l}} + m K_{\bar{i}} V_{j\bar{j}},$$

where

$$K_i = \partial_i K, \quad K_{\bar{i}} = \partial_{\bar{i}} K.$$

Remark 3.17. The covariant derivative introduced above combines two geometric structures.

First, the complex moduli space $\mathcal{M}_{cs}$ carries the Weil–Petersson metric. The Levi–Civita connection of this metric, with Christoffel symbols Γ_{ij}^l , ensures that derivatives of tensors with indices on $\mathcal{M}_{cs}$ transform tensorially under holomorphic coordinate changes

$$z^i \mapsto z'^i(z).$$

Second, the holomorphic top form Ω is not a function on $\mathcal{M}_{cs}$ but a section of the Hodge line bundle

$$\mathcal{L} \rightarrow \mathcal{M}_{cs}.$$

Objects involving Ω therefore carry a nontrivial line bundle weight. The additional terms proportional to $\partial_i K$ arise from the Chern connection on $\mathcal{L}$.

Thus the operator D combines the Levi–Civita connection on $\mathcal{M}_{cs}$ with the Chern connection on the Hodge line bundle $\mathcal{L}$.

3.1.3 The n -Point Couplings

Proposition 3.18 (Iterated Griffiths transversality). *Let Ω_n be the holomorphic top form on the Calabi–Yau manifold M_n . For any sequence of indices $i_1, \dots, i_r$ corresponding to directions in the complex moduli space one has*

$$\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n \in \mathcal{F}^{n-r}.$$

Proof. This follows from Griffiths transversality, which states that

$$\nabla_i^{\text{GM}} \mathcal{F}^p \subset \mathcal{F}^{p-1}.$$

Since $\Omega_n \in \mathcal{F}^n$, repeated application of ∇^{GM} lowers the filtration index by one at each step. $\square$

Definition 3.19 (Primary horizontal subspace). The derivatives of Ω_n generate the *primary horizontal subspace*

$$H_H^n(M_n) := \text{Span} \{ \nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n \mid r \geq 0 \} \subset H^n(M_n, \mathbb{C}).$$

Proposition 3.20. *Let $0 \leq r \leq n$. Then*

$$\langle \nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n, \Omega_n \rangle = \int_{M_n} (\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n) \wedge \Omega_n = \begin{cases} 0, & 0 \leq r < n, \\ C_{i_1 \dots i_n}(z), & r = n. \end{cases}$$

Proof. By the previous proposition

$$\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n \in \mathcal{F}^{n-r}.$$

Thus its Hodge components have type (p, q) with $p \leq n - r$. Since Ω_n has type $(n, 0)$, the wedge product

$$(\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n) \wedge \Omega_n$$

has type $(p + n, q)$. For $r < n$ one has $p < n$, so this form cannot have type (n, n) and therefore integrates to zero. For $r = n$ the component of type $(0, n)$ may contribute, giving a non-vanishing quantity which defines $C_{i_1 \dots i_n}(z)$. $\square$

Definition 3.21 (n -point coupling). The *holomorphic n -point coupling* is defined by

$$C_{i_1 \dots i_n}(z) = \int_{M_n} (\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_n}^{\text{GM}} \Omega_n) \wedge \Omega_n.$$

Proposition 3.22. The functions $C_{i_1 \dots i_n}(z)$ are holomorphic on $\mathcal{M}_{cs}$.

Proof. In the expression

$$\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_n}^{\text{GM}} \Omega_n$$

only the terms involving ordinary derivatives ∂_i change the Hodge type sufficiently to produce a component of type $(0, n)$. Terms involving Christoffel symbols preserve the Hodge type and therefore do not contribute to the pairing with Ω_n . Hence one may replace ∇^{GM} by ∂_i in the above expression, showing that $C_{i_1 \dots i_n}(z)$ depends holomorphically on the moduli ($\bar{\partial}C = 0$). $\square$

Remark 3.23. The holomorphicity of the n -point couplings has a more intrinsic explanation in terms of the holomorphic structure of the cohomology bundle

$$\mathcal{H}^n = R^n \pi_* \mathbb{C} \otimes \mathcal{O}_{\mathcal{M}_{cs}}.$$

The Gauss–Manin connection induces the holomorphic structure on $\mathcal{H}^n$ by

$$\bar{\partial} := \nabla_{(0,1)}^{\text{GM}}.$$

Since the holomorphic top form satisfies

$$\nabla_i^{\text{GM}} \Omega_n = 0,$$

it is a holomorphic section of $\mathcal{H}^n$. The flatness of the Gauss–Manin connection implies

$$[\nabla_j^{\text{GM}}, \nabla_i^{\text{GM}}] = 0.$$

Therefore

$$\nabla_j^{\text{GM}} (\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n) = 0,$$

so that all iterated derivatives

$$\nabla_{i_1}^{\text{GM}} \dots \nabla_{i_r}^{\text{GM}} \Omega_n$$

remain holomorphic sections of $\mathcal{H}^n$. Since the polarization pairing is flat with respect to the Gauss–Manin connection, the pairing with Ω_n is again holomorphic. Thus the holomorphicity of the couplings $C_{i_1 \dots i_n}$ ultimately follows from the flatness of the Gauss–Manin connection and the fact that Ω_n is a holomorphic section of the cohomology bundle.

Remark 3.24. The coupling $C_{i_1 \dots i_n}$, originally introduced by Bryant and Griffiths, transforms as a section of

$$\mathcal{L}^2 \otimes \text{Sym}^n(T_{1,0}^*(\mathcal{M}_{cs})).$$

Moreover there exists a gauge choice for Ω such that the functions $C_{i_1 \dots i_n}(z)$ become rational functions in algebraic coordinates on $\mathcal{M}_{cs}$. These functions are determined by the Picard–Fuchs differential ideal. They are regular at maximally unipotent (large radius) points and typically develop poles along the discriminant locus of the Picard–Fuchs system.

3.1.4 Action of the Covariant Derivatives on the Hodge Basis

Proposition 3.25. *For $\Omega_n \in \mathcal{L}$, $D_i \Omega_n = \chi_i \in H^{n-1,1}(M_n)$, we have*

$$[D_i, D_{\bar{j}}] \Omega_n = -G_{i\bar{j}} \Omega_n.$$

Proof. Here one uses the Chern connection on the line bundle $\mathcal{L}$. In a local holomorphic frame the $(0,1)$ part of the connection coincides with the holomorphic structure, hence

$$D_{\bar{j}} = \partial_{\bar{j}}, \quad D_i = \partial_i + K_i.$$

Since the holomorphic top form satisfies $\partial_{\bar{j}} \Omega_n = 0$, one obtains

$$[D_i, D_{\bar{j}}] \Omega_n = -\partial_{\bar{j}} K_i \Omega_n = -\partial_i \partial_{\bar{j}} K \Omega_n = -G_{i\bar{j}} \Omega_n.$$

□

Proposition 3.26. *Similarly the commutator acting on χ_k takes the form*

$$[D_i, D_{\bar{j}}] \chi_k = -G_{i\bar{j}} \chi_k + R_{i\bar{j}k}^p \chi_p,$$

where $R_{i\bar{j}k}^p$ denotes the curvature tensor of the Weil–Petersson metric.

Proof. The appearance of the curvature term can be understood from the fact that the forms

$$\chi_k = D_k \Omega_n$$

provide a natural identification

$$T\mathcal{M} \cong H^{n-1,1}(M_n).$$

Hence the index k should be interpreted as a tangent index on the moduli space. When the commutator acts on χ_k , besides the curvature contribution $-G_{i\bar{j}}$ coming from the Hodge line bundle $\mathcal{L}$, one also obtains the curvature of the tangent bundle of the moduli space, which is precisely the Weil–Petersson curvature tensor $R_{i\bar{j}k}^p$. □

Next we consider the derivative $D_i \bar{\chi}_{\bar{j}}$. By Griffiths transversality it can be expanded in the basis of $(1, n-1)$ forms $(\bar{\chi}_{\bar{k}})$ and the $(0, n)$ form $\bar{\Omega}_n$,

$$D_i \bar{\chi}_{\bar{j}} = \alpha_{i\bar{j}}^{\bar{k}} \bar{\chi}_{\bar{k}} + \beta_{i\bar{j}} \bar{\Omega}_n.$$

Remark 3.27. The above expansion does not follow directly from the naive form of Griffiths transversality. Recall that Griffiths transversality states that the Gauss–Manin connection satisfies

$$\nabla F^p \subset F^{p-1} \otimes \Omega_{\mathcal{M}_{CS}}^1,$$

which only controls the behavior of the Hodge *filtration*. In particular, it does not by itself determine how the connection acts on the individual (p, q) –components of the Hodge decomposition.

To obtain the stronger statement used here, one needs additional structure on the Hodge bundle. Namely, one must use the real structure on the underlying flat bundle $\mathcal{H} = R^n \pi_* \mathbb{R} \otimes \mathbb{C}$ together with the Hodge metric to recover the C^∞ decomposition

$$\mathcal{H} = \bigoplus_{p+q=n} E^{p,q}, \quad E^{p,q} = F^p \cap \overline{F^q}.$$

With respect to this decomposition the Gauss–Manin connection splits into components that preserve the (p, q) –type and components that lower the holomorphic degree. Consequently, when acting on a section of $E^{1,n-1}$ the $(1, 0)$ part of the connection can only produce components in $E^{1,n-1}$ and $E^{0,n}$, which leads to the expansion written above.

A detailed justification requires the analytic description of variations of Hodge structure and the decomposition of the flat connection relative to the Hodge metric.

Remark 3.28. The expansion of the $E^{1,n-1}$ component in terms of $\bar{\chi}_k$ uses the fact that the forms $\{\chi_i\}$ span $H^{n-1,1}(M_n)$ via the Kodaira–Spencer identification between complex structure deformations and $(n-1, 1)$ classes. Consequently the conjugate forms $\{\bar{\chi}_k\}$ span $H^{1,n-1}(M_n)$. Therefore any component of $D_i \bar{\chi}_j$ lying in $E^{1,n-1}$ can be written as a linear combination of the $\bar{\chi}_k$.

Proposition 3.29. *For*

$$D_i \bar{\chi}_{\bar{j}} = \alpha_{i\bar{j}}^{\bar{k}} \bar{\chi}_{\bar{k}} + \beta_{i\bar{j}} \bar{\Omega}_n,$$

we have $\alpha = 0$, $\beta = G$. In other words,

$$D_i \bar{\chi}_{\bar{j}} = G_{i\bar{j}} \bar{\Omega}_n.$$

Proof. Using the definition

$$g_{i\bar{j}} = -i^{n^2} \langle D_i \Omega_n, \bar{\chi}_{\bar{j}} \rangle,$$

one obtains

$$g_{i\bar{j}} = -i^{n^2} \langle \partial_i \Omega_n, \bar{\chi}_{\bar{j}} \rangle = i^{n^2} \langle \Omega_n, \partial_i \bar{\chi}_{\bar{j}} \rangle = i^{n^2} \langle \Omega_n, D_i \bar{\chi}_{\bar{j}} \rangle = i^{n^2} \beta_{i\bar{j}} \langle \Omega_n, \bar{\Omega}_n \rangle.$$

Since

$$i^{n^2} \langle \Omega_n, \bar{\Omega}_n \rangle = e^{-K},$$

this implies

$$\beta_{i\bar{j}} = G_{i\bar{j}}.$$

On the other hand, we claim that

$$\langle \chi_l, D_i \bar{\chi}_{\bar{j}} \rangle = 0.$$

Recall the commutator relation

$$[D_i, D_{\bar{j}}] \Omega_n = -G_{i\bar{j}} \Omega_n.$$

Taking complex conjugation gives

$$[D_{\bar{j}}, D_i] \bar{\Omega}_n = -G_{i\bar{j}} \bar{\Omega}_n.$$

Pairing both sides with χ_l yields

$$\langle \chi_l, [D_{\bar{j}}, D_i] \bar{\Omega}_n \rangle = -G_{i\bar{j}} \langle \chi_l, \bar{\Omega}_n \rangle.$$

Therefore

$$\langle \chi_l, [D_{\bar{j}}, D_i] \bar{\Omega}_n \rangle = 0.$$

Expanding the commutator gives

$$0 = \langle \chi_l, D_{\bar{j}} D_i \bar{\Omega}_n \rangle - \langle \chi_l, D_i D_{\bar{j}} \bar{\Omega}_n \rangle.$$

The first term vanishes because $D_i \bar{\Omega}_n = 0$, since $\bar{\Omega}_n$ depends only on the anti-holomorphic moduli coordinates. Hence

$$0 = -\langle \chi_l, D_i D_{\bar{j}} \bar{\Omega}_n \rangle.$$

Using the definition $\bar{\chi}_{\bar{j}} = D_{\bar{j}} \bar{\Omega}_n$ we obtain

$$\langle \chi_l, D_i \bar{\chi}_{\bar{j}} \rangle = 0.$$

Thus,

$$0 = \langle \chi_l, D_i \bar{\chi}_{\bar{j}} \rangle = \alpha_{i\bar{j}}^{\bar{k}} \langle \chi_l, \bar{\chi}_{\bar{k}} \rangle = i^{n^2} \alpha_{i\bar{j}}^{\bar{k}} G_{l\bar{k}} e^{-K}.$$

Since the Weil–Petersson metric is non-degenerate we conclude

$$\alpha_{i\bar{j}}^{\bar{k}} = 0.$$

Therefore

$$D_i \bar{\chi}_{\bar{j}} = G_{i\bar{j}} \bar{\Omega}_n.$$

□

Remark 3.30. In the previous proof it is necessary to discuss more carefully what is meant by taking the complex conjugate of the identity

$$[D_i, D_{\bar{j}}] \Omega_n = -G_{i\bar{j}} \Omega_n.$$

Here the complex conjugation is taken as an operation inside the vector space H^n . Therefore the right-hand side is straightforward:

$$\overline{-G_{i\bar{j}} \Omega_n} = -\overline{G_{i\bar{j}}} \bar{\Omega}_n.$$

However, the complex conjugation of the left-hand side requires a more careful analysis. The key point is that the operators themselves also change under complex conjugation. Indeed, focusing on the differential operator ∂_i , one has locally

$$\partial_i = \partial_{x_i} + i \partial_{y_i}.$$

Taking complex conjugation sends i to $-i$, and therefore the operator transforms as

$$\overline{\partial_i} = \partial_{x_i} - i \partial_{y_i} = \partial_{\bar{i}}.$$

Consequently the covariant derivative changes accordingly,

$$\overline{D_i} = D_{\bar{i}},$$

and this explains why the complex conjugation of the commutator involves the exchange of the holomorphic and anti-holomorphic derivatives.

Remark 3.31. Similarly one may consider the ansatz

$$D_i \chi_j = \alpha_{ij}^k \chi_k + \cdots,$$

and from

$$0 = \langle D_i \chi_j, \bar{\chi}_k \rangle = i \alpha_{ij}^k G_{k\bar{k}}$$

it follows that

$$\alpha_{ij}^k = 0.$$

Remark 3.32. We may see the expansion

$$D_i \chi_j = \alpha_{ij}^k \chi_k + \beta_{ij}^{\bar{k}} \bar{\chi}_{\bar{k}},$$

which relies crucially on the Calabi–Yau threefold case. In general, Griffiths transversality implies

$$D_i \chi_j \in H^{n-1,1} \oplus H^{n-2,2}.$$

For a Calabi–Yau threefold one has $n = 3$, so that

$$H^{n-2,2} = H^{1,2}.$$

Since $\bar{\chi}_{\bar{k}}$ span $H^{1,2}$ (as the complex conjugates of the basis χ_k of $H^{2,1}$), the $(n-2, 2)$ component of $D_i \chi_j$ can be expanded in this basis. For higher-dimensional Calabi–Yau manifolds this identification no longer holds, and the above ansatz does not apply.

Further determination of D_i on the remaining basis elements involves the n -point coupling and its factorization through the Frobenius algebra structure, which depends on the dimension of the Calabi–Yau manifold. We will specialize to the case $n = 3$ in the following.

3.2 Special Geometry on the Complex Moduli Space of CY 3-Folds

3.2.1 Covariant Derivatives of the Hodge Basis

Special geometry is the principal differential–geometric structure governing the variation of Hodge structure over the complex moduli space $\mathcal{M}_{cs}$ of Calabi–Yau threefolds. It is characterized by the existence of a holomorphic prepotential $\mathcal{F}$ from which both the Kähler potential of the Weil–Petersson metric on $\mathcal{M}_{cs}$ and the holomorphic triple couplings are derived. From the perspective of curve counting, all the data involved correspond to genus–zero quantities.

Since in $D_i \chi_j = \alpha_{ij}^k \chi_k + \beta_{ij}^{\bar{k}} \bar{\chi}_{\bar{k}}$, one has already seen that

$$\alpha_{ij}^k = 0,$$

it remains to evaluate the coefficient

$$\beta_{ij}^{\bar{k}}.$$

Using the bilinear pairing on H^3 , one obtains

$$\beta_{ij}^{\bar{p}} e^{-K} G_{\bar{p}k} = -i \langle \chi_i, D_j \chi_k \rangle = -i \langle \partial_i \Omega, D_j (\partial_k + K_k) \Omega \rangle = i \langle \Omega, \partial_i \partial_j \partial_k \Omega \rangle$$

The quantity

$$C_{ijk} = i\langle \Omega, \partial_i \partial_j \partial_k \Omega \rangle$$

is the holomorphic Yukawa coupling. Hence

$$\beta_{ij}^{\bar{p}} e^{-K} G_{\bar{p}k} = -iC_{ijk}.$$

Hence

$$\beta_{ij}^{\bar{p}} = -e^K C_{ijk} G^{k\bar{k}}$$

and we can summarize this together with the generic equations derived in the previous section, the Tian–Todorov theorem and the non-degeneracy of the metric and the 3-point functions as

Appl. D_i	spans
Ω	$H^{3,0}$
$D_i \Omega = (\partial_i + K_i) \Omega = \chi_i$	$H^{2,1}$
$D_i \chi_j = -ie^K C_{ijk} G^{k\bar{k}} \bar{\chi}_{\bar{k}}$	$H^{1,2}$
$D_i \bar{\chi}_{\bar{k}} = G_{i\bar{k}} \bar{\Omega}$	$H^{0,3}$
$D_i \bar{\Omega} = 0$	$\emptyset$

We can now evaluate the left hand side

$$[D_i, D_{\bar{j}}] \chi_k = G_{k\bar{j}} \chi_i - e^{2K} C_{j\bar{m}\bar{n}} G^{m\bar{m}} G^{n\bar{n}} C_{ikm} \chi_n.$$

Thus

$$R_{i\bar{j}l}{}^k = \delta_l^k G_{\bar{j}i} + \delta_i^k G_{\bar{j}l} - C_{\bar{j}}^{km} C_{ilm}.$$

The non-holomorphic quantities $C_{\bar{j}}^{kl}$ are defined as

$$C_{\bar{j}}^{kl} = e^{2K} C_{j\bar{k}\bar{l}} G^{k\bar{k}} G^{l\bar{l}},$$

with

$$C_{\bar{j}\bar{k}\bar{l}} = (C_{jkl})^*.$$

Hence

$$C_{\bar{j}}^{kl} \in \mathcal{L}^{-2} \otimes T_{0,1}^* \otimes \text{Sym}^2(T^{1,0}).$$

Remark 3.33. One should note that although

$$\Omega \in \mathcal{L},$$

its complex conjugate does not naturally lie in the inverse bundle $\mathcal{L}^{-1}$. Indeed, complex conjugation sends

$$H^{n,0}(M) \longrightarrow H^{0,n}(M),$$

so strictly speaking

$$\bar{\Omega} \in \bar{\mathcal{L}},$$

the complex conjugate line bundle of $\mathcal{L}$.

However, the Hodge metric

$$h(\Omega, \bar{\Omega}) = i^n \int_M \Omega \wedge \bar{\Omega}$$

provides a Hermitian pairing on $\mathcal{L}$, which induces a natural identification

$$\bar{\mathcal{L}} \cong \mathcal{L}^{-1}.$$

Under this identification one may regard

$$\bar{\Omega}$$

as a section of $\mathcal{L}^{-1}$, which is the convention commonly used in the literature on special geometry.

3.2.2 Homogeneous Period Coordinates and the Prepotential

Let's review some notions on Calabi–Yau threefolds first.

Definition 3.34 (Kähler potential). Let M be a Calabi–Yau threefold and Ω the holomorphic three-form. The Kähler potential on the complex structure moduli space $\mathcal{M}_{cs}$ is defined by

$$e^{-K} = i \int_M \Omega \wedge \bar{\Omega}.$$

Definition 3.35 (Yukawa coupling). The holomorphic triple coupling on $\mathcal{M}_{cs}$ is defined by

$$C_{ijk} = \int_M \Omega \wedge \partial_i \partial_j \partial_k \Omega,$$

where ∂_i denotes differentiation with respect to the local complex structure parameters on $\mathcal{M}_{cs}$.

Remark 3.36. Griffiths transversality implies that covariant derivatives are not required in the definition of the Yukawa coupling. Indeed,

$$\partial_i \Omega \in H^{2,1}(M),$$

so successive derivatives of Ω already produce the relevant Hodge components contributing to the integral pairing. Consequently

$$C_{ijk} \in \text{Sym}^3(T_{1,0}^*(\mathcal{M}_{cs})) \otimes \mathcal{L}^2.$$

Proposition 3.37 (Integral expression for the Weil–Petersson metric). *The Weil–Petersson metric on the complex structure moduli space can be written entirely in terms of period integrals. One has*

$$G_{i\bar{j}} = i^{-n^2} e^K \int_M D_i \Omega_n \wedge \overline{D_j \Omega_n}.$$

For a Calabi–Yau threefold this becomes

$$G_{i\bar{j}} = - \frac{\int_M \chi_i \wedge \bar{\chi}_{\bar{j}}}{\int_M \Omega \wedge \bar{\Omega}} = - \frac{\langle \chi_i, \bar{\chi}_{\bar{j}} \rangle}{\langle \Omega, \bar{\Omega} \rangle}.$$

Remark 3.38. The minus sign together with Corollary 1.71 implies that the eigenvalues of the non-degenerate metric $G_{i\bar{j}}$ are positive. Physically this is important because the metric multiplies the kinetic terms of the moduli fields associated with complex structure deformations.

Definition 3.39 (Homogeneous period coordinates). Let M be a Calabi–Yau threefold, and choose a symplectic basis

$$\{A^I, B_I\}_{I=0}^h, \quad h = h^{2,1}(M),$$

of

$$H_3(M, \mathbb{Z}).$$

Let

$$\{\alpha_I, \beta^I\}_{I=0}^h$$

be the dual basis of $H^3(M, \mathbb{Z})$, normalized by

$$\int_{A^J} \alpha_I = \delta_I^J, \quad \int_{B_J} \beta^I = \delta_J^I, \quad \int_M \alpha_I \wedge \beta^J = \delta_I^J.$$

The holomorphic three-form Ω admits an expansion

$$\Omega = X^I \alpha_I - F_I \beta^I,$$

where

$$X^I = \int_{A^I} \Omega, \quad F_I = \int_{B_I} \Omega.$$

The functions X^I are called the homogeneous period coordinates.

Proposition 3.40 (Local Torelli). *Locally, the period map is an immersion. Hence the homogeneous periods X^I may be used as homogeneous coordinates on the enlarged moduli space.*

Remark 3.41. Since Ω is only defined up to nonzero rescaling,

$$\Omega \mapsto \lambda \Omega,$$

the periods X^I are not ordinary local coordinates on $\mathcal{M}_{cs}$, but homogeneous coordinates on the enlarged space parametrizing pairs (M, Ω) . Thus the symbols

$$\partial_I := \frac{\partial}{\partial X^I}$$

are understood as derivatives on this enlarged space.

Proposition 3.42 (Transversality identity). *Griffiths transversality implies*

$$\langle \Omega, \partial_K \Omega \rangle = 0.$$

Equivalently,

$$F_K - X^I \partial_K F_I = 0.$$

Proof. From

$$\Omega = X^I \alpha_I - F_I \beta^I$$

we obtain

$$\partial_K \Omega = (\partial_K X^I) \alpha_I - (\partial_K F_I) \beta^I.$$

Since X^I are the coordinate functions on the enlarged moduli space,

$$\partial_K X^I = \delta_K^I,$$

hence

$$\partial_K \Omega = \alpha_K - (\partial_K F_I) \beta^I.$$

Using the symplectic pairing,

$$\langle \Omega, \partial_K \Omega \rangle = \int_M \Omega \wedge \partial_K \Omega = F_K - X^I \partial_K F_I.$$

On the other hand, Griffiths transversality gives

$$\partial_K \Omega \in H^{3,0} \oplus H^{2,1},$$

so

$$\langle \Omega, \partial_K \Omega \rangle = 0.$$

This proves the claim. □

Corollary 3.43. *One has*

$$2F_K = \partial_K(X^I F_I).$$

Proof. By the product rule,

$$\partial_K(X^I F_I) = F_K + X^I \partial_K F_I.$$

Using

$$F_K - X^I \partial_K F_I = 0,$$

we obtain

$$\partial_K(X^I F_I) = 2F_K.$$

□

Definition 3.44 (Prepotential). Define the prepotential by

$$F = \frac{1}{2} X^I F_I.$$

Proposition 3.45. *The periods F_I are determined by the prepotential:*

$$F_I = \partial_I F.$$

Proof. By definition,

$$\partial_I F = \frac{1}{2} \partial_I (X^J F_J) = \frac{1}{2} \cdot 2F_I = F_I.$$

□

Proposition 3.46 (Homogeneity). *The prepotential F is homogeneous of degree 2, and each F_I is homogeneous of degree 1. More precisely,*

$$F_K = X^I \partial_I F_K,$$

and

$$2F = X^I \partial_I F.$$

Proof. Differentiating F , we have

$$F_K - X^I \partial_K F_I = 0.$$

And using $\partial_K F_I = \partial_I F_K$, we obtain

$$F_K = X^I \partial_I F_K.$$

Multiplying

$$F_I = \partial_I F$$

by X^I gives

$$X^I \partial_I F = X^I F_I = 2F.$$

□

Remark 3.47. The homogeneity also follows from the rescaling symmetry

$$\Omega \mapsto \lambda \Omega,$$

which induces

$$X^I \mapsto \lambda X^I, \quad F_I \mapsto \lambda F_I, \quad F \mapsto \lambda^2 F.$$

Proposition 3.48 (Yukawa couplings in period coordinates). *The Yukawa couplings satisfy*

$$C_{ijk} = X^I \partial_i \partial_j \partial_k F_I - F_I \partial_i \partial_j \partial_k X^I.$$

Moreover, writing

$$X_i^I := \partial_i X^I,$$

one has

$$C_{ijk} = -X_i^I X_j^J X_k^K F_{IJK},$$

where

$$F_{IJK} := \partial_I \partial_J \partial_K F.$$

Proof. From

$$\Omega = X^I \alpha_I - F_I \beta^I$$

we get

$$\partial_i \partial_j \partial_k \Omega = (\partial_i \partial_j \partial_k X^I) \alpha_I - (\partial_i \partial_j \partial_k F_I) \beta^I.$$

Pairing with Ω , one finds

$$C_{ijk} = \langle \Omega, \partial_i \partial_j \partial_k \Omega \rangle = X^I \partial_i \partial_j \partial_k F_I - F_I \partial_i \partial_j \partial_k X^I.$$

Next, since $F_I = \partial_I F$, we may regard F_I as functions of the homogeneous variables X^I . By the chain rule,

$$\partial_i F_I = X_i^J F_{IJ}.$$

Differentiating once more,

$$\partial_i \partial_j F_I = (\partial_i X_j^J) F_{IJ} + X_j^J X_i^K F_{IJK}.$$

Differentiating a third time gives

$$\partial_i \partial_j \partial_k F_I = (\partial_i \partial_j X_k^J) F_{IJ} + (\partial_j X_k^J) X_i^K F_{IJK} + (\partial_i X_k^J) X_j^K F_{IJK} + (\partial_i X_j^K) X_k^J F_{IJK} + X_i^L X_j^K X_k^J F_{IJKL}.$$

Substituting this into

$$C_{ijk} = X^I \partial_i \partial_j \partial_k F_I - F_I \partial_i \partial_j \partial_k X^I$$

and using the homogeneity relations

$$X^I F_{IJ} = F_J, \quad X^I F_{IJK} = 0, \quad X^I F_{IJKL} = -F_{JKL},$$

the remaining term is

$$C_{ijk} = -X_i^I X_j^J X_k^K F_{IJK}.$$

□

Remark 3.49. The last formula shows that, up to a minus sign, the Yukawa coupling is obtained by pulling back the third derivative of the prepotential along the period map.

3.2.3 The Big Moduli Space

Returning to the relation

$$F_I = \frac{\partial F}{\partial X^I},$$

we may view the variables X^I as coordinates on a larger space, called the *big moduli space*.

Definition 3.50 (Yukawa couplings in the big moduli space). Let Ω be the holomorphic three-form on the Calabi–Yau threefold W . Then the Yukawa couplings can be written as

$$C_{IJK} = \int_W \partial_I \partial_J \partial_K \Omega \wedge \Omega = X^L F_{IJKL} = -F_{IJK}.$$

Definition 3.51 (Symplectic interpretation). Using the symplectic pairing on $H_3(W, \mathbb{R})$, the variables X^I may be viewed as *positions*. The dual variables are defined by

$$P_I := F_I = \frac{\partial F}{\partial X^I},$$

which play the role of the corresponding *momenta*.

Proposition 3.52 (Local Torelli). *Let*

$$X = (X_\alpha^I) = (D_\alpha X^I)$$

with the convention

$$D_0 X^I = X^I.$$

Then the matrix X_α^I of size $(b_3/2) \times (b_3/2)$ is generically invertible.

Proof. This follows from the local Torelli theorem, which states that the period map for Calabi–Yau threefolds is locally injective. Consequently the derivatives $D_\alpha X^I$ are generically linearly independent. □

We denote the components of the inverse matrix by X_I^α .

Definition 3.53 (Kähler potential on the big moduli space). The rigid supergravity metric on the big moduli space is derived from the Kähler potential

$$\mathcal{K} = \frac{i}{2}(\bar{X}^I F_I - X^I \bar{F}_I) = \frac{1}{2}e^{-K}.$$

Equivalently,

$$\mathcal{K} = \frac{i}{2} \vec{\Pi}^\dagger \Sigma \vec{\Pi},$$

where

$$\vec{\Pi} = \begin{pmatrix} X^I \\ F_I \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is the symplectic form.

Proposition 3.54 (Metric on the big moduli space). *The metric induced by the above Kähler potential is*

$$G_{IJ} = \partial_I \partial_{\bar{J}} \mathcal{K} = \frac{i}{2} (F_{IJ} - \overline{F_{IJ}}) = -\text{Im}(F_{IJ}).$$

Proposition 3.55 (Christoffel symbols). *Let G_{IJ} be the metric on the big moduli space. Then the Christoffel symbols of the Levi-Civita connection are*

$$\Gamma_{IJ}^K = \frac{i}{2} C_{IJ}^K = \frac{i}{2} C_{IJL} G^{KL}.$$

Proof. On the big moduli space, the metric is given by

$$G_{I\bar{J}} = \partial_I \partial_{\bar{J}} \mathcal{K} = \frac{i}{2} (F_{IJ} - \overline{F_{IJ}}).$$

Hence

$$\partial_I G_{J\bar{L}} = \frac{i}{2} \partial_I F_{JL} = \frac{i}{2} F_{IJL},$$

since F is holomorphic.

For a Kähler metric, the only nonzero Christoffel symbols of the Levi-Civita connection are

$$\Gamma_{IJ}^K = G^{K\bar{L}} \partial_I G_{J\bar{L}}.$$

Substituting the previous identity, we obtain

$$\Gamma_{IJ}^K = \frac{i}{2} G^{K\bar{L}} F_{IJL}.$$

Now, using

$$C_{IJL} = -F_{IJL},$$

we get

$$\Gamma_{IJ}^K = -\frac{i}{2} G^{K\bar{L}} C_{IJL}.$$

□

Definition 3.56 (Hodge frame induced by period coordinates). Viewing X^I as local coordinates on the big moduli space, define

$$\hat{\Omega}_I := \frac{\partial \Omega}{\partial X^I}.$$

Using the period expansion

$$\Omega = X^I \alpha_I - F_I \beta^I,$$

we obtain

$$\hat{\Omega}_I = \alpha_I - F_{IJ} \beta^J.$$

Introducing the normalized period matrix

$$\tau_{IJ} := F_{IJ},$$

this becomes

$$\hat{\Omega}_I = \alpha_I - \tau_{IJ} \beta^J.$$

Proposition 3.57. *The forms $\hat{\Omega}_I$ provide a frame for the bundle*

$$\mathcal{H}^{3,0} \oplus \mathcal{H}^{2,1}.$$

Proof. By Griffiths transversality one has

$$\partial_{X^I} \Omega \in H^{3,0} \oplus H^{2,1}.$$

Moreover, the forms $\hat{\Omega}_I = \partial_{X^I} \Omega$ are linearly independent.

$$\int_{A_J} \hat{\Omega}_I = \partial_{X^I} \int_{A_J} \Omega = \partial_{X^I} X^J = \delta_I^J.$$

Hence the matrix of A -periods of $\hat{\Omega}_I$ is the identity, which implies that the forms $\hat{\Omega}_I$ are linearly independent. Since their number equals

$$\dim(H^{3,0} \oplus H^{2,1}) = h^{2,1} + 1,$$

they provide a frame for the bundle $\mathcal{H}^{3,0} \oplus \mathcal{H}^{2,1}$. □

Definition 3.58 (Period matrix). Let $\{\omega_I\}$ be a basis of $H^{3,0} \oplus H^{2,1}$ and $\{A_I, B_I\}$ a symplectic basis of $H_3(W, \mathbb{Z})$. The period matrix of $\{\omega_I\}$ is defined as

$$\left(\int_{A_J} \omega_I, \int_{B_J} \omega_I \right).$$

Proposition 3.59 (Normalized period matrix). *With respect to the basis $\{\hat{\Omega}_I\}$, the period integrals are*

$$\int_{A_J} \hat{\Omega}_I = \delta_I^J, \quad \int_{B_J} \hat{\Omega}_I = \tau_{IJ}.$$

Hence the period matrix takes the normalized form

$$(I, \tau).$$

Remark 3.60. The conjugate forms

$$\bar{\bar{\Omega}}_I = \alpha_I - \bar{\tau}_{IJ}\beta^J$$

provide a frame for

$$\mathcal{H}^{2,1} \oplus \mathcal{H}^{0,3}.$$

Note that α_I and β^J are real forms.

Proposition 3.61 (Polarization). *The symplectic pairing between these frames is*

$$\langle \hat{\Omega}_I, \bar{\bar{\Omega}}_J \rangle = 2i \operatorname{Im}(\tau_{IJ}).$$

Remark 3.62. By the Riemann bilinear relations, the matrix $\operatorname{Im}(\tau_{IJ})$ has signature

$$(h^{2,1}, 1),$$

that is, it has $h^{2,1}$ positive eigenvalues and one negative eigenvalue.

Definition 3.63 (Modified Hodge frame). In order to obtain a negative definite bilinear form on a space of dimension $b_3/2$, we construct a frame for

$$\mathcal{H}^{3,0} \oplus \mathcal{H}^{1,2}$$

of the form

$$\hat{\Theta}_I := \alpha_I - \mathcal{N}_{IJ}\beta^J.$$

Here the matrix $\mathcal{N}_{IJ}$ is in general non-holomorphic and will be determined by orthogonality conditions.

Definition 3.64 (Projection to $H^{1,2}$). Let

$$\Xi_J^{(12)}$$

be the projection of $\bar{\bar{\Omega}}_J$ to $H^{1,2}$. Explicitly,

$$\Xi_J^{(12)} = \bar{\bar{\Omega}}_J - \frac{\langle \bar{\bar{\Omega}}_J, \Omega \rangle}{\langle \bar{\Omega}, \Omega \rangle} \bar{\Omega}.$$

Proposition 3.65. *The frame $\hat{\Theta}_I$ is required to satisfy*

$$\langle \hat{\Theta}_I, \Omega \rangle = 0, \quad \langle \hat{\Theta}_I, \Xi_J^{(12)} \rangle = 0.$$

Remark 3.66. Using the period expansion

$$\Omega = X^I \alpha_I - F_I \beta^I,$$

the first condition gives

$$-F_I + X^J \mathcal{N}_{IJ} = 0.$$

Proposition 3.67 (Determination of $\mathcal{N}_{IJ}$). *Writing*

$$N_{IJ} := \tau_{IJ} - \bar{\tau}_{IJ},$$

the above conditions determine

$$\mathcal{N}_{IJ} = \bar{\tau}_{IJ} + \frac{N_{IL} N_{JK} X^L X^K}{N_{KL} X^K X^L}.$$

Remark 3.68 (Interpretation). The matrix τ_{IJ} is the period matrix associated with the Griffiths intermediate Jacobian, while

$$\mathcal{N}_{IJ}$$

is the period matrix of the Weil intermediate Jacobian. We will talk about it more formally later.

Remark 3.69 (Physical meaning). In $N = 2$ supergravity the matrix $\mathcal{N}_{IJ}$ determines the gauge kinetic terms of the vector fields.

3.2.4 Inhomogeneous Flat Coordinates on the Small Moduli Space

Definition 3.70 (Projective and Inhomogeneous Coordinates). Let

$$X^I = \int_{A_I} \Omega, \quad I = 0, 1, \dots, h$$

be the A -cycle periods of the holomorphic $(n, 0)$ form Ω , where $\{A_I\}$ forms a symplectic basis of non-intersecting cycles.

The *projective coordinates* on the moduli space are given by X^I . Passing to inhomogeneous coordinates we define

$$t^i = \frac{X^i}{X^0}, \quad i = 1, \dots, h.$$

Remark 3.71 (Choice of the inhomogeneous coordinates). The inhomogeneous coordinates

$$t^i = \frac{X^i}{X^0}$$

are defined only on the open locus where

$$X^0 \neq 0.$$

Since the periods

$$X^I = \int_{A_I} \Omega$$

are holomorphic functions on the complex moduli space, there is no global guarantee that X^0 never vanishes. Thus the coordinates t^i are defined only on a local patch of the moduli space.

In practice one chooses the symplectic basis of cycles so that near the large complex structure limit one of the periods satisfies

$$X^0 = 1 + O(z),$$

where z denotes a local coordinate on the moduli space. In particular X^0 is non-vanishing in a neighborhood of this point, so that the ratios t^i provide well-defined local coordinates.

Definition 3.72 (Prepotential in Inhomogeneous Coordinates). Let $F(X)$ denote the homogeneous prepotential of degree two in the projective coordinates X^I .

In inhomogeneous coordinates t^i we define the holomorphic function

$$\mathcal{F}(t) = \frac{F(X)}{(X^0)^2}.$$

Proposition 3.73 (Special Geometry Relations in Flat Coordinates). *In the inhomogeneous flat coordinates t^i the Kähler potential and the Yukawa couplings are given by*

$$e^{-K} = i |X_0|^2 \left[(\bar{t}^i - t^i)(\mathcal{F}_i + \bar{\mathcal{F}}_i) + 2(\mathcal{F} - \bar{\mathcal{F}}) \right],$$

$$C_{ijk} = \mathcal{F}_{ijk}.$$

Here

$$\mathcal{F}_i = \partial_i \mathcal{F} = \frac{\partial \mathcal{F}}{\partial t^i}.$$

Remark 3.74 (Geometric Meaning of the Coordinates). The ratios

$$t^i = \frac{X^i}{X^0}$$

are determined by the A -cycle periods

$$X^I = \int_{A_I} \Omega.$$

Thus the flat coordinates on the small moduli space are constructed from the periods of the holomorphic top form Ω with respect to a symplectic basis of cycles.

Remark 3.75 (Integrability condition for the prepotential). Previously one derives a differential relation among the periods in inhomogeneous but not necessarily flat coordinates. In terms of the connection on the Hodge bundle this relation can be written as the curvature identity (see Section 3.2.1)

$$-\partial_{\bar{j}} \left(G^{k\bar{m}} \partial_i G_{l\bar{m}} \right) = -\partial_{\bar{j}} \Gamma_{i\bar{l}}^k = R_{i\bar{j}l}^k = \delta_l^k G_{\bar{j}i} + \delta_i^k G_{\bar{j}l} - C_{\bar{j}}^{km} C_{ilm}.$$

This equation expresses a compatibility relation between the Kähler metric $G_{i\bar{j}}$ and the Yukawa couplings C_{ijk} .

If there exists a holomorphic prepotential $\mathcal{F}(t)$ in flat coordinates such that

$$C_{ijk} = \partial_i \partial_j \partial_k \mathcal{F}(t),$$

then the metric and the Yukawa couplings are determined by $\mathcal{F}(t)$ through

$$e^{-K} = i \left[(t^i - \bar{t}^i)(\mathcal{F}_i + \bar{\mathcal{F}}_i) + 2(\mathcal{F} - \bar{\mathcal{F}}) \right], \quad C_{ijk} = \mathcal{F}_{ijk}.$$

In this situation the above curvature relation is automatically satisfied. Conversely, the curvature relation can be interpreted as a necessary compatibility (integrability) condition ensuring that the geometric data $(G_{i\bar{j}}, C_{ijk})$ can locally be derived from a single holomorphic function $\mathcal{F}(t)$.

Proposition 3.76 (Period vector in terms of the prepotential). *Let $t^i = X^i/X^0$ be the inhomogeneous coordinates and let*

$$\mathcal{F}(t) = \frac{F(X)}{(X^0)^2}$$

be the holomorphic prepotential in these coordinates.

Then the period vector

$$\vec{\Pi} = \begin{pmatrix} F_I \\ X^I \end{pmatrix}$$

can be written as

$$\vec{\Pi} = X^0 \begin{pmatrix} 2\mathcal{F}(t) - t^i \partial_i \mathcal{F}(t) \\ \partial_i \mathcal{F}(t) \\ 1 \\ t^i \end{pmatrix}.$$

Remark 3.77 (Expansion of the holomorphic three-form). Let $\{\alpha_I, \beta^I\}$ be a symplectic basis of $H^3(M, \mathbb{C})$. The holomorphic three-form can be expanded as

$$\Omega = X^I \alpha_I - F_I \beta^I.$$

Choosing the gauge $X^0 = 1$ in the Kähler line bundle $\mathcal{L}$ and denoting the corresponding form by $\hat{\Omega}$, we obtain

$$\hat{\Omega} = \alpha_0 + t^a \alpha_a - \partial_a \mathcal{F} \beta^a - (2\mathcal{F} - t^a \partial_a \mathcal{F}) \beta^0.$$

Definition 3.78 (Hodge-adapted basis of $H^3(M, \mathbb{C})$). In the gauge $X^0 = 1$ one can complete $\hat{\Omega}$ to a basis of $H^3(M, \mathbb{C})$ by defining

$$\begin{aligned} \hat{\chi}_a &= \alpha_a - \partial_a \partial_b \mathcal{F} \beta^b - (\partial_a \mathcal{F} - t^b \partial_b \partial_a \mathcal{F}) \beta^0, \\ \hat{\chi}^a &= -\beta^a + t^a \beta^0, \\ \hat{\Omega}^0 &= \beta^0. \end{aligned}$$

Remark 3.79. Although the forms $\hat{\chi}_a$ seem to correspond to the $(2, 1)$ variations of the holomorphic three-form, they are actually not of pure Hodge type in general.

Indeed, writing

$$\Omega = X^0 \hat{\Omega},$$

and using

$$D_i \Omega = (\partial_i + K_i) \Omega = \chi_i,$$

we obtain

$$\frac{1}{X^0} D_i \Omega = \partial_i \hat{\Omega} + (\partial_i \log X^0 + \partial_i K) \hat{\Omega}.$$

Since

$$\frac{1}{X^0} D_i \Omega \in H^{2,1},$$

the ordinary derivative $\partial_i \hat{\Omega}$ generally contains an additional component along $\hat{\Omega}$, i.e. a $(3, 0)$ contribution. Hence

$$\hat{\chi}_i := \partial_i \hat{\Omega}$$

lies in

$$F^2 = H^{3,0} \oplus H^{2,1},$$

rather than in $H^{2,1}$ itself.

Proposition 3.80 (Holomorphic Picard–Fuchs system). *In the basis*

$$(\hat{\Omega}_0, \hat{\chi}_b, \hat{\chi}^b, \hat{\Omega}^0)$$

the holomorphic derivatives act as

$$\partial_a \begin{pmatrix} \hat{\Omega}_0 \\ \hat{\chi}_b \\ \hat{\chi}^b \\ \hat{\Omega}^0 \end{pmatrix} = \begin{pmatrix} 0 & \delta_a^c & 0 & 0 \\ 0 & 0 & C_{abc} & 0 \\ 0 & 0 & 0 & \delta_a^b \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \hat{\Omega}_0 \\ \hat{\chi}_c \\ \hat{\chi}^c \\ \hat{\Omega}^0 \end{pmatrix}.$$

Remark 3.81 (Relation to the Gauss–Manin connection). The above first order system can be interpreted as the holomorphic first order form of the Picard–Fuchs differential equations.

Let

$$(\hat{\Omega}_0, \hat{\chi}_b, \hat{\chi}^b, \hat{\Omega}^0)$$

be the chosen basis of H^n . In this basis the holomorphic derivatives act by

$$\partial_a v = C_a v,$$

where v denotes the column vector of coefficients of a section expanded in this basis, and C_a is the matrix appearing in Proposition 3.80.

Equivalently, the Gauss–Manin connection takes the form

$$\nabla^{GM} = \partial_a + C_a,$$

where the operator on the right hand side is understood to act on the coefficient vector of a section written in the above basis. In this sense C_a is the connection matrix of the Gauss–Manin connection in this frame.

The matrix C_a is nilpotent and may be viewed as a special case of the nilpotent connection matrix of the Gauss–Manin operator.

However, the structure above is much more restrictive than that of a general Gauss–Manin system, since it relies on the existence of the unique holomorphic $(n, 0)$ form and on the special geometry relations.

Remark 3.82 (From the Gauss–Manin system to the Picard–Fuchs equation). Let

$$\Pi = \int_{\Gamma} \hat{\Omega}_0$$

be a period, where $\Gamma \in H_n(X, \mathbb{Z})$ is a flat cycle. Integrating the system of Proposition 3.80 over Γ gives the first order system for the period vector

$$(\Pi, \Pi_a, \Pi^a, \Pi^0) = \left(\int_{\Gamma} \hat{\Omega}_0, \int_{\Gamma} \hat{\chi}_a, \int_{\Gamma} \hat{\chi}^a, \int_{\Gamma} \hat{\Omega}^0 \right),$$

namely

$$\begin{aligned} \partial_a \Pi &= \Pi_a, \\ \partial_a \Pi_b &= C_{abc} \Pi^c, \\ \partial_a \Pi^b &= \delta_a^b \Pi^0, \\ \partial_a \Pi^0 &= 0. \end{aligned}$$

In the one–parameter case these equations reduce to

$$\partial \Pi = \Pi_1, \quad \partial \Pi_1 = C(z) \Pi^1, \quad \partial \Pi^1 = \Pi^0, \quad \partial \Pi^0 = 0.$$

Eliminating the intermediate variables Π_1, Π^1, Π^0 gives

$$\partial^2 \Pi = C(z) \Pi^1,$$

$$\partial \left(\frac{1}{C(z)} \partial^2 \Pi \right) = \Pi^0,$$

and therefore

$$\partial^2 \left(\frac{1}{C(z)} \partial^2 \Pi \right) = 0.$$

Thus the period Π satisfies a fourth order linear differential equation. This scalar equation is the Picard–Fuchs equation associated with the Gauss–Manin system.

References

- [1] A. Klemm, *The B-Model Approach to Topological String Theory on Calabi-Yau n-Folds*, in *B-Model Gromov–Witten Theory*, E. Clader and Y. Ruan (eds.), Trends in Mathematics, Birkhäuser/Springer, Cham, 2018, pp. 79–397.
- [2] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge Studies in Advanced Mathematics, Vol. 76, Cambridge University Press, Cambridge, 2002.
- [3] P. Griffiths and J. Harris, *Principles of Algebraic Geometry*, John Wiley & Sons, New York, 1978.
- [4] D. A. Cox and S. Katz, *Mirror Symmetry and Algebraic Geometry*, Mathematical Surveys and Monographs, Vol. 68, American Mathematical Society, 1999.